The Standard Model from a Cubic Lattice

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Abstract

The Standard Model arises as the unique low-freedom representation of a finite bijection on a $d = 3$ simple-cubic lattice governed by the lattice wave equation. Two qualifications are stated at the outset because they bound every claim below. First, the cubic lattice is not itself derived: among the fourteen Bravais lattices, simple cubic is selected as the one compatible with an observed SU(3) factor (only the cubic crystal system supplies a three-dimensional point-group irrep; §4.6, Theorem 7b), so the lattice is an empirically-anchored structural commitment, forced by the observed gauge group rather than by the substratum alone. Second, $d = 3$ is fixed by empirical filters (propagating gravity, stable matter, boundary-entropy concordance; §3.2), not derived from the construction. *Given* those commitments, an extensive amount follows by representation theory with no further freedom, and that is this paper's content: the gauge group SU(3) × SU(2) × U(1), three chiral generations, the Higgs as a composite scalar in the singlet staggered taste, anomaly cancellation with the observed hypercharges, and the absolute generation count locked at exactly three by coupling-degree minimization jointly with the condensate-stabilizer accounting of Theorems 5 and 7 — each step proved end-to-end as a chain of theorems on the equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$. The foundational result that quantum mechanics emerges as the description of any embedded observer under structural conditions C1–C3 is established in a companion paper [Main], where it is stated as an identification via the Barandes correspondence rather than a parameter-free derivation. The gauge coupling strengths follow quantitatively: the structural induced coupling $1/\alpha_0 = 23.25$ (computed analytically from the one-loop staggered vacuum polarization), combined with a universal threshold $\delta_0 = 10.0$ (determined by the U(1) row; a 2-loop staggered VP rebuild indicates δ_0 is not reproduced by strict 2-loop PT and is accordingly treated as empirical input — see §6.2) and a C_2-dependent threshold parameterized by a logarithmic resummation, reproduces all three SM couplings at M_Z via standard SM renormalization group running. The structural predictions are the specific value $1/\alpha_0 = 23.25$, the universality of δ_0 across gauge groups, and the analytic 1-loop computation of the $A \cdot B$ coefficient, consistent with the fit under the $1/N^2$ finite-size assumption and inconclusive across finite-size models (§6.2.1). Deriving the C_2-dependent threshold parameters from first-principles lattice perturbation theory is an open computational task that would upgrade the SU(2) and SU(3) couplings from retrodictions to strict predictions. Quanti-

tative predictions for SM observables include the Cabibbo angle $\lambda = 1/(\pi\sqrt{2})$ to 0.04%, the Wolfenstein $A = \sqrt{2/3}$ to 0.23σ, the Koide angle $\theta_0 = 2/9$ to 0.02%, all three PMNS angles within 1.1σ — with $\sin^2\theta_{12}$ confirmed by the post-JUNO global fit at 0.07σ — six fermion masses from a single empirical input to better than 1%, and the top mass matching the weak-scale relation $m_t = v/\sqrt{2}$ $(y_t(m_t) \approx 1)$ to 0.9% — a retrodiction whose fixed-point mechanism is open: one-loop running shows $y_t(M_{\mathrm{Pl}}) = 1$ yields $m_t \approx 207$ GeV, so the weak-scale coincidence is not delivered by the compositeness-scale normalization (§7.2). These predictions are forced by the cubic-lattice geometry and contain no parameters fit to the predicted quantities; they are best understood as a tight compression of the Standard Model onto the cubic-lattice commitment — a claim about the non-independence of the SM's structural facts — rather than as a derivation of the Standard Model from first principles (see §8.3, *Derivation status: the three tiers*).

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1. Introduction

The Standard Model contains approximately nineteen free parameters: three gauge couplings, nine fermion masses, four CKM angles, two Higgs parameters, and the QCD vacuum angle. None of these are derived from a deeper principle within the SM itself, and despite five decades of effort, no successful explanation of their values has emerged from grand unification, supersymmetry, string compactifications, or other extensions. This paper presents a derivation of the Standard Model that fixes the gauge group, the matter content, the discrete symmetries, and the gauge coupling values from a single structural input — a finite bijection on a $d = 3$ cubic lattice — with the primary chain proved end-to-end as a sequence of theorems and the quantitative predictions matching observation with no parameters fit to the predicted quantities.

Prerequisites. This paper treats as established the following result from a companion work [Main]: an observer embedded in a deterministic bijection on a finite configuration space, coupled to a hidden sector that is non-trivial (condition C1), slow (C2: $\tau_S \ll \tau_B$), and high-capacity (C3: $N_H \gg N_V$), necessarily describes the visible sector using unitarily evolving quantum mechanics; the three conditions are necessary as well as sufficient. The technical bridge from P-indivisible classical stochastic processes on finite configuration spaces to unitary quantum evolution on finite Hilbert spaces is Barandes' stochastic-quantum correspondence [28, 29], used as the companion paper's primary tool. We take the characterization theorem and the stochastic-quantum bridge as inputs and address the question they leave open: given that quantum mechanics is the observer's necessary description under C1–C3, what quantum field theory does the observer see when the substratum dynamics is the wave equation on a $d = 3$ cubic lattice?

The present paper reproves none of [Main]'s foundational results. Where a theorem below invokes a specific input — the P-indivisibility of reduced dynamics on finite bijections (used in §2.1), the Born rule $T_{ij}(t) = |U_{ij}(t)|^2$ (used in §5's detailed-balance chain), or the C2-necessity argument (used in §8.2's genericity theorem) — the invocation is stated as a known result and the reader is directed to [Main] for the proof. The SM-derivation chain developed here depends on these inputs but is otherwise self-contained.

This paper answers in the affirmative. The wave equation on a $d = 3$ cubic lattice — itself uniquely selected among second-order reversible nearest-neighbor dynamics by center independence, spatial isotropy, and linearity — determines the Standard Model's structure through three complementary chains of theorems. *Bottom-up:* the wave equation factors into staggered Dirac fermions (Susskind), center independence enforces exact chiral symmetry which mandates the Higgs mechanism, and the staggered tastes give exactly three fermion generations plus one composite Higgs doublet. *Gauge emergence:* coupling-degree minimization gives $K = 2d = 6$ internal components per site, the cubic rotation group fixes the eigenvalue multiplicities of the coupling matrix to $(3, 2, 1)$, and background independence promotes the commutant to local gauge invariance, yielding $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ uniquely. *Top-down:* the six anomaly conditions force the observed hypercharges, the trace-out makes SU(2) chiral while SU(3) remains vector-like, and exact T-invariance of the underlying bijection forces $\bar{\theta} = 0$ at the kinematic level of the lattice construction. This does not dissolve the strong CP problem, and §5.5 shows why it cannot be carried over to the Standard Model as it stands: the same premise that sets $\bar{\theta} = 0$ — detailed balance at every time scale — also makes the two Yukawa matrices real in a common basis, which forces the Jarlskog invariant to vanish. A construction with no CKM CP violation is not the observed theory, so either the inference from T-invariance to real Yukawas fails, in which case $\bar{\theta} = 0$ loses its derivation, or it holds and the construction is excluded. §5.5 sets out both branches. The gauge coupling values at M_Z follow quantitatively from a one-parameter chain whose single free parameter δ_0 is treated as empirical input — a 2-loop staggered vacuum-polarization rebuild indicates it is not reproduced by strict 2-loop perturbation theory and is accordingly fixed by the U(1) row (§6.2).

The status of these results admits a useful three-tier classification.

Tier 1 — Structural foundations. The minimal mathematical object is established (a finite set with a deterministic bijection of bounded coupling degree, partitioned into visible and hidden sectors), and the spatial dimension $d = 3$ is shown to be the unique self-consistent dimension by three independent filters: propagating gravity ($d \geq 3$), stable matter ($d \leq 3$), and the boundary entropy concordance $\rho_s/\rho_{\mathrm{crit}} = 2/(d - 1)$ (which equals unity only for $d = 3$). Renormalizability of the emergent Yang-Mills coupling at $d = 3$ is a downstream consequence rather than an independent filter (§3.2). These results are theorems about the equivalence class $[(S, \varphi)]/\mathcal{G}_{\mathrm{sub}}$; they do not depend on which specific dynamics is selected.

Tier 2 — The primary derivation chain. Center independence, spatial isotropy, and linearity uniquely select the wave equation. The wave equation determines staggered Dirac fermions, $K = 2d = 6$ from coupling-degree minimization, multiplicities $(3, 2, 1)$ from the cubic rotation group, local gauge invariance from background independence, chirality from the trace-out, the unique anomaly-free hypercharges, three generations from cubic symmetry, and one Higgs doublet. Each step is proved; the chain runs through twenty-nine theorems and lemmas. The gauge coupling prediction extends the chain quantitatively, with a structural input $1/\alpha_0 = 23.25$ (one-loop staggered vacuum polarization, analytic) and a universal threshold $\delta_0 = 10.0$ (empirically fixed by the U(1) row; a 2-loop staggered VP rebuild indicates δ_0 is not reproduced by strict 2-loop PT and is accordingly treated as empirical input — see §6.2). Deriving the C_2-dependent threshold from first-principles lattice PT is an open computational task.

Tier 3 — Redundant confirmations. Asymptotic freedom requiring $N_c \geq 3$, the minimal chiral group constraint, and the Jordan-Chevalley projection (Appendix A) provide independent second-route checks on the Tier 2 results without being load-bearing in the primary chain.

The paper is organized as follows. §2 sets out the minimal structure on which the framework's results depend. §3 establishes background independence and derives $d = 3$ from the three self-consistency filters. §4 develops the emergent quantum field theory through the bottom-up, gauge-emergence, and top-down chains. §5 derives $\bar{\theta} = 0$ from T-invariance and detailed balance at the kinematic level of the construction; §5.5 shows that the same premise forces the Jarlskog invariant to vanish, and sets out what follows. §6 extends the chain quantitatively through the gauge coupling prediction. §7 collects the quantitative predictions for the CKM matrix, the lepton sector, the fermion mass chain, the PMNS angles, and the Higgs mass. §8 discusses structural realism, observer genericity, the structural-vs-bijection-level distinction, and the neutrino sector. §9 concludes — including a synthesis with the cosmological application developed in [GR] and the substratum-level reconstruction theorem developed in [Substratum].

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2. The Minimal Structure

2.1 What the theorems require

Deterministic bijectivity. The P-indivisibility result used below (see Prerequisites) requires a bijection on a finite set: bijectivity guarantees recurrence (φ^N = id), and recurrence produces the non-monotonic distinguishability that defines P-indivisibility. The entire framework rests on QM emerging from marginalizing deterministic dynamics — if the dynamics were already stochastic, there would be nothing to derive. Determinism is the core of the framework and cannot be weakened. However, determinism does not require a lattice. A continuous Hamiltonian system on a compact phase space is also deterministic and bijective (by Liouville's theorem). The lattice is one way to implement determinism on a finite set; it is not the only way.

Finite boundary entropy. The gap equation [GR, §3] requires a finite number of degrees of freedom across the partition boundary: $\mathrm{S} = \mathrm{A}/\varepsilon^2$. This follows from the holographic entropy bound [1] applied to any finite-area surface, not from lattice structure. What matters is that the boundary carries finitely many modes — not that the bulk is a lattice.

Bounded coupling degree (locality). This assumption does the most work. The area law (§3.1) follows from the spatial Markov property, which requires range-1 coupling. Without bounded coupling, long-range correlations could produce volume-law entropy, breaking the Jacobson route to GR. Locality does not require a regular lattice. Any bounded-degree graph has the spatial Markov property for range-1 dynamics. The emergent-QM, dimension, and gravity theorems (Theorems 1–6) work on any bounded-degree graph with the right statistical properties; the gauge-group derivation is where the substratum's crystallography becomes visible, the $(3, 2, 1)$ multiplicities reading out the octahedral point group of the simple-cubic structure (§2.2, §4.6).

Statistical homogeneity and isotropy. The Myrheim-Meyer dimension [2] requires a well-defined notion of spacetime dimension, which emerges from the causal order statistics on a homogeneous, isotropic structure. The dispersion relation (§3.1) uses translation invariance. The dynamics selection (§4.1) uses isotropy. Exact lattice regularity is sufficient but not necessary for these. A random graph with statistical homogeneity and isotropy at large scales produces the same emergent-QM, dimension, and gravity results (Theorems 1–6); the $(3, 2, 1)$ gauge decomposition is the direct read-out of the cubic point group, which statistical isotropy alone does not supply (§2.2, §4.6).

Non-trivial coupling (C1) and slow bath with capacity (C2, C3). These are conditions on the partition, not on the lattice. They hold for any system with the right partition geometry, regardless of whether the underlying space is a lattice, a graph, or something else.

2.2 What the theorems do not require

A *regular lattice*. Any bounded-degree graph with statistical isotropy suffices for emergent QM, the emergent dimension, and the gravitational sector (Theorems 1–6). Within OI's universality class, the *gauge-group structure* and the quantitative predictions of §7 additionally require the simple-cubic Bravais lattice: the multiplicities $(3, 2, 1)$ are fixed by the octahedral point group on the six link directions (§4.5–4.6, §8.3), and a generic isotropic graph preserves only the spectral mode count $2d_s = 6$, not the cubic decomposition. The cubic Bravais type is an empirically-selected input, forced by the observed SU(3) factor (§8.3). The gauge group itself is a universality-class invariant, shared with any Standard-Model-reproducing class through different machinery ([Structure §2.3, §14.4]); the cubic-group commutant of §4.6 is OI's specific realization of it. A *specific alphabet size q.* All proofs work for any q ≥ 2; no prediction depends on q. A *specific dimensionality d.* The proofs work for any d ≥ 1; self-consistency selects d = 3 (§3.2). The *wave equation.* It is derived from center independence, isotropy, and linearity (§4.1), each of which follows from the structural properties listed above.

2.3 The minimal object

The theorems require exactly: a deterministic bijection on a finite set, whose state space factors into local degrees of freedom coupled by a bounded-degree graph with statistical isotropy, partitioned into visible and hidden sectors satisfying C1–C3. Everything else — the regular lattice, the alphabet, the dimensionality, the wave equation, quantum mechanics, general relativity — is either derived or irrelevant. The minimal object is the triple (S, φ, V): a finite set S, a bijection φ: S → S with bounded-degree coupling, and a partition V ⊂ S of the degrees of freedom into visible and hidden sectors.

2.4 Definition and the factorization problem

Let S = S_1 × S_2 × ... × S_N be a product of finite sets (the local state spaces), and let φ: S → S be a bijection. The *coupling graph* G_φ is the undirected graph on vertices {1, ..., N} where i and j are connected if the i-th component of φ(s) depends on the j-th component of s for at least one state s ∈ S. The coupling graph is not an additional structure imposed on the system — it is read off from φ.

What determines the product decomposition? An abstract finite set S does not come with a canonical factorization. The answer is that the factorization is determined by φ itself, as the one that *minimizes the coupling degree.* For the wave equation, numerical verification confirms this: the natural spatial factorization is the *unique minimizer* of coupling degree — 0% of random factorizations achieve it, and 100% have strictly higher coupling.

Theorem (factorization uniqueness for translation-invariant dynamics). *Let φ be a bijection on $S = (\mathbb{Z}/q\mathbb{Z})^N$ satisfying: (a) translation invariance, (b) range-1 coupling on the natural factorization, and (c) genuinely bidirectional coupling. Then the natural factorization is the unique minimizer of coupling degree among all product decompositions $S = S_1 \times \cdots \times S_M$ with $|S_k| = q$ for all k.* The key step is that the wave equation's addition $x_{i-1} + x_{i+1} - x_i^{\text{prev}}$ creates irreducible algebraic dependence among all input components (for $q > 2$), so any relabeling that splits or merges natural sites increases the coupling degree.

Extension to general graphs. The translation-invariant proof extends to any bounded-degree graph with polynomial growth exponent d and statistical isotropy. The argument proceeds in three steps. First, §3.2 establishes (via Gromov's theorem) that any such graph is quasi-isometric to $\mathbb{Z}^d$, so the asymptotic dimension is d. Second, the wave equation on a d-dimensional graph has $2d$ independent low-energy propagation modes (from heat kernel asymptotics: the spectral dimension equals d, giving d momentum-like parameters over two orientations each). Third, the integer-valued coupling degree $\delta(K)$ depends on the graph only through its spectral properties (heat kernel decay exponent, spectral dimension), which vary continuously under quasi-isometric deformations; since K is integer-valued and the minimizer is $K = 2d$ for $\mathbb{Z}^d$, it cannot change without a discontinuous jump in the spectral dimension — which quasi-isometries preserve. The factorization uniqueness is therefore a property of the equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$, not of the specific graph.

2.5 Space as coupling structure

Every "spatial" property used in the derivation chain is a property of G_φ. The *area* of a region V is the number of edges crossing from V to its complement. The *dimension* is extracted from the causal partial order via the Myrheim-Meyer estimator [2]. The *area law* follows from the spatial Markov property on any graph with range-1 dynamics. The *dispersion relation* requires translation invariance. The regular cubic lattice is one graph with these properties; it is not the only one.

2.6 The dissolution of the container problem

The coupling graph is not embedded in a pre-existing space. There is no ambient manifold in which the graph "lives." The vertices are labels for degrees of freedom; the edges record which degrees of freedom are dynamically coupled. Any spatial embedding is a representation for human convenience, not a physical fact. The coupling graph is not made of material. The vertices are not atoms of space, and the edges are not physical connections. A degree of freedom is a component of the state — it is an abstract mathematical entity, not an object. Asking "what is site i made of?" is a category error: it is a label for a variable, not a name for a thing.

The most common objection to discrete models of spacetime is the container problem: if space is a lattice, what does the lattice sit in? On the structural reading, the container problem does not arise. The coupling graph is defined by the bijection φ, not by embedding in a manifold. Two sites are neighbors because φ couples them — because the state at one affects the future of the other. This is a dynamical fact, not a spatial margin. Locality is defined by the dynamics, and space emerges from locality. At no point does anything need to be "inside" anything else. The relationship is analogous to a social network: two people are "connected" not because they occupy adjacent points in physical space, but because they interact. The topology is defined by the interactions, not by geography.

2.7 The alphabet as gauge freedom

Every prediction of the OI framework is independent of the alphabet size q. The gap equation $\hbar = c^3\varepsilon^2/(4G)$ contains no q. The Bekenstein-Hawking formula contains no q. The dispersion relation, area law, P-indivisibility, center independence, and Einstein's equations are all q-independent. Numerical confirmation: on a 1D ring of L = 40 sites, the wave equation produces significant P-indivisibility at every q tested from 2 to 64. No experiment, even in principle, could measure q.

Two systems (S, φ) and (S', φ') with different alphabet sizes q and q' are *physically equivalent* if they produce the same emergent transition probabilities. The analogy to electromagnetic gauge invariance is precise: the gauge potential A_μ is not physical; the field strength F_$\mu\nu$ is. In the OI framework, the mod-q state space is not physical; the coupling structure and its emergent consequences are. Operationally this means the *absolute scale* of the field value is unobservable — only the coupling structure and emergent transition probabilities are (the discrete analogue of field-normalization redundancy in ordinary field theory). The transformations realizing this are the alphabet's **additive-group automorphisms** — the size q, rescalings $x \mapsto \lambda x$ (for λ a unit), and shifts $x \mapsto x + c$ — not arbitrary relabellings of $\mathbb{Z}/q\mathbb{Z}$. A nonlinear relabelling preserves the orbit structure (hence the transition probabilities under relabelling) but changes the fluctuation dispersion, so it is not a symmetry of the physical observables; the gauge group is the additive-automorphism group, a proper subgroup of all permutations. This amplitude-scale (additive-automorphism) gauge is what the §4.1 linearity argument invokes; whether it is entailed by the q-size statement above, or is an additional principle, is an identified open structural question.

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3. Background Independence and the Selection of d = 3

3.1 Background independence

The companion paper [Main] treats the coupling graph as fixed. In general relativity, the spacetime geometry is dynamical. If space is the coupling graph, background independence requires the graph to evolve with the state: s(t+1) = φ_{s(t)}(s(t)), where each φ_s is a bijection but G_{φ_s} varies with s.

Bijectivity is automatic. The wave equation is second-order: x_i(t+1) = f(neighbors of i at time t) − x_i(t−1) mod q. The phase-space map F: (x(t−1), x(t)) → (x(t), x(t+1)) has an explicit inverse that uses x(t) to determine which graph to apply. Bijectivity of the phase-space map is automatic for any second-order reversible dynamics, regardless of whether the coupling is state-dependent. The P-indivisibility proof applies without modification.

The derivation chain survives under three constraints. (i) *Local graph-dependence:* G(x) at site i depends only on x_j within a bounded range of i. (ii) *Center-independent graph-dependence:* G(x) at site i does not depend on x_i, preserving center independence. (iii) *Statistical isotropy:* G(x) is statistically isotropic on large scales for typical configurations.

Under these constraints: the area law holds, center independence is preserved, the gap equation is unchanged, and the dynamics selection theorem applies at each step.

Explicit construction. On a ring of L ≥ 4 sites with alphabet ℤ/qℤ, define the right-neighbor assignment $R(i,x) = i + 2 \bmod L$ if $x_{(i+1) \bmod L} = 0$, and $R(i,x) = i + 1 \bmod L$ otherwise, with left neighbor $L(i) = i - 1 \bmod L$ fixed. The dynamics is $x_i(t+1) = (x_{L(i)}(t) + x_{R(i,x(t))}(t) - x_i(t-1)) \bmod q$. This satisfies: constraint (i) with range 1 ($R(i,x)$ depends only on x_{i+1}); constraint (ii) exactly ($R(i,x)$ does not depend on x_i); constraint (iii) by translation invariance. Bijectivity is verified by explicit inverse: $u_i = (v_{L(i)} + v_{R(i,v)} - w_i) \bmod q$, since v determines all neighbor assignments. Center independence, P-indivisibility, and graph regularity (undirected degree bounded in [3, 2] independently of the state) all hold for any L and q.

Local gauge invariance from background independence. With state-dependent coupling, the coupling matrix M (§4.4 below) becomes site-dependent: $M = M(\mathbf{n}, \hat{e}_j)$. At every site, $M(\mathbf{n}, \hat{e}_j)$ has K = 6 components with eigenvalue multiplicities (3, 2, 1). Since M varies from site to site, the commutant U(3) × U(2) × U(1) acts independently at each site. Under a site-dependent transformation $G(\mathbf{n})$:

$$\phi(\mathbf{n}) \to G(\mathbf{n})\,\phi(\mathbf{n}), \qquad M(\mathbf{n}, \hat{e}_j) \to G(\mathbf{n})\,M(\mathbf{n}, \hat{e}_j)\,G(\mathbf{n} + \hat{e}_j)^{-1}$$

The wave equation is invariant. This is local gauge invariance. The link variable $M(\mathbf{n}, \hat{e}_j)$ transforms as a gauge connection: the plaquette product $P(\mathbf{n}, j, k) = M(\mathbf{n}, \hat{e}_j)\,M(\mathbf{n} + \hat{e}_j, \hat{e}_k)\,M(\mathbf{n} + \hat{e}_k, \hat{e}_j)^{-1}\,M(\mathbf{n}, \hat{e}_k)^{-1}$ transforms in the adjoint ($P \to G(\mathbf{n})\,P\,G(\mathbf{n})^{-1}$), so $\mathrm{Re}\,\mathrm{Tr}(P)$ is gauge-invariant — it is the Wilson plaquette action [4], now derived rather than postulated.

Einstein's equations as self-consistency. The Jacobson thermodynamic argument [5], applied to local causal horizons on the state-dependent graph, requires three inputs — each proved here for the wave equation.

Lemma (Area law). *The entanglement entropy of a spatial region V in the wave equation scales as $S(V) = \eta\,|\partial V|$. This holds for both the linear wave equation over ℝ and the mod-q wave equation over ℤ/qℤ.*

Proof. Over ℝ: the wave equation is a system of coupled harmonic oscillators with a quadratic Hamiltonian determined by the adjacency matrix. The area-law theorem for Gaussian systems applies directly [3]. Over ℤ/qℤ: the wave equation has range $R = 1$, so for any $i \in V^\circ$ (interior of V), all neighbors lie inside V. With uniform initial conditions: $V^\circ \perp V^c \mid \partial V$ (spatial Markov property). By the chain rule: $I(V; V^c) = I(\partial V; V^c) \leq H(\partial V) \leq |\partial V| \cdot \log_2 q$. The mutual information scales as boundary area, not volume. □

Lemma (Relativistic dispersion). *The wave equation has exact dispersion relation* $\cos\omega = \cos k$.

Proof. Substituting $x_j(t) = A\exp(i(kj - \omega t))$: $e^{-i\omega} + e^{i\omega} = e^{-ik} + e^{ik}$, giving $\cos\omega = \cos k$. For small k: $\omega = |k| + O(k^3)$ — relativistic propagation with $v = 1$. □

Scope of the emergent-Lorentz claim (what is and is not established). The dispersion lemma establishes that the leading low-energy behavior is relativistic, with the first correction entering at $O(k^3)$ in the frequency (equivalently $O(\varepsilon^2 k^2)$ in ω^2); there is no linear Lorentz-violating term. This is necessary for emergent Lorentz invariance but is not the full statement. Three points. (i) *Rotational sector.* The leading rotational-symmetry-breaking operator on the cubic lattice transforms in a nontrivial representation of the octahedral group O_h (the E_g irrep of the symmetric rank-2 tensors), so it is forbidden in any cubic-invariant action; the lowest *allowed* anisotropy is the $O(k^4)$ operator, which is irrelevant by power counting (dimension six in $d = 3 + 1$). Spatial isotropy at leading order is therefore protected by the cubic symmetry already required for the gauge derivation (§4.5), not an independent assumption — though it remains a statement about the bare action. (ii) *Boost sector.* The lattice possesses a preferred frame (the rest frame in which update steps are simultaneous), and the lattice dispersion is not boost-invariant at the cutoff scale; boost invariance can hold only as an emergent infrared property. The deviation of the dispersion surface from boost-invariance is $O(k^4)$ (dimension six, irrelevant) with no lower-order term, so the boost sector is, at tree level, in the same position as the rotational sector: Lorentz violation is irrelevant and suppressed by $(E/M_{\mathrm{Pl}})^2$. (iii) *Radiative stability (open).* The tree-level irrelevance of both the rotational and boost Lorentz-violating operators does not by itself guarantee emergent Lorentz invariance in the interacting theory: radiative corrections can in principle generate a lower-dimension (relevant or marginal) Lorentz-violating operator that is not Planck-suppressed — the Lorentz fine-tuning problem familiar from effective-field-theory treatments of Planck-scale Lorentz violation. The standard field-theoretic resolution (supersymmetric suppression of the offending divergences) is not available here, as the framework is not supersymmetric. Cubic symmetry must be checked at loop level, not assumed: the lattice-field-theory analysis of Davoudi and Savage (Phys. Rev. D **86**, 054505) shows that on a hyper-cubic lattice the renormalization mixing into lower-dimensional operators with power-divergent coefficients is mixing among *angular-momentum-covariant* operators of definite J — an operator-identification effect — while the genuinely rotational-symmetry-violating contributions remain suppressed by $\mathcal{O}(a^2)$ at every order (unconditionally in $\lambda\phi^4$; in gauge theory when the gauge fields are smeared comparably to the matter fields). Since the E_g operator is an octahedral non-singlet, its forbiddenness *does* extend to all orders for any dynamics that preserves the octahedral symmetry of the action and measure — radiative corrections of an octahedral-invariant theory generate only octahedral-invariant operators. The loop-level protection of the rotational sector therefore reduces to a single question: whether the emergent theory preserves the octahedral symmetry exactly (treated in §3.1 (*A present-day constraint, and the rotational sector's distinct status*) and [Ch. 19 §19.3.7]). The mechanism by which rotational-symmetry restoration is known to hold at all orders is operator *smearing* — a definition of the effective operators that filters ultraviolet modes over a physical region — under which the rotational-violating contributions remain suppressed by $\mathcal{O}(a^2)$ at every order in $\lambda\phi^4$, and in gauge theory provided the gauge fields are likewise smeared. Whether the framework's substratum-to-observer coarse-graining map (§4.7.1.1) realizes a sufficient smearing of this type is now characterized, in the negative: the map's role in assigning gauge representations is fixed (§4.7.1.2), and an exact surface-Green's-function extraction of its boundary spectral measure on the linear substratum yields a power-suppressed local exponent ($dN/d\omega \propto \omega^4$; the $+1/+2$ estimator controls validated), invariant over thirty orders of magnitude in the slow-bath ratio and under generic quenched deformation — the map is neither an ultraviolet filter nor an infrared amplifier, the spectral-measure counterpart of the $R \approx \varepsilon$ kernel-width result of §3.1 (*A present-day constraint, and the rotational sector's distinct status*). The emergent-Lorentz claim should therefore be read as: relativistic to leading order with no linear violation (established at tree level, consistent with the one-loop and two-loop staggered checks of §6), with radiative stability resting not on operator smearing — which the characterized map does not supply — but on loop-level octahedral invariance of the emergent theory (the mechanism developed below and in [Ch. 19 §19.3.7]); the interacting (condensate-dressed) extension of the characterization is the stated residual. A second, structural ingredient is available within the framework itself: at $M = \mu I$ the substratum dynamics carries an exact custodial U(6) relating the gauge sectors (Theorem 5, revised; §4.6), broken only spontaneously by the equivariant condensate, so sector-differential Lorentz-violating coefficients admit no independent counterterm at the regulator level. The resulting sector-differential boost-violation is then controlled by the heavy cross-charged mass M_X generated at the condensate gap: a one-loop lattice computation of the boost-projected self-energy curvature, after absorbing the universal tree-level cone, finds the differential coefficient obeys a quadratic-with-logarithm decoupling law of the form $\delta c \sim \Delta C_2\,(\alpha_0/4\pi)\,(M_X/M_{\rm Pl})^2\,[\,c_L \ln(M_{\rm Pl}/M_X) + c_0\,]$ rather than a non-decoupling constant, so the heavy sector decouples as M_X rises. The decoupling *form* is established (the law survives both a scalar- and a fermion-line check, and fails — reverting to a non-decoupling logarithm — precisely when the cone is not absorbed, confirming the absorption is load-bearing); the normalization is bounded by the framework's own representation content: ΔC_2 is a difference of total gauge Casimirs across species, giving $\Delta C_2 \in [0.58, 2.08]$ — order unity. The representation-side point values are now fixed per bound: the clock-comparison pair (nucleon–electron, chirality-averaged) gives $\Delta C_2 \approx 0.9$–1.1 across the two U(1)-normalization conventions, and the electron coefficient — measured against the *induced* photon, whose boost coefficient is the vacuum-polarization-weighted average of the charged fermions', $w_f \propto N_c Q_f^2$ — gives $\Delta C_2 \approx 0.6$–0.7. The induced architecture produces one structural alignment worth recording: the photon coefficient lies within $\Delta C_2 \lesssim 0.03$ of the loop-weighted species mean (a weighted average cannot stray far from itself), so the photon-sector channel is suppressed roughly two orders below generic when referenced to that mean, with its residual sensitivity confined to the absorbed-cone reference definition — the one place the reference choice materially matters, owned by the kernel computation. With the computed kernel constants ($c_L = -9.4$, $d = -0.88$, superseding the order-unity placeholders) and the point values, the tightest ceiling refines from the worst-case figures below to $M_X \lesssim 4 \times 10^7$ GeV — a factor-two class shift that leaves the window's decade structure intact. What remains open on the normalization side is the absorbed-cone reference (photon channel only); the mechanism's remaining quantitative item is the condensate-gap determination of M_X itself.

Baryon safety of the cross-charged sector, the pinned reference, and the closed form of the window. Three further results complete the accounting. First, the cross-charged sector is baryon-safe at the gauge level, structurally: all matter carriers are mixed-index objects in $V_K \otimes V_K^*$ (§4.7.1.2), on which the U(6) adjoint action conserves index flow per block — baryon number, realized as V_3-flow/3, is conserved identically by all thirty-six generators, and the diquark vertices that gauge-mediated proton decay requires are absent because they need same-index carriers ($V \otimes V$), which the framework does not have. The cross-charged bosons are Pati–Salam-class vector leptoquarks, not SU(5)-class X/Y: the $\mathrm{Hom}(V_3, V_1)$ modes carry exactly the U_1 vector-leptoquark quantum numbers $(\mathbf{3}, \mathbf{1}, 2/3)$ — an identity of the forced hypercharge assignment, $Y_Q - Y_L = Y_u = 2/3$, obtained by two independent routes. The M_X floor is therefore leptoquark phenomenology rather than proton decay: $\sim 1.5 \times 10^6$ GeV ($K_L \to \mu e$ class) if the cross couplings are generation-off-diagonal, 10^3–10^4 GeV (collider and precision) if generation-diagonal — the generation structure being solution-level, like M_X itself. The refined window is $[10^3\text{–}10^6,\ \sim 4 \times 10^7]$ GeV: between one and a half and four and a half decades. Scope: this argument covers the gauge sector; baryon violation through non-gauge operators (the condensate and Yukawa sectors, higher-dimension operators) is a separate accounting item, not excluded here. Second, the absorbed-cone reference is pinned by the framework's own architecture rather than left as a scheme choice: spacetime geometry is classical-substratum-level ([GR §6.2]) and takes no emergent-loop corrections, so the gravitational-wave cone *is* the tree cone — gravity-referenced bounds (the GW170817 class) therefore constrain each species' full coefficient ($\Delta C_2^{\text{eff}} = \sum_f w_f C_2^f \approx 1.6$ for the induced photon, ceiling 4×10^{11} GeV, still the loosest), while matter-matter bounds constrain differences, as computed above. The same classical priority of geometry that dissolves the cosmological-constant problem pins the Lorentz-violation reference. Third, with M_X classified as solution-level (set by the equivariant block splittings, §4.7.1.1, not fixed at the lattice scale), the boost conditional reaches its closed form: it is discharged into one solution-level parameter confined to a one-and-a-half-to-four-and-a-half-decade window, baryon-safe, reference-pinned, and falsifiable from both sides — the same closure the framework accepts for the mass eigenvalues, and the terminal state available to it short of Layer 2(b) input. With $\alpha_0 = 1/23.25$ and the worst-case ΔC_2, the resulting ceilings are $M_X \lesssim 9 \times 10^7$ GeV against the tightest (clock-comparison) multi-species bound, 7.5×10^9 GeV against the electron SME coefficient, and 1.1×10^{12} GeV against photon-sector bounds, while the floor is set by cross-charged (leptoquark-like) phenomenology at 10^3–10^6 GeV: a window of two to five orders of magnitude remains. The mechanism therefore requires the cross-charged condensate scale to be intermediate, $M_X \lesssim 10^8$ GeV rather than GUT- or Planck-scale; an independent determination of M_X from the condensate-gap chain above this ceiling would falsify it. The bound-to-coefficient mapping and the cross-charged phenomenology floor are treated here at order of magnitude only. We situate this in the literature. The radiative Lorentz-fine-tuning problem is general to Planck-scale Lorentz violation (Collins, Perez, Sudarsky, Urrutia, and Vucetich, Phys. Rev. Lett. **93**, 191301), and its standard field-theoretic resolution — supersymmetric suppression of the offending divergences — is unavailable here, as the framework is not supersymmetric. The one non-supersymmetric protection mechanism consistent with the framework's structure is renormalization-group emergence: Lorentz-violating couplings flowing to zero at an infrared-attractive Lorentz-invariant fixed point (Chadha and Nielsen, Nucl. Phys. B **217**, 125 (1983); Bednik, Pujolàs, and Sibiryakov, JHEP **11** (2013) 064). This mechanism is natural in the present setting because the trace-out of [Main] is itself a coarse-graining toward the infrared, and it is the same mechanism on which emergent-quantum-mechanics-from-lattice constructions rely (e.g. arXiv:2207.09465); its documented limitation is that the inter-species velocity flow is only logarithmically fast, leaving a residual fine-tuning against the tightest multi-species bounds. Independently, the causal-set discreteness results (Bombelli, Henson, and Sorkin, *Discreteness without symmetry breaking*) establish that a finite-valency, locally-coupled discrete structure cannot carry exact Lorentz invariance microscopically — only a random, statistically isotropic structure can, and then only non-locally — which is consistent with the framework's concession of a preferred lattice frame and reinforces that Lorentz invariance here must be wholly emergent (an infrared property of the coarse-grained theory), never microscopic.

A present-day constraint, and the rotational sector's distinct status. The radiative-stability discussion above acquires a mild empirical handle from ultra-high-energy data, but the handle applies unevenly across the two subsectors, and the distinction is important. Ultra-high-energy observations now constrain the quadratic ($n = 2$, dimension-six) Lorentz-violating coefficient in the *isotropic* dispersion $E^2 = m^2 + k^2[1 + c_6(k/M_{\mathrm{Pl}})^2]$: the absence of neutrino splitting for KM3-230213A gives $\Lambda_{n=2} \gtrsim 1.1 \times 10^{19}$ GeV $\approx M_{\mathrm{Pl}}$ (Satunin, arXiv:2502.09548), and the GZK consistency of the Auger/HiRes proton spectra bounds $\delta_{n=2} \lesssim 10^{-23}$ at Lorentz factor $\sim 2 \times 10^{11}$. In the framework's parameterization these require the *isotropic / sector-differential* coefficient to satisfy $c_6 \lesssim \mathcal{O}(1)$, consistent with an order-unity coefficient at the lattice scale; this is genuine present-day exposure for the boost-adjacent sector treated above. It does **not**, however, control the rotational E_g anisotropy, and that sector is in a more precarious position. The rotational operator is a custodial-U(6) singlet — it carries spatial indices but is a gauge-sector singlet — so the power-law $(M_X/M_{\mathrm{Pl}})^2$ decoupling that suppresses the sector-differential boost violation does not apply to it; its protection is instead the cubic symmetry itself, operating at the level of operator *dimension* rather than coefficient. The leading rotational-violating operator is the E_g ($\ell = 2$) anisotropy, an octahedral non-singlet; the lowest octahedral-invariant anisotropy is the $\ell = 4$ cubic harmonic, dimension six. A direct computation of the emergent tree-level dispersion confirms this places the violation at dimension six: the fractional anisotropy $[\omega^2(\hat{n}_1) - \omega^2(\hat{n}_2)]/|k|^2$ scales as $(\varepsilon k)^2 = (E/M_{\mathrm{Pl}})^2$, with no dimension-four (k-independent) term. Whether this survives loops reduces, as in §3.1 (*Scope of the emergent-Lorentz claim*, point (iii)), to whether the emergent theory preserves the octahedral symmetry: octahedral-invariant dynamics cannot generate the E_g operator at any order. This corrects an earlier reading of Davoudi and Savage: the power-divergent operator mixing that arises without ultraviolet smearing is mixing among *angular-momentum-covariant* operators of definite J, not the generation of a relevant rotational-violating operator, which they find remains suppressed by $\mathcal{O}(a^2)$ at every order. The kernel width R/ε therefore sets the *coefficient* of the dimension-six rotational operator — an additional $(\varepsilon/R)^2$ suppression when $R \gg \varepsilon$ — and not its dimension; even $R/\varepsilon = \mathcal{O}(1)$ leaves the rotational violation Planck-suppressed. The differentiating angular signature of any residual — the $\ell = 4$ cubic harmonic aligned to the cosmic rest frame ([Main §3.5]), correlated across species — is recorded as the conditional forward signature F14 ([appendix-A §A.5]); being dimension six, its amplitude is $\sim (E/M_{\mathrm{Pl}})^2$ and unobservable at accessible energies. One structural consideration bears on the coefficient. The smearing kernel is the traced-out hidden-sector propagator, and a direct computation of its position-space width gives $R \approx \varepsilon$, flat as the slow-bath capacity ratio τ_B/τ_S is varied over thirty orders of magnitude: the framework's (large) observational incompleteness is incompleteness about the hidden *state*, not about spatial scale, and a state-average does not damp ultraviolet spatial modes. This fixes the dimension-six operator's coefficient at order unity rather than further-suppressed, but does not lift it from dimension six, so it is not by itself an exposure. The interpretation — that embedded observation does not actively *smear* the rotational sector, for the same reason the trace-out selects a rest frame and generates quantum mechanics — is developed in [Ch. 19 §19.3.7]; on the present reading it does not need to, because octahedral symmetry and locality already place the violation at dimension six. The open question that remains is correspondingly narrower, and is not a sidereal-bound fine-tuning: whether the emergence preserves octahedral symmetry *exactly* at loop level, or introduces an octahedral-breaking effect — for instance an effective momentum-shell structure in the trace-out — not visible in the octahedral-invariant checkerboard partition. *[Status note, 2026-08-04: the octahedral-invariance premise has a second dependency, more direct than the trace-out. The effective operator is octahedral-invariant only if the condensate Σ is, and the O-invariance of the dynamics does not descend to the solution:*

the framework's own gauge mechanism is spontaneous breaking of a dynamical symmetry by that condensate (§4.6). Theorem 5 takes Σ *to be O-equivariant, and that equivariance is a hypothesis on the solution rather than a consequence of the dynamics. It is not secured by the observed gauge group alone: a condensate with the same block multiplicities* $(3, 2, 1)$ *— hence the same stabilizer* $\mathrm{U}(3) \times \mathrm{U}(2) \times \mathrm{U}(1)$ *— but with eigenspaces not aligned to the O-irreps generates an* E_g *anisotropy at dimension four rather than dimension six. What secures the alignment is the geometric identification of the blocks with the link-direction irreps, used independently in the link-carrier construction (§4.7.1.2) and in the inter-generation prescription of §7.1. In the other direction the premise is robust: an equivariant condensate leaves the* E_g *channel exactly null at every block splitting, and a staggered spatial modulation of the condensate does not lift it, so the rotational sector is insensitive to the condensate's spatial structure and turns only on its internal alignment.]* That loop-level question, not the kernel width, is the arbiter. Two results bear on it. If the microscopic action, the measure, and the partition are each exactly O_h-invariant about the observer's centre, the effective action obtained by marginalizing the hidden sector is O_h-invariant to all orders (substitute $H \to g^{-1}H$ in the defining integral for $g \in O_h$; the Jacobian is unity), and every ingredient of the checkerboard trace-out — the dispersion, the position-space partition, and the timescale-selected momentum region $\{\omega(k) < \omega_{\text{cut}}\}$ — is O_h-invariant, so no direction-dependent shell is available to break the symmetry. A numerical check on the rounded interacting dynamics is consistent with this: with the extraction validated against the analytic tree coefficients $(1/3,\ -1/135,\ -1/36)$ to the percent level and the allowed $\ell = 4$ cubic harmonic detected at high significance, the E_g coefficient is zero within errors at the 2×10^{-5} level in the perturbative regime — a discrimination of order 10^3 against the allowed anisotropy — and consistent with zero in structure-factor measurements of the thermalized strong-coupling state. What remains is larger-volume strong-coupling statistics and the check that the $O(\varepsilon/R_H)$ observer-position boundary anisotropy is $\ell = 4$-aligned. This tree-level premise — that the trace-out commutes with the lattice octahedral group — is itself grounded in the factorization criterion that fixes the gauge content ([Main §3.1, §3.5]): an observer-centered partition breaks translations, boosts, and rotations about distant points (each maps visible boundary sites into the hidden sector), but the octahedral group about the observer's own centre is not of this kind, and the horizon-isotropic visible region is spherical and hence octahedrally symmetric, so this O_h preserves the visible/hidden factorization and descends to an exact symmetry of the effective theory. The criterion that selects the gauge symmetries (factorization-preserving visible-internal symmetries), the one that protects the rotational sector (a factorization-preserving spatial symmetry, hence dimension six), and the one that exposes the boost sector (boosts are factorization-breaking) are a single criterion; the rotational-protected/boost-exposed asymmetry is therefore structural rather than incidental.

Clarification of the protection mechanisms (relationship and remaining gap).

Two of the ingredients above are not independent: operator smearing is the coarse-graining-toward-the-infrared realization of the renormalization-group-emergence mechanism, so the question "does the substratum-to-observer map smear sufficiently?" and the question "does the theory flow to a Lorentz-invariant infrared fixed point?" are two descriptions of the same requirement. The custodial U(6) ingredient is logically separate and stronger where it applies: it removes sector-*differential* Lorentz-violating counterterms at the regulator level, and the resulting differential coefficient obeys the established power-law-with-logarithm decoupling in M_X — a genuine power-law suppression, not merely the logarithmically-slow RG flow. The remaining gap is sharper than "radiative stability in general": the custodial argument controls the *sector-differential* (inter-species) boost-violation that the multi-species bounds constrain most tightly, but the *overall* (sector-common) Lorentz-violating operator in the rotational sector has no analogous power-law ally and rests on the smearing/RG-emergence mechanism alone, whose inter-species flow is only logarithmically fast. The rotational sector is therefore the less-protected of the two sub-problems. Finally, the preferred frame conceded in (ii) is not free-floating: on identification with the observed universe it is the frame in which the substratum update steps are simultaneous, which is operationally the cosmic rest frame (the CMB frame); the framework makes no claim to derive this identification, but records that the preferred frame, if physical, is not an additional tunable structure but the cosmological rest frame already present in the data.

Lemma (Lattice Bisognano-Wichmann). *For the wave equation, whose low-energy effective theory is Lorentz-invariant, the entanglement Hamiltonian for a half-space bipartition has the Bisognano-Wichmann form. The reduced state for a Rindler observer is thermal at $T_U = \hbar\kappa/(2\pi c k_B)$, with corrections at $O(\varepsilon^2\kappa^2/c^2)$.*

Proof. The mod-q wave equation is a coupled harmonic oscillator system — the class for which the lattice BW theorem is proved analytically [6]. Its low-energy dispersion is relativistic (the preceding lemma). Thermality of the Unruh effect is robust against UV modifications of the dispersion relation, with corrections scaling as $O(\varepsilon^2\kappa^2/c^2)$. □

Theorem (Discrete Einstein equation). *For the state-dependent wave equation on a bounded-degree graph G(x) satisfying constraints (i)–(iii), the Jacobson thermodynamic argument produces:*

$$\kappa_{OR}(i, j; G(x)) = \alpha \cdot \mathcal{T}_{ij}(x, G(x))$$

where κ_{OR} is the Ollivier-Ricci curvature [7], $\mathcal{T}_{ij}$ is the discrete stress-energy, and $\alpha = 8\pi G/(c^4\eta)$.

Proof. (i) *Area law.* The entanglement entropy scales as $S(V) = \eta\,|\partial V|$ (area-law lemma above), holding for any bounded-degree graph with range-1 dynamics.

(ii) *Unruh temperature.* The lattice Bisognano-Wichmann lemma gives $T = \hbar\kappa/(2\pi c k_B)$ at local causal horizons. (iii) *Clausius relation.* At each edge (i,j): the area variation under graph deformation is captured by κ_{OR} (which measures, via optimal transport, the deviation from flat space), giving $\delta A/A = -\kappa_{OR}\,\delta\lambda$. Substituting into $\delta Q = T\,\delta S$: $\mathcal{T}_{ij}\,\delta\lambda\,\delta A = T\cdot\eta\cdot(-\kappa_{OR}\,\delta\lambda\,A)$, which rearranges to $\kappa_{OR} = \alpha\cdot\mathcal{T}_{ij}$. (iv) *Continuum limit.* Van der Hoorn et al. [8] prove $\kappa_{OR}\to$ Ric on random geometric graphs converging to a Riemannian manifold. Kelly et al. [9] prove $\sum\kappa_{OR}\to\int R\sqrt{g}\,d^dx$. Together: the discrete Einstein equation converges to $G_{\mu\nu}+\Lambda g_{\mu\nu} = (8\pi G/c^4)\,T_{\mu\nu}$. □

Every orbit is self-consistent by construction: the state determines the graph, which determines the entropy and temperature, and the Clausius relation constrains the graph's response to energy flux — paralleling continuum GR where the Einstein equations are the equations of motion, not constraints on separate dynamics. The mathematical framework is unchanged: the full dynamics F(u, v) = (v, WE_{G(v)}(v) − u) is a single bijection on the finite phase space. The ontological object is still (S, F, V).

Lattice corrections to Einstein's equations. The derivation invokes continuum results in the final step (Jacobson). Decomposing Jacobson's argument into its four inputs and checking each independently: (i) $\delta S = \eta\,\delta A/\varepsilon^2$ (area-law entropy) — proved for Gaussian systems over $\mathbb{R}$ [3] and for any nearest-neighbor dynamics over $\mathbb{Z}/q\mathbb{Z}$ via the spatial Markov property (area-law lemma above). No Lorentz invariance enters. **Exact on the lattice.** (ii) $T = \hbar\kappa/(2\pi c k_B)$ (Unruh temperature) — requires the Bisognano-Wichmann theorem. On the lattice, the BW form is proved analytically for coupled harmonic oscillator systems [6] and shown robust against UV modifications of the dispersion relation with corrections at $O(\varepsilon^2\kappa^2/c^2)$. **Approximate.** (iii) $\delta Q = \int T_{\mu\nu}k^\mu d\Sigma^\nu$ (energy flux) — a definition on the emergent manifold. No Lorentz invariance required. **Exact.** (iv) The Raychaudhuri equation — kinematic: it holds on any pseudo-Riemannian manifold as a consequence of the definition of curvature. **Exact.** Since inputs (i), (iii), and (iv) are lattice-exact, corrections enter only through input (ii):

$$G_{\mu\nu}+\Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}\times\left[1+\mathcal{O}\left(\frac{\varepsilon^2\kappa^2}{c^2}\right)\right]$$

For the cosmological horizon, $\varepsilon\kappa/c = 2l_P\cdot H/c\sim 10^{-61}$, giving corrections of order 10^{-122}. For solar-mass black holes, $\varepsilon\kappa/c\sim 10^{-38}$, giving corrections of order 10^{-76}. The corrections are negligible for any horizon much larger than the Planck scale.

The $\eta = 1/4$ coefficient. The Bekenstein-Hawking formula $S = A/(4l_P^2)$ follows from $\varepsilon = 2l_P$ (the gap equation [GR, §3]): one entropy unit per cell of area $\varepsilon^2 = 4l_P^2$, giving $S = A/\varepsilon^2 = A/(4l_P^2)$. This cannot be confirmed via entanglement entropy because of the species problem — the entanglement entropy coefficient is species-dependent (Srednicki found $\eta_{\text{ent}}\approx 0.295$ per scalar field), while the BH entropy is species-independent. The thermal matching route avoids this by counting classical boundary degrees of freedom (one per ε^2 cell, independent of field content) rather than any particular field's vacuum entanglement.

3.2 Why d = 3

The theorems work for any $\mathrm{d} \geq 1$. Three independent self-consistency filters converge on $\mathrm{d} = 3$.

Propagating gravity requires d $\geq$ 3. The Weyl tensor vanishes for $\mathrm{d} \leq 2$. The Jacobson derivation requires propagating gravitational degrees of freedom.

Stable matter requires d $\leq$ 3. The Coulomb potential in d dimensions gives unstable atoms for $\mathrm{d} \geq 4$ (Ehrenfest [10]; the non-existence of normalisable bound states of the d-dimensional Coulomb Hamiltonian for $\mathrm{d} \geq 4$ is the corresponding quantum-mechanical statement). Gravitational orbits are unstable for $\mathrm{d} \geq 4$ [11]. Without stable matter, no observers exist to define the partition.

The intersection of $\mathrm{d} \geq 3$ and $\mathrm{d} \leq 3$ is $\mathrm{d} = 3$, already uniquely. The remaining filter is a further consistency check rather than a logically independent constraint.

On circularity. The stable-matter filter invokes the existence of observers ("without stable matter, no observers exist to define the partition"), and a reasonable objection is that this imports a known fact about our $\mathrm{d} = 3$ world — that stable matter exists — to select $\mathrm{d} = 3$, rather than deriving $\mathrm{d} = 3$ from the substratum. If the selection rested *only* on this filter, the objection would have force: it would be an anthropic selection presented as a derivation. The objection is answered by the third filter, which is purely structural and observer-independent. The boundary-entropy concordance below is a property of the partition geometry alone — it is computed from the Friedmann equation, the shell theorem, and horizon-mode counting ([GR §7.2]), none of which mention observers or matter stability — and it equals unity only at $\mathrm{d} = 3$. This filter selects $\mathrm{d} = 3$ without invoking the existence of observers at all. The logical structure is therefore: propagating gravity ($d \geq 3$) and stable matter ($d \leq 3$) together fix $d = 3$, and the circularity worry attaches only to the anthropic-flavored stable-matter step. That worry is answered by the boundary-entropy concordance — a non-anthropic filter computed from partition geometry with no reference to observers or matter stability — which independently excludes $d \geq 4$ by equalling unity only at $d = 3$. A referee who rejects the stable-matter filter as circular is therefore left with propagating gravity plus the boundary-entropy concordance, which still yield $d = 3$: the anthropic step is *substitutable*, not indispensable. This does not make the selection a pure a priori derivation — the concordance filter uses the observed near-flatness, so $d = 3$ remains a self-consistency selection that consumes d-dependent emergent physics (§8.3), independently anchored by the observed gauge group — but it does defeat the specifically anthropic form of the circularity objection.

Boundary-entropy concordance (d = 3 gives $\rho_s = \rho_{\mathbf{crit}}$). In d spatial dimensions, the boundary entropy ratio is:

$$\frac{\rho_s}{\rho_{\text{crit}}} = \frac{2}{d-1}$$

This equals unity only for d = 3. For d = 2: overclosure. For d ≥ 4: gravitational deficit with no source.

d	ρ_s / ρ_crit	Status
2	2	Overclosure
3	**1**	**Exact concordance**
4	2/3	Deficit

The d = 3 case is derived explicitly in [GR §7.2] from the Friedmann equation, shell theorem, and horizon-mode counting. The general-d formula is obtained by running the same construction with Friedmann in d spatial dimensions, $H^2 = 16\pi G_d\rho/[d(d-1)]$.

Self-contained statement of the concordance ratio. Because the circularity rebuttal above relies on this filter being structural and observer-independent, the origin of the ratio $\rho_s/\rho_{\text{crit}} = 2/(d-1)$ is stated here rather than only cited. Two densities are compared, both computed from the partition geometry alone. (a) The *critical density* is read off the d-dimensional Friedmann equation $H^2 = 16\pi G_d\,\rho/[d(d-1)]$ at spatial flatness, giving $\rho_{\text{crit}} = d(d-1)H^2/(16\pi G_d)$ — this is the energy density at which the spatial sections are flat, a property of the gravitational dynamics with no reference to observers or matter content. (b) The *boundary (horizon) entropy density* ρ_s is the energy density associated with the horizon-mode count via the holographic area law: the number of boundary modes scales as the horizon area $A \sim r_H^{d-1}$, the energy per mode is set by the horizon temperature $T_H \sim 1/r_H$, and dividing the resulting boundary energy by the enclosed volume $V \sim r_H^d$ gives $\rho_s \sim r_H^{d-1} \cdot r_H^{-1}/r_H^d = r_H^{-2} \sim H^2$ with a d-dependent geometric coefficient. Carrying the coefficients (horizon area formula, mode counting, and the Friedmann normalization, all in d spatial dimensions) yields the ratio $\rho_s/\rho_{\text{crit}} = 2/(d-1)$. Neither density uses the existence of observers, stable matter, or any anthropic input; both are computed from the gravitational dynamics and the holographic mode count, which is why this filter is independent of the stable-matter filter and answers the circularity objection. *What is imported from [GR]:* the horizon-mode counting and the area-law coefficient are derived in [GR §3, §7.2]; this paragraph states the structure of the ratio and its observer-independence but relies on [GR] for the coefficient derivation, which a referee checking the concordance filter would verify there. The qualitative result — that the ratio depends only on d and equals unity solely at

$d = 3$ — does not depend on the coefficient details, only on the scaling $\rho_s \sim H^2$ and $\rho_{\text{crit}} \sim H^2$ with the stated d-dependence.

Downstream consequence: renormalizability at d = 3. At the selected d = 3, the emergent Yang-Mills gauge theory (derived in §4.4–§4.6) has dimensionless coupling: $[g] = (3-d)/2 = 0$. The emergent QFT is therefore renormalizable in 3+1 dimensions. This is a property of the derived structure at d = 3, not an independent filter on d: at d $\neq$ 3, the coupling-degree minimizer would give a different $K = 2d$, the rotation-group decomposition of $2d$ link directions would give different multiplicities, and the resulting gauge theory would have a different operator structure — so asking whether "that" Yang-Mills is renormalizable at that d is not a d-independent question.

The intersection of the three filters — d $\geq$ 3, d $\leq$ 3, and d = 3 concordance — is d = 3 uniquely. Removing the concordance filter still gives d = 3 from the other two; adding concordance sharpens to a consistency check that the framework passes exactly rather than by a numerical coincidence.

Integer dimension is derived, not presupposed. The self-consistency argument might appear to presuppose that the coupling graph has a well-defined integer dimension. In fact, every bounded-degree graph falls into one of three growth classes: exponential ($|B(r)| \sim k^r$), fractal ($|B(r)| \sim r^\alpha$ with non-integer α), or integer-polynomial ($|B(r)| \sim r^d$ with integer d). Self-consistency excludes the first two.

Exponential growth fails because the wave equation on such graphs does not produce Lorentz-invariant dispersion (the spectral gap doesn't close correctly), so Jacobson's derivation of Einstein's equations fails. Additionally, $d_{\text{eff}} \to \infty$ is incompatible with stable matter ($d \leq 3$).

Fractal growth fails because Jacobson's argument requires the Raychaudhuri equation, which requires geodesic congruences on a manifold — fractal spaces don't support them. The staggered fermion decomposition requires a hypercubic BZ structure ($\eta \in \{0,1\}^d$), which doesn't exist on fractal graphs. The cubic group decomposition of $2d$ link directions — which constrains the gauge structure (§4.6) — requires cubic lattice symmetry, absent in fractals.

Only *integer-polynomial growth* survives. By Gromov's theorem (finitely generated groups of polynomial growth are virtually nilpotent) and statistical isotropy, the graph is quasi-isometric to $\mathbb{Z}^d$ for integer d. The three filters then select $d = 3$.

Corollary ($d = 3$ is necessary, not contingent). *Any bijection whose coupling graph has $d \neq 3$ produces internal contradictions in its emergent description. The coupling graph dimension is not a free parameter or a property of a specific φ — it is the unique value compatible with self-consistency.*

Relation to the anthropic principle. This is not a standard anthropic argument. The standard argument invokes a multiverse of dimensions and selection bias: many d exist; we observe $d = 3$ because only there can observers exist. The present argument is different: the framework's definition requires an observer as a constitutive element (the partition is defined relative to an observer). The requirement that such an observer exist, combined with the framework-internal concordance calculation, constrains d. The observer is not selecting from a pre-existing landscape but is a structural prerequisite for the definition to have content.

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4. The Emergent Quantum Field Theory

The wave equation on a d = 3 lattice determines not just QM and GR but the specific QFT the observer sees. This section derives the Standard Model's structure through three complementary routes: bottom-up from the lattice dynamics to fermionic matter (§§4.1–4.3), gauge emergence from multi-component dynamics (§§4.4–4.6), and the staggered species / consistency constraints that fix the matter content and discrete symmetries (§§4.7–4.12).

4.1 The wave equation as lattice Klein-Gordon

The OI lattice is a d-dimensional hypercubic lattice Λ = ℤ^d with spacing ε = 2 l_p. The dynamics on this lattice is not a free choice — it is uniquely determined by three requirements.

Theorem (Dynamics selection). *Among all second-order reversible nearest-neighbor dynamics on a d-dimensional lattice of alphabet size q, the requirements of (i) center independence, (ii) spatial isotropy, and (iii) linearity uniquely select the discrete wave equation. This holds for any $q \geq 2$ and $d \geq 1$.*

Remark (scope). The nearest-neighbor restriction is an assumption beyond bounded coupling degree (A3): center-independent families with next-nearest-neighbor couplings exist and tune the low-energy speed. Nearest-neighbor is adopted as the minimal-coupling commitment consistent with Theorem 6's factorization; the dispersion — and the 4/45 coefficient of §6.4 — is conditional on it.

Proof. The update function has the form $x_i(t{+}1) = (f(\text{neighbors of } i \text{ at time } t) - x_i(t-1)) \bmod q$. Center independence requires that f not depend on x_i itself: f reduces to a function of the $2d$ neighboring values only. Spatial isotropy requires f to be symmetric under all permutations of the neighbors corresponding to lattice symmetries. Linearity over $\mathbb{Z}/q\mathbb{Z}$ requires $f(a+a') = f(a)+f(a')$. The unique function satisfying all three is $f = \alpha(x_1 + x_2 + \cdots + x_{2d}) \bmod q$ for some constant α. The propagation speed is $v = \alpha$, so $\alpha = 1$ gives the maximum lattice-scale speed. This is the discrete wave equation. □

Each requirement has a physical justification.

Center independence is necessary for emergent QM. The mechanism is information screening. For the checkerboard partition, the adjacency matrix A is bipartite: $A_{VV} = 0$ (no visible–visible edges). In the center-independent (CI) case, the update $x_i(t+1) = [h_{i-1}(t) + h_{i+1}(t) - x_i(t-1)] \bmod q$ does not contain $x_i(t)$. Therefore $X_V(t+1)$ carries no information about $X_V(t)$; conditioning on $X_V(t+1)$ constrains $H(t)$ but does not screen $X_V(t)$ from the future. Since $H(t+1)$ depends on $X_V(t)$, the path $X_V(t) \to H(t+1) \to X_V(t+2)$ bypasses the present: the process is non-Markovian. In the center-dependent (CD) case, the update $x_i(t+1) = [x_i(t) + h_{i-1}(t) + h_{i+1}(t) - x_i(t-1)] \bmod q$ contains $x_i(t)$ explicitly. Conditioning on $(X_V(t), X_V(t+1))$ gives $|V|$ linear equations in $|H|$ unknowns over $\mathbb{Z}/q\mathbb{Z}$; generically (for q prime) the hidden state $H(t)$ is fully determined, so the augmented process is Markov. Center coupling makes the present an explicit function of the visible past, allowing it to determine the hidden state and screen the past from the future. Without center coupling, the hidden sector retains unscreened correlations — which is P-indivisibility.

Spatial isotropy is required for Lorentz invariance: without rotational symmetry, the dynamics has a preferred spatial direction, breaking the isotropy that Lorentz invariance demands.

Linearity is selected by four independent criteria. First, *amplitude-scale gauge invariance* (a sharpening of the q-gauge principle of §2.7). Linearity is **equivalent** to the linearized fluctuation dispersion being independent of the background field amplitude: demanding amplitude-independence forces the per-neighbour Jacobian to be constant, hence the update affine, hence (with masslessness fixing the constant) linear. The discrete statement is exact — *all first finite-differences over $\mathbb{Z}/q\mathbb{Z}$ are base-point-independent iff the rule is affine as a function* (Lemma, linearity-equivalence). Linearity is therefore **necessary given that the field-value (amplitude) scale is gauge** — equivalently, that the alphabet's additive automorphisms $x \mapsto \lambda x + c$ are unphysical (operationally: the embedded observer has no access to the absolute field-value scale — see [Substratum §4, Theorem 24 Remark]). This is a distinct (and stronger) principle than the *q-size* gauge freedom of §2.7: q-independence of predictions does not by itself entail invariance under rescaling the field at fixed q. The necessity is conditional on amplitude-scale gauge; whether §2.7's q-gauge entails it is identified as an open structural question (see §2.7; amplitude-scale gauge is stated authoritatively in [Substratum §4, Theorem 24, Remark]). Second, over $\mathbb{R}$, nonlinear center-independent dynamics have propagation speed $v < 1$, so linearity gives maximum speed. Third, over $\mathbb{Z}/q\mathbb{Z}$, among all linear CI dynamics $f = \alpha(l + r) \bmod q$ with prime $q \geq 5$, $\alpha = 1$ uniquely maximizes P-indivisibility. Fourth, modified (nonlinear) dispersion relations break the thermality of the Unruh effect — the KMS condition holds only for linear dispersion $\omega = |k|$. Since the Jacobson derivation (§3.1) requires thermal equilibrium at horizons, only the wave equation supports the full GR chain. All four criteria converge on the same dynamics.

$$\phi(n, t+1) = \phi(n-1, t) + \phi(n+1, t) - \phi(n, t-1)$$

Theorem 1. *The OI wave equation is the massless lattice Klein-Gordon equation. Center independence is equivalent to the vanishing of the lattice mass term.*

Proof. The general linear update is $\phi(n, t+1) = \alpha\,\phi(n, t) + \beta[\phi(n-1, t) + \phi(n+1, t)] + \gamma\,\phi(n, t-1)$. Reversibility: $\gamma = -1$. Center independence: $\alpha = 0$. Isotropy + unit speed: $\beta = 1$. The d'Alembertian $\Box_{\text{lat}}\phi = -\alpha\,\phi = 0$: massless KG. □

4.2 Factorization into Dirac operators

Theorem 2 (Susskind [12]). $D_{st}^2 = -\frac{1}{4}\Box_{lat}$. *The OI wave equation is equivalent to the massless staggered Dirac equation.*

The bosonic and fermionic descriptions are related by an exact change of variables on the lattice.

Remark (measure reduction). Theorem 2 is an operator identity: it identifies D_{st}^2 with $-\frac{1}{4}\Box_{\text{lat}}$ acting on lattice fields. For the standard staggered ChPT machinery used in §6 and §7.5 to apply with its conventional measure factors (e.g., the $1/(16\pi^2 f^2)$ measure factor of Aubin–Bernard SχPT), one needs a stronger statement: that the substratum's path-integral measure on real scalar bijections $\varphi : S \to S$ reduces, under the Susskind change of variables, to the standard Berezin pseudofermion measure on staggered fermion fields in the relevant correlator sector. This measure-equivalence statement is established at Level 2 rigor as follows.

The substratum is real-valued and, on linear dynamics with energy conservation, has a stationary distribution that is Gaussian on energy shell at large L (sphere-Gaussian projection theorem). The substratum two-point function is then $\langle\varphi(x)\varphi(y)\rangle = [\Box_{\text{lat}}^{-1}]_{xy}$. The standard MC code (cf. §7.5) does not use Grassmann variables directly; it uses the *pseudofermion* representation, in which fermion determinants are rewritten as bosonic Gaussian integrals over auxiliary fields with covariance $(D_{\text{st}}^\dagger D_{\text{st}})^{-1}$. With $D_{\text{st}}^\dagger = -D_{\text{st}}$ (anti-Hermiticity of the staggered Dirac operator), $D_{\text{st}}^\dagger D_{\text{st}} = -D_{\text{st}}^2 = +\frac{1}{4}\Box_{\text{lat}}$ by Theorem 2, so the pseudofermion covariance is $4\,\Box_{\text{lat}}^{-1}$. Both substratum and pseudofermion measures are bosonic Gaussian; their covariances differ by a factor of 4 absorbable into field normalization. The measure equivalence is therefore established up to (i) the standard sphere-Gaussian projection at large L (classical), and (ii) cycle ergodicity for the specific bijection determining which energy shell is sampled.

The cycle ergodicity assumption deserves explicit acknowledgment. Pure linear dynamics on a cubic lattice is *integrable*, not chaotic: the lattice frequencies $\omega(\vec{k}) = 2\sin(|\vec{k}|/2)$ are not linearly independent over $\mathbb{Q}$, so orbits fill a sub-torus of the energy shell rather than the full shell. The Gaussianization argument therefore has two natural interpretations: (way out i) the cubic-lattice + finite-set discretization introduces small but non-zero non-linearity (from rounding rules in the bijection update), providing the chaos needed for full-shell sampling; or (way out ii) the "Gaussian on energy shell" claim is shorthand for "Gaussian on the sub-torus actually sampled," which is a smaller phase space and would modify the substratum-pseudofermion covariance match. Way out (i) is the standard interpretation for lattice-substrate frameworks. Numerical verification of way out (i) for OI's specific bijection has been carried out via two-trajectory separation with per-site RMS amplitude measurement inside the spreading cone (the correct Lyapunov-relevant diagnostic, distinguishing chaotic amplitude amplification from Lieb-Robinson ballistic spreading). The measurement gives a chaotic Lyapunov exponent $\lambda_{\text{max}} = 0.0092 \pm 0.0001$ per timestep in the thermodynamic limit, N_{levels}-independent for $N_{\text{levels}} \geq 16$, with exponential amplitude growth in the cone confirmed against power-law alternatives at high statistical significance (exponential-vs-power-law residual ratio ~ 2.2 across $\log(\|\bar{\Delta\phi}\|/\delta) \in [0, 1.0]$). Way out (i) is therefore *empirically demonstrated* rather than merely adopted: the rounded bijection is genuinely chaotic, the integrable linear dynamics having been broken by the rounding rule's threshold-crossing nonlinearity. The cycle-ergodicity question for OI's specific bijection is closed at chat-track numerical scale; analytical derivation of the λ_{max} coefficient and the structural source of N_{levels}-independence remains open and would elevate this result to Level 2(a) in the bridge-gap classification of §7.5. Four further diagnostics on the rounded bijection fall on the chaotic side: spectral continuity of local observables (broadband, versus the linear control's discrete peaks), thermalization (autocorrelation decay to the noise floor, versus persistent coherent oscillation in the linear control), comparative cycle-structure consolidation in an exactly enumerable one-dimensional analog (the rounded map's dominant cycle is $\sim 400\times$ larger than the linear control's), and orbit non-fragmentation in direct three-dimensional sampling (no recurrences or trajectory merges). The quantitative rate is preparation-ensemble- and protocol-dependent: at an infinite-temperature ensemble the separation shows an incubation-then-avalanche profile with an effective rate several times the value quoted above, so the quoted λ_{max} is specific to its preparation ensemble and fit protocol. The qualitative chaos, which is what the ergodicity interpretation requires, is robust across protocols.

The two roles of the dynamics, stated explicitly. The substratum update carries two structurally distinct pieces, and the framework's results divide cleanly between them. The *exact* dynamics is the linear rule $f = \sum_{\text{nn}} x - x(t-1)$ over $\mathbb{Z}/q\mathbb{Z}$; the finite-state arithmetic (the mod-q wraparound / threshold-crossing at the alphabet boundary) is a bounded nonlinearity that is present in any faithful finite realization but vanishes in the continuum/large-q limit. The *structural* chain — the wave-equation uniqueness (§4.1), the Susskind factorization (Theorem 2), the $(3, 2, 1)$ decomposition and the gauge group (§4.6), and the amplitude-scale-gauge argument for linearity (§4.1) — reads only the linear part: it is exact modular-linear algebra and does not invoke the threshold nonlinearity. The *ergodic/measure* properties — full-shell sampling for the Gaussianization of §4.2, the chaotic mixing underlying operational measurement independence ([Main] §3.3), and the ETH-conditioning of the C2 necessity direction ([Main]

§3.4) — read the threshold nonlinearity, which is what breaks the integrability of the strictly-linear map (way out (i) above). The two are not in tension: they are different features of one object, consumed by different parts of the framework, and the ergodic properties ride on a bounded correction that leaves the structural (continuum-limit) content untouched. The one commitment this makes explicit is that the amplitude-scale gauge of §4.1 is exact at the level of the structural chain (where the threshold nonlinearity is absent) and holds up to the bounded finite-state correction at the level of the realized bijection; a full account of how large that correction is on each ergodic observable is the open residual the $\lambda_{\max}$ measurement begins to quantify.

4.3 Center independence and chiral symmetry

Theorem 3. *Center independence of the lattice dynamics is equivalent to exact chiral symmetry of the emergent staggered fermions.*

Proof. The staggered mass term squares to $\Box_{\text{lat}}\phi = -4m_{\text{lat}}^2\phi$: center dependence. Conversely, center independence gives $D_{\text{st}}\chi = 0$, invariant under $\chi \to \varepsilon\chi$ (chiral symmetry). $\Box$

Remark (scope of the center-dependence–mass identification). The specific self-term $C = 2(1-d)$ is not a mass term: it is the diagonal of the massless 4D lattice Laplacian, whose standard 4D staggered factorization has exact chiral symmetry. The hypercubic discretization is excluded not by this lemma but by the generation count — a 4D taste structure gives 2^4 corners and the wrong number of generations (§4.7) — and by $K = 2d$ with the $(3, 2, 1)$ decomposition requiring the spatial cubic group (§4.5–4.6).

The chain: P-indivisibility → center independence → chiral symmetry → Higgs mechanism [13, 14]. One algebraic condition ($\alpha = 0$) produces quantum mechanics, chiral fermions, and the Higgs boson.

4.4 Multi-component dynamics and gauge structure

Each site now carries a K-component vector $\phi(\mathbf{n}, t) \in (\mathbb{Z}/q\mathbb{Z})^K$. The general second-order linear update is $\phi(\mathbf{n}, t+1) = C\,\phi(\mathbf{n}, t) + \sum_j M^{(j)}[\phi(\mathbf{n}+\hat{e}_j, t) + \phi(\mathbf{n}-\hat{e}_j, t)] + D\,\phi(\mathbf{n}, t-1)$, where $C, D, M^{(j)} \in \text{Mat}(K)$. Reversibility requires $D = -I_K$; center independence requires $C = 0$; spatial isotropy requires $M^{(j)} = M$ for all j. This uniquely selects the matrix wave equation:

$$\phi(\mathbf{n}, t+1) = M \sum_{j=1}^{d} \left[\phi(\mathbf{n}+\hat{e}_j, t) + \phi(\mathbf{n}-\hat{e}_j, t)\right] - \phi(\mathbf{n}, t-1)$$

where $M \in \text{Mat}(K)$ is the sole free parameter.

Theorem 4 (Mass spectrum). *Each eigenvalue μ_a of M determines a dispersion relation $\cos\omega_a = \mu_a \cos k$. Massless iff $\mu_a = 1$. Stability requires $|\mu_a| \le 1$.*

Proof. Diagonalizing $M = V\,\mathrm{diag}(\mu_1, \dots, \mu_K)\,V^{-1}$, each eigenmode decouples. Substituting the plane wave $\phi_a \propto e^{i(kn-\omega t)}$ into the a-th decoupled equation gives $e^{-i\omega} + e^{i\omega} = \mu_a(e^{-ik} + e^{ik})$, hence $\cos\omega_a = \mu_a \cos k$. For $|\mu_a| > 1$, there exist real k with $|\mu_a \cos k| > 1$, giving complex ω (exponential growth). □

Theorem 5 (Gauge group, revised). *Let $\Sigma = \langle\phi\phi^\dagger\rangle$ be the O-equivariant two-point condensate of the solution; by Schur's lemma $\Sigma = a\,P_{T_1} + b\,P_E + c\,P_{A_1}$. The global gauge group is the stabilizer $G = C_{\mathrm{U}(K)}(\Sigma)$. For generic (distinct) block values, $G = \mathrm{U}(3) \times \mathrm{U}(2) \times \mathrm{U}(1)$ — equivalently, the unitary group of the bicommutant of the cubic-group action. The coupling matrix itself is scalar, $M = \mu I$ (forced jointly by Theorems 6 and 7); the commutant formula of earlier versions applies to Σ, not M.*

Proof. With $M = \mu I$ the matrix wave equation is preserved by every $U \in \mathrm{U}(K)$: the substratum carries an exact custodial U(6), extended to local invariance by the induced-link construction. The gauge group of the emergent theory is the subgroup unbroken by the solution: U stabilizes the condensate iff $[U, \Sigma] = 0$. By Schur's lemma applied to the equivariant block form of Σ, the stabilizer is $\mathrm{U}(3)\times\mathrm{U}(2)\times\mathrm{U}(1)$ at generic block values; the 22 broken generators' link modes acquire masses at the condensate scale and decouple (§4.6). □

The gauge group and condensate block structure are the same information: modes within one condensate block share a gauge symmetry. (With $M = \mu I$ all substratum modes are massless and share one cone; mass distinctions are state-level.)

4.5 K = 2d from coupling-degree minimization

Theorem 6 ($K = 2d$). *For any bounded-degree graph with polynomial growth exponent d and statistical isotropy, the wave equation (selected by center independence, isotropy, and linearity) has a unique minimal-coupling factorization with $K = 2d$ components, where the minimization is restricted to the observable substratum — factorizations in which every component has at least one spatial coupling (no inert components, $\delta \geq 1$ everywhere — fully decoupled components carry no observable degrees of freedom and can be removed without changing the physics). Minimality confines K to the link-attached family $K = 2d \cdot m$ (Step 3); the multiplicity $m = 1$ is fixed jointly by the condensate-stabilizer accounting of Theorems 5 and 7, as Step 3 records. The "observable substratum" qualifier is a physical commitment, not a mathematical restriction: any sector that fails $\delta \geq 1$ is by definition unobservable to the embedded observer of [Main, §3], so the theorem applies to the full content visible from within.*

Proof (translation-invariant case). The matrix wave equation on $\mathbb{Z}^d$ updates the K-component vector at site $\mathbf{n}$ using exactly 2d independent spatial inputs: $\{\phi(\mathbf{n}\pm\hat{e}_j, t)\}_{j=1}^{d}$. Define the *internal coupling degree* δ(K) as the maximum number of spatial input channels on which any single component's update depends.

Step 1 (K = 2d achieves δ = 1). With K = 2d, assign component a to link direction $a \in \{\pm\hat{e}_1, \dots, \pm\hat{e}_d\}$. If M is diagonal in this basis, then $\phi_a(\mathbf{n}, t+1) = \mu_a \cdot \phi_a(\mathbf{n}+\hat{e}_{j(a)}, t) - \phi_a(\mathbf{n}, t-1)$: each component depends on exactly one spatial input. Therefore $\delta(2\text{d}) = 1$.

Step 2 ($K < 2d$ forces $\delta \geq 2$). By the pigeonhole principle, at least one component must receive contributions from $\lceil 2d/K \rceil \geq 2$ spatial input channels. These channels carry independent data from distinct neighboring sites, so $\delta(\text{K}) \geq 2$.

Step 3 ($\delta = 1$ confines K to multiples of $2d$; the residual multiplicity is fixed by the gauge chain). With K > 2d components each required to have $\delta \geq 1$ but only 2d distinct spatial channels available, by pigeonhole at least two components share a channel. Sharing alone does not force $\delta \geq 2$: two components that read the *same scalar datum* of the shared neighbor and nothing else are linearly dependent dynamical variables and are excluded, but two components reading *distinct* components of the shared neighbor's vector remain dynamically independent at $\delta = 1$ (checked directly on the m-fold link-diagonal factorization at $K = 12$ and $K = 18$: no inert component, and the stacked coupling has trivial nullspace). What $\delta = 1$ does force is that every component attach to exactly one link direction; statistical isotropy then requires the attachment to be O-covariant, and since O acts transitively on the $2d$ signed directions, every direction carries the same number m of components. Coupling-degree minimality therefore confines the component count to the family $K = 2d \cdot m$, $m \geq 1$, and does not by itself select $m = 1$. The multiplicity is fixed by the gauge chain instead. The emergent gauge group is the stabilizer of the condensate in $\mathrm{U}(K)$ (Theorem 5), which for any Hermitian condensate is $\prod_i \mathrm{U}(k_i)$ with $\sum_i k_i = K$; the cubic multiplicities $(3, 2, 1)$ of Theorem 7 sum to six, so the gauge sector of §4.6 arises precisely at $K = 6$, while $K = 6m$ with $m > 1$ carries m commuting copies of it — a generic equivariant condensate on $V_K \otimes \mathbb{R}^m$ has stabilizer $[\mathrm{U}(3) \times \mathrm{U}(2) \times \mathrm{U}(1)]^m$. The selection of $m = 1$ is therefore of the same kind as the selection of $d = 3$ (§3.2): a discrete structural parameter fixed by consistency with the observed single gauge sector, not by minimality alone.

Uniqueness. Steps 1–3 show $\delta = 1$ iff $K = 2d \cdot m$; within that family, $m = 1$ — hence $K = 2d$ — is fixed by the stabilizer accounting above.

Extension to general graphs. For a bounded-degree graph G with polynomial growth exponent d and statistical isotropy: (i) G is quasi-isometric to $\mathbb{Z}^d$ (§3.2, Gromov's theorem); (ii) the wave equation on G has $2d$ independent low-energy propagation modes, determined by the spectral dimension $d_s = d$ (from heat kernel asymptotics $p_t(x, x) \sim t^{-d/2}$ on graphs with polynomial growth exponent d); (iii) the minimizer is fixed by the *spectral mode count* $2d_s$, not by the local vertex degree, and d_s is a quasi-isometry invariant. This last point requires care, because the $\mathbb{Z}^d$ proof of Steps 1–3 counts the $2d$ nearest-neighbor link *directions* (the vertex degree), and vertex degree is *not* a quasi-isometry invariant — a bounded-degree graph quasi-isometric to $\mathbb{Z}^d$ may have locally varying degree, so "K = number of link directions = $2d$" does not transfer to a general graph by integrality alone. What does transfer is the count of independent low-energy propagation channels, which is $2d_s$ (Step ii): the on-diagonal heat-kernel decay exponent — equivalently the long-time return probability of the random walk — is invariant under quasi-isometry for graphs of bounded geometry (Pittet–Saloff-Coste; Kanai; Grigor'yan–Telcs), so $2d_s$ is a genuine quasi-isometric invariant whereas the local degree is not. The minimal-coupling factorization of the *coarse-grained* low-energy dynamics therefore has $K = 2d_s$ channels regardless of local degree fluctuations, which are renormalization-irrelevant. This step is consequently conditional on the same premise as the framework's emergent-Lorentz claim (§3.1, point iii): that the substratum-to-observer coarse-graining preserves the relevant low-energy structure. On the simple-cubic lattice — the case the gauge derivation of §4.6 actually uses — vertex degree and spectral mode count coincide ($2d = 2d_s = 6$) and Steps 1–3 are a complete proof with no such conditionality; the qualification applies only to the general-graph generalization. □

This is the per-site analog of the global factorization principle (§2.4): at the global level, $S = S_1 \times \cdots \times S_N$ is the unique factorization minimizing coupling degree — defining "space"; at the per-site level, K components minimize internal coupling degree — defining "gauge structure." The objection that "nature need not minimize coupling degree" mistakes the logical role. The framework does not postulate a preference for minimal coupling — coupling-degree minimization is the mathematical definition of "site" given a bijection, and the factorization is unique (§2.4). A system with $K = 8$ or $K = 4$ would correspond to a *different bijection* φ', not to the same φ with a non-minimal factorization. Since the dynamics is selected by center independence, isotropy, and linearity (§4.1), and its factorization is unique, $K = 2d = 6$ is a theorem about the wave equation on $\mathbb{Z}^3$, not a selection rule.

4.6 Cubic decomposition: multiplicities (3, 2, 1)

Theorem 7. *The $2d = 6$ nearest-neighbor link directions of the $d = 3$ cubic lattice decompose under the rotation group O as $6 = T_1(3) \oplus E(2) \oplus A_1(1)$. The equivariant (isotypic) block multiplicities of the internal space are (3, 2, 1); the stabilizer of a generic equivariant condensate is accordingly the gauge group* $\mathrm{U}(3) \times \mathrm{U}(2) \times \mathrm{U}(1) \supset \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$.

Proof. Characters at each conjugacy class of O (24 elements): E: $\chi = 6$; $8C_3$: $\chi = 0$; $3C_2$: $\chi = 2$; $6C_2$': $\chi = 0$; $6C_4$: $\chi = 2$. Character inner products give $n(A_1) = 1$, $n(E) = 1$, $n(T_1) = 1$, with $n(A_2) = n(T_2) = 0$. By Schur's lemma and isotropy, any equivariant operator is block-scalar; in particular the condensate is $\Sigma = \mathrm{diag}(aI_3, bI_2, c)$, while the coupling matrix itself is scalar (Remark below). □

The 6D representation in question is the natural action of O on the six nearest-neighbor link vectors $\{\pm\hat{e}_1, \pm\hat{e}_2, \pm\hat{e}_3\}$ of the cubic lattice — the same 6D space on which the coupling matrix $M(\mathbf{n}, \hat{e}_j)$ of §4.4 acts. The decomposition therefore fixes the multiplicities of M's eigenspaces uniquely; there is no freedom in "which 6D rep" is meant.

The max-speed constraint, applied to the scalar coupling, requires $\mu = 1$: all six modes propagate at the lattice speed of light — one cone, exact. The physical identifications: T_1 (spatial vector) $\to$ color, E (quadrupole) $\to$ weak isospin, A_1 (scalar) $\to$ hypercharge. Local gauge invariance follows from background independence (§3.1).

Remark (joint consistency of Theorems 4, 6, 7). Theorem 6's $\delta = 1$ factorization (coupling diagonal in the link basis) and Theorem 7's equivariant block form are jointly consistent only at $M = \mu I$: the only O-equivariant matrices diagonal in the link basis are scalars. The max-speed constraint and the masslessness of the substratum modes (center independence $\Rightarrow$ chiral symmetry, Theorem 3) then give $\mu = 1$: all six components propagate at the lattice speed — one cone, exact. Mass and gauge-group distinctions are properties of the state (the condensate Σ of Theorem 5), not of the update.

Remark (robustness and the cubic commitment). Theorem 7 uses the character table of the octahedral group O, the exact symmetry of the simple cubic (SC) lattice.

Stability under small perturbations. Integer multiplicities cannot change continuously. Small perturbations of the coupling graph that preserve local O symmetry shift the eigenvalues of M continuously but leave the multiplicities $(3, 2, 1)$ intact. $K = 2d$ (Theorem 6) is stable in the same sense under quasi-isometric deformations that preserve the spectral dimension.

Dependence on the Bravais lattice. The $(3, 2, 1)$ decomposition depends on the discrete O action on the six NN link vectors of SC. Under the continuous group O(3), the same six vectors span two copies of the vector representation — multiplicities $(3, 3)$, not $(3, 2, 1)$ — because O(3) does not preserve the discrete set of six vectors as a distinguished orbit. The three-factor split requires the discrete subgroup.

Theorem 7b (Bravais-lattice uniqueness). *Among all 14 three-dimensional Bravais lattices, the simple cubic (SC) lattice is the unique candidate compatible with the Standard Model gauge group* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ *under the commutant construction of Theorem 5. Furthermore, the inter-generation BZ-distance prescription of §7.1 produces the observed Cabibbo angle* $\lambda = 1/(\pi\sqrt{2})$ *uniquely on SC; BCC and FCC produce* λ *values discrepant with observation by* 42% *and* 29% *respectively.*

The proof has two parts: (a) elimination of all non-cubic Bravais lattices via the crystallographic point group's irrep dimensions, and (b) elimination of BCC and FCC within the cubic system via the explicit O-decomposition of the NN link representation on each, with the SC Cabibbo prediction as quantitative confirmation.

Among the three cubic Bravais lattices, only SC produces the SM gauge group under the commutant construction of Theorem 5. Computing the O-decomposition of the NN link representation on each:

Lattice	NN	O-decomposition of NN rep	Gauge group (commutant)
SC	6	$T_1 \oplus E \oplus A_1$	$SU(3) \times SU(2) \times U(1)$ ✓
BCC	8	$A_1 \oplus A_2 \oplus T_1 \oplus T_2$	$SU(3)^2 \times U(1)^2$
FCC	12	$A_1 \oplus E \oplus T_1 \oplus 2T_2$	$SU(3)^3 \times SU(2) \times U(1)$

SC is uniquely compatible with the observed Standard Model gauge structure. BCC gives two non-abelian $SU(3)$ factors with no $SU(2)$; FCC gives three $SU(3)$ factors and three additional $U(1)$s. Neither matches observation at the group-theoretic level, independent of any numerical prediction.

The Cabibbo angle on the three lattices, computed using the inter-generation BZ distance prescription of §7.1 on each lattice's natural triplet sector (SC: X-points; BCC: H-points; FCC: L-points), with equal-NN normalization:

Lattice	Triplet separation	Cabibbo λ
SC	$\pi\sqrt{2}$	0.2251
BCC	$\pi\sqrt{6}$	0.1299
FCC	2π	0.1592

Observed $\lambda = 0.2250 \pm 0.00067$ [20]. The SC value matches at 0.04% (0.12σ); BCC and FCC miss by 42% and 29% respectively. The SC match is tied to the specific BZ geometry corresponding to the SC gauge group.

The §2 structural reading that "any bounded-degree graph with statistical isotropy suffices" therefore refers to the existence-and-emergent-QM part of the chain (Theorems 1–6, §§3.1–3.4), not to the specific gauge group or quantitative predictions of §§4.6 and 7. These require the SC Bravais structure; the framework treats this as a structural commitment forced by the observed SM gauge group and its quantitative consequences, rather than as a free choice.

Extension to non-cubic Bravais lattices. The uniqueness of SC among cubic Bravais lattices extends to a stronger statement: **among all 14 three-dimensional Bravais lattices, SC is the unique candidate compatible with an $SU(3)$ gauge factor**. The reason is a standard crystallographic fact: among the 32 crystallographic point groups, the 3-dimensional irreducible representations appear only in the cubic crystal system (point groups T, T_h, T_d, O, O_h). Point groups of the hexagonal, trigonal, tetragonal, orthorhombic, monoclinic, and triclinic systems act on $\mathbb{R}^3$ reducibly as (axial direction) + (perpendicular plane), and therefore have irreps only of dimensions 1 and 2 — no 3-dimensional irrep. Theorem 5's commutant construction requires a 3-dimensional irreducible block in the M matrix to produce an $SU(3)$ factor, which requires a 3-dimensional irrep of the point group acting on the NN vectors. No non-cubic Bravais lattice supplies this. The binding constraint is in fact the SU(2) rather than the SU(3): a 3-dimensional irrep is necessary for the color factor but is shared by the tetrahedral group A_4 and the (non-crystallographic) icosahedral group A_5, so it does not by itself single out the cube. The full ladder $(3, 2, 1)$ is equivalent to the point group's irreducible-degree set containing $\{1, 2, 3\}$, realized among finite subgroups of O(3) only by $S_4 \cong O$ — A_4 has a 3 but no genuine 2-dimensional irrep (its apparent doublet is a complex-conjugate pair of 1-dimensional irreps, commutant $\mathrm{U}(1)^2$), and A_5 has 3-dimensional irreps but no 2-dimensional one. The octahedral group is the unique finite rotation group carrying both a genuine 2- and a 3-dimensional irrep, and is the point group of the cubic system; it is specifically the SU(2) factor that forces the octahedral structure. Combined with the result above that BCC and FCC within the cubic system produce the wrong gauge structure, SC is uniquely compatible with the SM gauge group among all 14 Bravais lattices.

Remark (completeness of the irrep catalog). The octahedral group O has five irreducible representations: A_1, A_2, E, T_1, T_2. All five are accounted for in the framework, either by identified physical role or by derived absence:

- A_1: the U(1) factor of the 6-link decomposition (Theorem 7); the singlet-taste scalar that hosts the Higgs (Theorems 8, 11); appears in $\mathrm{Sym}^2(T_1)$ in §7.2 and §8.4.
- E: the SU(2) factor of the 6-link decomposition; controls neutrino mass splittings in $\mathrm{Sym}^2(T_1)$ (§8.4); provides the Koide angle $\theta_0 = 2/9$ via the quadratic Casimir $C_2(T_1)$ normalized by the lattice bandwidth d^2 (§7.2).
- T_1: the SU(3) factor of the 6-link decomposition; the generation space from staggered tastes (Theorems 8, 10); the neutrino flavor basis (§8.4).
- T_2: appears in $\mathrm{Sym}^2(T_1) = A_1 \oplus E \oplus T_2$ and **controls the off-diagonal structure** of the neutrino mass matrix (§8.4); the resulting mixing angles enter the structural PMNS prediction of §7.3, where TBM (from cubic-group representation theory) plus U-parity-graded charged-lepton corrections at scale λ give all three angles.
- A_2: the pseudoscalar/sign representation; does not appear in the 6-link decomposition ($n(A_2) = 0$ in the character calculation above), does not appear in $\mathrm{Sym}^2(T_1)$, and plays no structural role in the framework. Its absence reflects the parity-even nature of nearest-neighbor coupling on the cubic lattice.

No irrep of O is unaccounted for.

Remark (the substratum-emergent layer split). Part III distinguishes between two layers: the *substratum* — the bijection (S, φ) on the cubic lattice, with its coupling matrix $M(\mathbf{n}, \hat{e}_j)$ and the wave-equation dynamics — and the *emergent* (observer) level — the unitary QFT that an embedded observer reports after the trace-out of [Main]. The SM gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ lives at the substratum: it is the commutant (stabilizer) of the O-equivariant condensate Σ (Theorem 5; the coupling matrix M itself is scalar, $M = \mu I$), a literal algebraic structure on the lattice, existing before any trace-out is performed. The SM gauge *couplings* $\alpha_1, \alpha_2, \alpha_3$ at M_Z live at the emergent level: they arise from the induced coupling $1/\alpha_0 = 23.25$ produced by the fermion determinant (§6.1), followed by thresholds and RG running. The framework's structural commitment is therefore: *group structural, couplings emergent.* This split explains why the gauge group is rigid (no parameter freedom — forced by the commutant structure of Σ) while the gauge couplings admit quantitative predictions at all (they depend on fermion-loop dynamics in the emergent theory, with the substratum's role being to fix $N_f = 6$ and $T(R) = 1/2$ structurally). The same split clarifies the asymmetry noted in §4.7.1.1 between gauge bosons (substratum objects) and fermions (observer-level objects).

4.7 Staggered species, generations, and the Higgs

The Nielsen-Ninomiya theorem [15] requires 2^{d+1} Weyl species. Staggered reduction gives $N_{\text{taste}} = 2^{\lfloor (d+1)/2 \rfloor} = 4$ Dirac tastes in d+1 = 4.

Theorem 8. *Under the cubic group O:* $4 = \mathbf{1} \oplus \mathbf{3}$. *One singlet taste and three triplet tastes.*

Proof. The 2^d = 8 BZ corners pair into 4 taste pairs under $\eta \leftrightarrow \mathbf{1} - \eta$. The pair {(0,0,0), (1,1,1)} is O-invariant (singlet A_1). The three axis-aligned pairs form irreducible T_1. □

Remark (carrier of the T_1 action). The irreducible T_1 is carried by the signed vector action of the spin-taste matrices: the j-th triplet pair is represented by γ^j (Theorem 9), which transforms as a genuine vector under O. The unsigned corner labels alone carry only the axis-permutation action, which decomposes as $A_1 \oplus E$, since sign flips act trivially on $\{0, \pi\}$ momentum components. The distinction is representation-theoretically real but does not affect diagonal-channel observables: both actions induce the same S_3 permutation on the axis label, so the generation-channel masses of §7.2 — and the Z_3-Fourier amplitude extraction built on them — are insensitive to it. One quantity is *not* covered by this insensitivity and should be flagged: the Koide amplitude and angle depend on the quadratic Casimir $C_2(T_1) = 2$ (§7.2), and the value of that invariant depends on which representation the mass-splitting operator transforms in. The insensitivity established here is to the *permutation* structure of the diagonal channel; the identification that the mass-splitting operator transforms in the signed T_1 (carrying $C_2 = 2$) rather than in the unsigned $A_1 \oplus E$ is a separate, load-bearing input to the Casimir chain. It is supplied by the identification of the generation triplet with the γ^j-carried tastes, and its derivation from the substrate mass operator's transformation law is the same open task as the amplitude closure of §7.2 (the $A^2 = C_2$ dynamical derivation); the pre-registered A^2 measurement tests both together, since a mis-identified carrier would present as a wrong amplitude.

Theorem 8b (Taste coupling structure). *On the $d = 3$ lattice with the nor-*

malized wave equation, each BZ corner η has coupling $\mu(\eta) = \sigma(\eta)/d$ where $\sigma = \sum_j \cos(\pi\eta_j)$:

$$\mu_\Gamma = 1, \quad \mu_X = \tfrac{1}{3}, \quad \mu_M = -\tfrac{1}{3}, \quad \mu_R = -1$$

The staggered taste pairs have couplings: singlet (Γ, R): $|\mu| = 1$; triplet (X_j, M_j): $|\mu| = 1/3$, identical for all $j = 1, 2, 3$.

Proof. Direct computation: $\sigma(\Gamma) = 3$, $\sigma(X) = 1$, $\sigma(M) = -1$, $\sigma(R) = -3$. Dividing by $d = 3$ gives the couplings. The triplet members are related by cubic symmetry (O permutes spatial axes), hence identical. □

Theorem 9 (Spin-taste correspondence). *(a) The singlet taste produces a spin-0 field. (b) The triplet taste produces spin-1/2 fields.*

Proof. (a) The singlet pairs Γ(0,0,0) = I_4 (scalar) and Γ(1,1,1) = $\gamma^1\gamma^2\gamma^3$ (pseudoscalar), both SO(3)-invariant: J = 0. (b) The j-th triplet pairs γ^{j} (vector) with −iΣ_j (spin generator): J = 1/2. □

Corollary (Spin-statistics from the lattice). Singlet → spin-0 → boson (Higgs). Triplet → spin-1/2 → fermion (quarks and leptons).

Theorem 10 (Three degenerate fermion sectors). *The three triplet staggered tastes provide three degenerate spin-1/2 sectors related by cubic symmetry, one per SM generation. The SM matter representations are placed on links (§4.7.1.2), where the link-carrier space $V_K \otimes V_K^*$ admits cross-charged tensor-product representations such as $(\mathbf{3}, \mathbf{2})$ without algebraic obstruction. Anomaly cancellation (Theorem 15) then uniquely fixes the hypercharge assignment as the sole consistent completion.*

Proof. (i) Three spin-1/2 sectors and one spin-0 sector (Theorem 9). (ii) The three sectors are related by cubic symmetry (O permutes spatial axes), hence degenerate. (iii) Theorem 15 below shows that, given a chiral $SU(3)\times SU(2)\times U(1)$ gauge structure, the SM matter content is the unique anomaly-free assignment with the observed family pattern [16]. The framework therefore derives three SM generations as a forced consequence: cubic symmetry fixes three degenerate sectors (step ii), the link-carrier construction of §4.7.1.2 places matter on links where cross-charged representations are admissible (no phenomenological input required), and anomaly cancellation (step iii) fixes the hypercharge assignment uniquely. □

Scope. The SM gauge representations of the matter fields arise from the link-carrier space $V_K \otimes V_K^*$, where cross-charged representations such as $(\mathbf{3}, \mathbf{2})$ exist as ordinary tensor-product subspaces; see §4.7.1.2 for the closure. Anomaly cancellation (Theorem 15) then fixes their hypercharges as the unique consistent assignment. The absolute generation count is locked at exactly three by coupling-degree minimality jointly with the condensate-stabilizer accounting of Theorems 5 and 7: Theorem 6 confines the component count to $K = 2d \cdot m$ and the stabilizer accounting fixes $m = 1$, giving $K = 6$ and exactly three spin-1/2 staggered tastes (Step 3, §4.7.1.2).

Theorem 11 (Higgs sector, consistency). *Given a Higgs doublet* $(\mathbf{1}, \mathbf{2}, +\frac{1}{2})$ *in the SM, the singlet taste provides a spin-0 sector whose* E *projection is consistent with hosting it; the* T_1 *and* A_1 *projections do not admit Yukawa couplings to the SM fermion doublets and play no role in the low-energy theory.*

Proof. The singlet taste is spin-0 (Theorem 9a) and decomposes under the gauge group as $T_1(\mathbf{3}) \oplus E(\mathbf{2}) \oplus A_1(\mathbf{1})$ (Theorem 7). Of these three projections, only $E(\mathbf{2})$ admits a gauge-invariant Yukawa coupling of the form $\bar{Q}_L \cdot \Phi \cdot u_R$: the T_1 projection is excluded on color grounds and the A_1 projection on weak grounds. The E projection is therefore the natural host for the SM Higgs doublet. □

Scope. This theorem does **not** derive the existence of a Higgs doublet, nor its hypercharge, from lattice structure. The Higgs doublet is taken as phenomenological input; the singlet taste is shown to accommodate it.

Consequences. The framework accommodates exactly one Higgs doublet via the singlet taste, consistent with the SM.

4.7.1 The fermion embedding question: diagnosis and closure The framework as written derives the gauge group $SU(3) \times SU(2) \times U(1)$ from the commutant construction (Theorems 5, 7, 14) and uses anomaly cancellation (Theorem 15) as a uniqueness check on the SM matter assignment, given a chiral $SU(2)$ structure. The matter representations and hypercharge assignments are derived — via the link-carrier construction of §4.7.1.2 and anomaly cancellation (Theorem 15) — rather than taken as input. The framework's claim on the fermion sector is therefore: *the SM gauge group is forced by first principles; the SM matter content, its gauge representations, and its hypercharges are the unique anomaly-free consequence of placing matter on links in the link-carrier space* $V_K \otimes V_K^*$. The residual gap is narrower: the assignment of SM representations to specific tastes rests on a uniqueness argument via anomaly cancellation rather than a direct geometric derivation of the representations themselves, and is noted as the residual scope.

Whether a stronger derivation exists — via a geometric embedding of (taste $\times$ sublattice $\times$ spinor) indices into SM gauge representations such that $Q_L = (\mathbf{3}, \mathbf{2}, +\frac{1}{6})$ emerges as an irreducible tensor-product representation — is open. One specific candidate construction (extending V_K by an O-equivariant taste factor and redefining the gauge group as the commutant of an extended operator) is ruled out: O has no irreps of dimension > 3, so no O-equivariant operator on $V_K \otimes V_{\text{taste}}$ admits a 6-dimensional irreducible eigenspace with the required tensor-product structure. Other constructions — breaking O-equivariance on the taste factor, replacing the commutant definition with a bimodule construction, or placing fermions outside the gauge-defining carrier space — are not ruled out, but none has been produced.

The neutrino mass mechanism (Dirac vs. Majorana, presence or absence of ν_R) is not fixed by the lattice structure as currently understood and is left open. The absolute generation count is addressed in §4.7.1.2: the $K = 2d \cdot m$ confinement of Theorem 6 together with the $m = 1$ stabilizer accounting (Theorems 5, 7) locks the count at three as a derived consequence rather than an assignment.

4.7.1.1 Gauge bosons from the Schur complement, and the level-confusion diagnosis The Schur-complement trace-out used in Theorem 13 to establish chirality contains more structural information than that theorem extracts. Pursuing it carefully yields a clean retrodiction of the gauge boson content of the Standard Model from $\mathrm{End}(V_K)$, and simultaneously sharpens the fermion-embedding question of §4.7.1 into a precise and tractable form — a form that §4.7.1.2 then answers via the link-carrier construction.

The gauge sector closes. With $D_{LL} = D_{RR} = 0$ (Theorem 13(ii)), the Schur-complement effective action takes the form $\mathcal{L}_{\text{eff}} = \bar{\psi}_L \, K_{\text{eff}} \, \psi_L + \cdots$ with kernel $K_{\text{eff}} = D_{LR} \, G_{RR} \, D_{RL} : W_L \to W_L$. The kernel is a linear map on the visible single-excitation space, i.e., an element of $\mathrm{End}(W_L) = W_L^* \otimes W_L$ — *not* of $\Lambda^2 W_L$ or $\mathrm{Sym}^2 W_L$. The two ψ_L factors are distinguishable (one barred, one unbarred), and bilinears of distinguishable objects pair W_L^* against W_L via the trace pairing.

Restricting to the internal index, the gauge content of K_{eff} is given by the decomposition of $\mathrm{End}(V_K) = \mathrm{End}(V_3 \oplus V_2 \oplus V_1)$ under $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$. The diagonal blocks give

- $\mathrm{End}(V_3) = \mathbf{8} \oplus \mathbf{1}$ — the SU(3) adjoint (gluons) plus a singlet,
- $\mathrm{End}(V_2) = \mathbf{3} \oplus \mathbf{1}$ — the SU(2) adjoint (W bosons) plus a singlet,
- $\mathrm{End}(V_1) = \mathbf{1}$ — the U(1) hypercharge boson.

The off-diagonal blocks $\mathrm{Hom}(V_i, V_j)$ for $i \neq j$ would carry cross-charged “gauge bosons” mixing different commutant factors. These are absent from the Standard Model. The substratum dynamics is locally U(6) invariant and does not exclude them; they are removed spontaneously: the O-equivariant condensate Σ breaks $\mathrm{U}(6) \to \mathrm{U}(3) \times \mathrm{U}(2) \times \mathrm{U}(1)$ (Theorem 5, revised), and the 22 cross-charged modes acquire masses at the condensate gap scale M_X — set by the equivariant block splittings, a solution-level quantity not fixed at the lattice scale — by the Higgs mechanism, decoupling from the low-energy spectrum: K_{eff} inherits the block structure of the stabilizer, and the off-diagonal Homs are gapped rather than absent. The Schur-complement kernel therefore retrodicts the SM gauge boson content cleanly: the adjoints of SU(3), SU(2), and U(1) together with the corresponding singlets, and nothing else.

The fermion sector does not close, and the obstruction is sharper than it looks. The natural next move is to ask whether the same Schur-complement structure also produces the SM matter content. It does not, and the failure mode is structurally informative.

The visible single-excitation space is $W_L = V_K \otimes V_{\text{taste}} \otimes S_L$, with $V_K = V_3 \oplus V_2 \oplus V_1$ by Theorem 5. Tensor products distribute over direct sums, so

$$W_L = (V_3 \otimes V_{\text{taste}} \otimes S_L) \oplus (V_2 \otimes V_{\text{taste}} \otimes S_L) \oplus (V_1 \otimes V_{\text{taste}} \otimes S_L).$$

A single excitation in W_L lives in exactly one summand and therefore carries exactly one nontrivial gauge charge: it is a color triplet *or* a weak doublet *or* a hypercharge singlet, never two simultaneously. The left-handed quark doublet Q_L, by contrast, transforms as $(\mathbf{3}, \mathbf{2}, +\frac{1}{6})$ — an irreducible representation of the joint gauge group that lives in $V_3 \otimes V_2$, not in $V_3 \oplus V_2$. There is no element of a direct sum that carries simultaneous nontrivial labels under summands meant to act independently. As a statement about the bijection-level carrier V_K, this is a hard algebraic fact and it cannot be circumvented by reorganizing $\Lambda^k W_L$ for any k.

Diagnosis: a level confusion between bijection-level and observer-level structure. The obstruction is fatal only under a hidden assumption that the present framework is required to reject. The assumption is that the observer-level single-fermion Hilbert space *is* $W_L = V_K \otimes V_{\text{taste}} \otimes S_L$, with the gauge group acting on it via the bijection-level decomposition $V_K = V_3 \oplus V_2 \oplus V_1$. This identifies two distinct objects:

1. The bijection-level carrier at a site, which is the finite alphabet of values the lattice variable φ can take, equipped after-the-fact with the complex inner-product structure $V_K = \mathbb{C}^6$ on which M acts and Theorem 5 diagonalizes.
2. The observer-level single-fermion Hilbert space, which is whatever an embedded observer's coarse-graining produces when it organizes lattice deviations from a vacuum into states it calls "one fermion."

In standard QFT, where quantum mechanics is axiomatic, these are the same object by fiat: one writes down a single-particle Hilbert space and gauge representations act on it directly. In the present framework that identification is not available. Quantum mechanics is *derived* from the bijection (S, φ) via the C1–C3 conditions of [Main], and the derivation runs through the observer's coarse-graining of the lattice, not through the lattice itself. The notion of "a state in V_K with definite gauge label" is a structure that exists at the bijection level. The notion of "one fermion as seen by an observer" is a structure that exists at the observer level. There is no theorem in the present paper, and none in [Main], asserting that these two structures coincide, and the foundational commitment to emergent QM positively requires that they need not.

Once this distinction is made, the direct-sum-vs-tensor-product obstruction reorganizes. At the bijection level, $V_K = V_3 \oplus V_2 \oplus V_1$ is correct and the commutant $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ acts block-diagonally on it. At the observer level, the single-fermion Hilbert space $\mathcal{H}_{\text{obs}}$ is the image of bijection-level configurations under the (still-to-be-characterized) coarse-graining map, and the gauge group acts on $\mathcal{H}_{\text{obs}}$ via the *induced* action of the bijection-level commutant. This induced action need not be block-diagonal in any natural basis of $\mathcal{H}_{\text{obs}}$, and a single observer-level fermion can perfectly well be a vector in $\mathcal{H}_{\text{obs}}$ whose preimage in the bijection-level structure has support in multiple V_K summands. That is exactly what the representation $(\mathbf{3}, \mathbf{2})$ encodes: a state with a definite color label and a definite weak-isospin label, which are two different observables on $\mathcal{H}_{\text{obs}}$ both inherited from the bijection-level structure but neither equal to "which V_K summand are you in."

This reframing is consistent with — and arguably required by — the way Theorem 13 already handles chirality. That proof treats the gauge group as acting on V_K *coupled to* the spinor/sublattice grading, not as a pure internal rotation on a 6-dimensional space: SU(3) is vector-like because color commutes with the grading, SU(2) is chiral because its action survives only on the left-handed external legs after the trace-out. Theorem 13 is already implicitly working at the observer level, where the gauge action is induced rather than literal. §4.7.1.1 simply makes this commitment explicit and applies it to the matter sector.

The asymmetry is structural, not embarrassing. The reframing also explains *why* the gauge sector closes from $\mathrm{End}(V_K)$ while the matter sector does not close from V_K. Gauge bosons are properties of the coupling matrix M itself — they live at the bijection level, where the direct-sum decomposition of V_K is the literal structure and $\mathrm{End}(V_K)$ is the literal kernel produced by the Schur complement. Fermions are observer-level objects: they are what an embedded observer reports when it coarse-grains lattice deviations into single-particle states. The two sectors live at two different layers of the framework, and the layer separation is exactly what "QM is emergent" means. The asymmetry between a closed gauge-boson retrodiction and a fermion-embedding question that requires a layer-corrected treatment is therefore not a defect; it is the expected signature of a framework in which gauge fields are structural and matter is emergent. §4.7.1.2 shows how the layer-corrected treatment resolves both the embedding and the generation count.

The correct successor question — characterizing the coarse-graining map from bijection-level configurations to the observer-level single-fermion Hilbert space $\mathcal{H}_{\text{obs}}$ such that the induced gauge action realizes exactly one SM generation per T_1 taste — is answered in §4.7.1.2 via the link-carrier construction.

4.7.1.2 Closure: Matter on Links The closure is a two-step corollary of theorems already proved, using constructions distributed across §3.1 and §4.4.

Step 1: The link is the carrier. The local gauge invariance established in §3.1 and §4.4 establishes that the coupling matrix $M(\mathbf{n}, \hat{e}_j)$ on a nearest-neighbor link is itself the gauge connection: under a local gauge transformation G, it transforms as $M(\mathbf{n}, \hat{e}_j) \to G(\mathbf{n})\, M(\mathbf{n}, \hat{e}_j)\, G(\mathbf{n} + \hat{e}_j)^{-1}$, with one gauge index at each endpoint. The link variable therefore lives in $V_K^{(\mathbf{n})} \otimes V_K^{(\mathbf{n}+\hat{e}_j)*}$, not in a single V_K. Furthermore, Theorem 7 establishes that the multiplicities $(3, 2, 1)$

themselves come from the cubic decomposition of the *$2d = 6$ nearest-neighbor link directions* under O, not from any site-internal structure: V_K was a link-direction space from the start. The site-level reading of $V_K = V_3 \oplus V_2 \oplus V_1$ that produced the direct-sum obstruction is therefore not the framework's natural reading; the link reading is.

On a link, the relevant carrier for matter is $V_K \otimes V_K^*$, which contains the cross-charged tensor products $V_3 \otimes V_2$, $V_3 \otimes V_1$, $V_2 \otimes V_1$, and so on. The left-handed quark doublet $Q_L \in (\mathbf{3}, \mathbf{2})$ has a natural home in $V_3 \otimes V_2 \subset V_K \otimes V_K^*$, with no algebraic obstruction. The direct-sum-vs-tensor-product problem dissolves: it was an artifact of misidentifying the site-internal V_K with the matter carrier, when the framework's own gauge-connection construction places the carrier on links.

Step 2: Anomaly cancellation completes the assignment. Once the matter carrier is $V_K \otimes V_K^*$ on links, the question reduces to which subspaces of $V_K \otimes V_K^*$ host propagating chiral fermions. Theorem 13 fixes the chirality structure: SU(2) is chiral, SU(3) is vector-like, with left-handed external legs after the Schur-complement trace-out. Theorem 15 then fixes the matter content uniquely: given chiral $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ with fermions in fundamental or singlet representations, the six anomaly conditions force exactly the SM hypercharge assignment $(Y_Q, Y_u, Y_d, Y_L, Y_e) = (1/6, 2/3, -1/3, -1/2, -1)$, with no free parameters beyond the overall normalization and the $Y_u \leftrightarrow Y_d$ relabeling (Theorem 15). Theorem 10 then distributes this content across the three T_1 staggered tastes as three identical generations, and Theorem 11 places the Higgs in the E projection of the singlet taste.

The closure. Combining these: (i) the local gauge invariance construction establishes that gauge connections live on links with carrier $V_K \otimes V_K^*$; (ii) cross-charged SM representations such as $(\mathbf{3}, \mathbf{2})$ exist in $V_K \otimes V_K^*$ as ordinary tensor-product subspaces, with no direct-sum obstruction; (iii) Theorem 13 fixes chirality; (iv) Theorem 15 fixes the unique anomaly-free hypercharge assignment; (v) Theorems 8–11 distribute the content across tastes as three generations plus one Higgs doublet. The §4.7.1 question is fully answered: the SM matter content is forced by the link-level carrier structure, the chirality theorem, and anomaly cancellation, jointly. No additional construction is required; the resolution was already distributed across the existing chain of theorems and was hidden only by the site-internal reading of V_K that §4.7.1.1's level-confusion diagnosis identified as incorrect.

Step 3: The absolute generation count is locked by minimality jointly with the stabilizer accounting. Theorem 6 confines the internal component count on the $d = 3$ lattice to $K = 2d \cdot m$ (every component link-attached, equal multiplicity per direction), and the condensate-stabilizer accounting of Theorems 5 and 7 fixes $m = 1$: only $K = 6$ carries the gauge sector once rather than in m commuting copies. The ad hoc addition of duplicate fundamental fields is thereby excluded — by the gauge chain, not by minimality alone. This single unique field possesses exactly $2^3 = 8$ BZ corners, which stagger-reduce to exactly 4

Dirac tastes. Theorem 8 decomposes these into exactly 1 scalar and 3 spin-1/2 tastes. Therefore, the single-excitation link-carrier space $V_K \otimes V_K^* \otimes V_{\text{taste}}$ contains exactly three copies of each SM gauge representation. The absolute generation count is not assigned; it is a rigid, derived consequence of $K = 6$ — the minimality confinement of Theorem 6 together with the single-gauge-sector accounting of Theorems 5 and 7.

The earlier $\mathrm{End}(V_K)$ result for the gauge sector now fits cleanly into this picture. Gauge bosons live in $\mathrm{End}(V_K)$ because they are properties of M itself, viewed as an operator on the carrier at one endpoint of a link; matter lives in $V_K \otimes V_K^*$ because it propagates *along* a link, picking up one gauge index at each end. Both sectors live at the link level. The asymmetry between "gauge bosons close from $\mathrm{End}(V_K)$" and "matter does not close from V_K" was an artifact of asking the matter question at the wrong layer; once both questions are asked at the link level, both close, and the closure is forced by theorems already in §§4.4–4.9.

Theorem 12 (Partition = spinor decomposition). *The OI checkerboard partition coincides with the staggered spinor decomposition: visible and hidden sectors carry complementary spinor components.*

Proof. The checkerboard assigns site $\mathbf{n}$ to the visible sector iff $(-1)^{\sum n_\mu} = +1$ (even sublattice). In the staggered-to-Dirac reconstruction, the even sublattice carries spinor components with $\Gamma(\eta)$ matrices of even rank (including I_4), which give the left-handed components under Lorentz decomposition. The odd sublattice carries the right-handed components. The partition structure and the spinor decomposition are the same object. □

Remark (the checkerboard is not a free choice). The checkerboard partition is not selected by hand to produce chirality — it is the bipartite structure of the lattice itself. The wave equation is nearest-neighbor, so the cubic lattice is bipartite: even sites couple only to odd sites and vice versa. This bipartite structure is a property of the dynamics (range-1 coupling on a hypercubic graph), not of the partition. The staggered-to-Dirac reconstruction identifies the two sublattices with left- and right-handed spinor components — a standard result in lattice QFT [12, 17], not an OI construction. Theorem 12 observes that the OI partition (visible vs. hidden) and the staggered partition (left vs. right) are the same decomposition. The physical partition (the cosmological horizon) traces out a spatial region containing both sublattices; at scales large compared to ϵ, any macroscopic region contains equal numbers of even and odd sites. Chirality in the emergent description is a property of how the staggered-to-Dirac reconstruction organizes the lattice variables into spinor components — it follows from the bipartite structure of the dynamics, not from a choice of which sites to trace out.

4.8 Chirality from the trace-out

Theorem 13 (Chirality). *The emergent gauge coupling of the visible sector is chiral: SU(2) is chiral while SU(3) remains vector-like.*

Proof. (i) *Chirality = sublattice parity.* The staggered-to-Dirac reconstruction [17, 12] assigns spinor components to hypercube corners $\eta \in \{0,1\}^{d+1}$ via $\chi(y+\eta\epsilon) = \frac{1}{4}\Gamma(\eta)_{\alpha\beta}\psi_\beta(y)$, where $\Gamma(\eta) = \gamma_0^{\eta_0}\cdots\gamma_d^{\eta_d}$. Under γ_5: $\Gamma(\eta) \to (-1)^{|\eta|}\Gamma(\eta)$. Sites with $|\eta|$ even are left-handed; $|\eta|$ odd are right-handed. This equals the sublattice parity $\varepsilon(\mathbf{x}) = (-1)^{\sum x_\mu}$. (ii) $D_{LL} = D_{RR} = 0$. The staggered Dirac operator couples only across sublattices: $\{D_{\text{st}}, \varepsilon\} = 0$ (chiral symmetry, Theorem 3), so the even-even and odd-odd blocks vanish identically. (iii) *Trace-out removes right-handed fields.* The OI checkerboard partition assigns even sites (left-handed) to V and odd sites (right-handed) to H (Theorem 12). The trace-out marginalizes over H: all operators in the emergent description act on the visible (left-handed) Hilbert space. (iv) *Left-handed effective Lagrangian.* Since $D_{LL} = 0$, the gauge coupling in the full theory is purely $L \leftrightarrow R$. After integrating out R (the hidden sublattice) via the Schur complement — a standard procedure in lattice QFT [12, 17] — the effective coupling has the form $\mathcal{L}_{\text{eff}} = \bar{\psi}_L\,(D_{LR}\,G_{RR}\,D_{RL})\,\psi_L + \cdots$, where G_{RR} is the hidden-sector propagator. Both external legs are left-handed. (v) *SU(3) vector-like; SU(2) chiral.* The chirality projection P_L acts on the spin/sublattice index and commutes with the internal (K-component) index. SU(3) acts on internal $T_1(3)$ components: both L and R carry color, so the trace-out preserves color coupling → **vector-like**. SU(2) acts on internal $E(2)$ components, but the effective coupling has only L external legs → **chiral**. □

Remark (chirality without a continuum limit). In standard lattice QCD, staggered chirality is a property of the regularization that maps to physical chirality only via the continuum limit $a \to 0$ — a procedure that requires the lattice to be a regulator approximating an underlying continuum theory. The OI lattice has no continuum limit because $\epsilon = 2\,l_p$ is the fundamental scale, not a regulator. This raises the question of how lattice chirality becomes physical chirality without the standard universality argument.

The resolution: physical observers don't take continuum limits either. Every experimentally probed scale satisfies $\lambda \gg \epsilon$, so what observers see is the *infrared effective theory* obtained by integrating out modes near the lattice scale. This effective theory inherits the sublattice parity $\varepsilon(\mathbf{x}) = (-1)^{\sum x_\mu}$ as a $\mathbb{Z}_2$ grading because the hidden-sublattice modes do not propagate to IR scales — they are integrated out at the lattice scale via the Schur complement (step (iv) of the proof). The resulting low-energy action

$$\mathcal{L}_{\text{eff}} = \bar{\psi}_L\,(D_{LR}\,G_{RR}\,D_{RL})\,\psi_L + (\text{gauge}) + (\text{Yukawa})$$

has manifestly left-handed external states. SU(2) acts on internal E components and couples only to ψ_L, giving chiral SU(2). SU(3) acts on internal T_1 components and commutes with the sublattice grading (color is internal, not spatial), giving vector-like SU(3). The standard lattice-QCD objection to identifying staggered chirality with physical chirality — that the doublers carry "wrong" chirality and contaminate the continuum theory — *inverts* in OI: the doublers are the physical generations (the $2^3 = 8$ Brillouin-zone corners stagger-reduce to 4 Dirac tastes — 3 generations plus the Higgs — Theorem 9), so the $\mathrm{U}(1)_\varepsilon$ symmetry IS the physical chirality, not an artifact to be removed.

Independent verification via anomaly cancellation. A chiral gauge theory is consistent only if its anomalies cancel. Theorem 15 below shows that the OI matter content cancels all six anomalies and determines the SM hypercharges $Y_Q = 1/6$, $Y_u = 2/3$, $Y_d = -1/3$, $Y_L = -1/2$, $Y_e = -1$ (uniquely in the sense stated there). This is an independent consistency check on the lattice-scale chirality identification: if the identification of sublattice parity with physical chirality were wrong, the resulting "chiral" theory would generically have nonzero anomalies, and the framework's prediction of the SM hypercharges would fail. Anomaly cancellation working out, with the unique SM values, is independent evidence that the lattice-scale chirality identified in step (i) is the correct physical chirality at IR scales.

Remark (relation to spatial sublattice grading). The ε used in Theorem 13's proof is the 4D parity $\varepsilon_{4D}(x) = (-1)^{\sum_{\mu=0}^{d} x_\mu}$. By the Bridge Theorem of §7.5, this is equivalent to the 3D spatial sublattice grading $\varepsilon_{3D}(\vec{x}) = (-1)^{x_1+x_2+x_3}$ for static observables: $\varepsilon_{4D} = (-1)^{x_0}\varepsilon_{3D}$, with $(-1)^{x_0}$ constant on each time slice. The framework-essential identification is therefore $\gamma_5 \leftrightarrow \varepsilon_{3D}$, with the time-direction sign factor reducing to a sign convention. This is consistent with the broader pattern that time enters OI's emergent 4D structure as bookkeeping rather than as fundamental Clifford structure (cf. §7.5 *Implication for the framework's $d = 3$ commitment*).

4.8.1 Evasion of Nielsen–Ninomiya

The Nielsen–Ninomiya theorem forbids a single Weyl fermion on the lattice under four hypotheses: (NN1) **locality** of the action (couplings decay faster than any power of separation), (NN2) **hermiticity** of the lattice Hamiltonian, (NN3) **translation invariance** under the lattice group, and (NN4) the action is a **bilinear in fermionic fields carrying a definite, conserved chirality charge** $U(1)_\chi$. Under NN1–NN4 the spectrum is forced to contain equal numbers of left- and right-handed Weyl modes — the doubling phenomenon — so a chiral gauge theory cannot be regulated on a lattice in the conventional sense.

OI manifestly satisfies NN1–NN3. The wave equation is nearest-neighbor (NN1, range exactly ϵ); the bijection φ is real and the dynamics is T-invariant, so the induced operator is hermitian (NN2); and $(S, \varphi)/\mathcal{G}_{\text{sub}}$ is invariant under the cubic translation group (NN3). Were NN4 also to apply, OI would inherit the no-go conclusion and chiral SU(2) would be impossible. It does not apply, and the reason is structural rather than evasive.

NN4 presupposes that the **fundamental** degrees of freedom are Grassmann fields $\psi(x)$ entering the action bilinearly, with chirality γ_5 defined ab initio on those fields. In OI the fundamental object is a *real scalar bijection* $\varphi : S \to S$ governed by the lattice wave equation. There are no fermions at the bijection level, no Grassmann variables, no γ_5, and consequently no $U(1)_\chi$ on which an obstruction could be formulated. Dirac structure is *derived*: Theorem 2 (Susskind) shows $D_{\rm st}^2 = -\frac{1}{4}\Box_{\rm lat}$, so staggered Dirac fermions appear as a *factorization* of the scalar wave operator, not as fields posited by an action principle. Theorem 12 then identifies the staggered spinor decomposition with the bipartite checkerboard partition of the lattice. Chirality only crystallizes in the third step, Theorem 13: the Schur-complement trace-out of the odd (right-handed) sublattice produces an effective theory whose external legs are uniformly left-handed, with $D_{LL} = D_{RR} = 0$ enforced by sublattice parity.

The NN premise therefore **fails to apply** at the level on which OI is defined — it is not violated within its scope of applicability. NN1–NN4 are conditions on lattice fermionic actions; the OI fundamental action is bosonic. By the time fermions and chirality exist as derived structures (post-Theorems 2, 12, 13), the trace-out has already broken the bilinear-fermionic-action setting that NN4 requires. The doubler-inversion remark of §4.8 makes the dual point from the opposite direction: the "wrong-chirality" doublers that NN forces upon a fermionic lattice action become, in OI, the physical generations (Theorem 9), so the very modes the no-go theorem treats as obstructions are reinterpreted as observed matter content. NN is not circumvented by a clever fermion formulation; it is bypassed because OI never enters its hypothesis space.

Remark (locality of the effective chiral operator, and its independence from the §3.1 (*Scope of the emergent-Lorentz claim*) smearing question). The "bypass" reading above is the strong one — that NN4 has no referent — and it is worth separating it explicitly from a weaker fallback that a referee might expect, namely that the effective theory escapes NN only by being *non-local* (the route taken by perfect-action and Ginsparg–Wilson formulations, where chiral symmetry is bought at the price of a non-local action). Whether OI needs that weaker fallback turns on the locality of the Schur-complement effective operator $K_{\rm eff} = D_{LR}\, G_{RR}\, D_{RL}$ of Theorem 13(iv): a long-ranged G_{RR} would make $K_{\rm eff}$ non-local, and the evasion would then be "non-locality escapes NN" rather than "NN does not apply." On the free substratum the question resolves in favour of the strong reading. The odd-sublattice self-term in the bipartite wave operator is the time part $2\cos\omega$, not a frozen spatial constant; on the physical dispersion surface $2\cos\omega = t(\mathbf{k})/d$, the effective kernel reduces to $K_{\rm eff}(\mathbf{k}) = t(\mathbf{k})/d = \frac{2}{d}\sum_j \cos k_j$ — a degree-one trigonometric polynomial, i.e. a strictly nearest-neighbour (finite-range) kernel in position space, with no interior poles. (An apparent interior divergence on the $t(\mathbf{k}) = \pm 4$ surface arises only in the frequency-frozen static approximation, which replaces $2\cos\omega$ by a spatial constant; it is an artifact of that approximation and is absent once the dispersion is imposed.) The free effective chiral operator is therefore local, and the NN evasion is the strong form. The same Schur-complement structure yields a second exact property. With the site-diagonal grading $\Gamma = (-1)^{x+y+z}$ and the nearest-neighbour operator anticommuting ($\Gamma A \Gamma = -A$), the effective kernel obeys $\Gamma_V K_{\rm eff}(E)\,\Gamma_V = -K_{\rm eff}(-E)$ for *any* site partition — a one-line identity from the block-restriction of Γ. At the mirror point the kernel is exactly chiral and the staggered-mass projection vanishes identically: the trace-out preserves the bipartite grading and cannot gap the taste multiplet at the linear level, for the horizon partition or any other. This has been verified at machine precision in full $d = 3$ with two-directional controls (a grading-breaking deformation confined to the hidden sector is detected; grading-preserving disorder is not). The interacting (condensate-dressed) extension is the residual, shared with §3.1 (*Scope of the emergent-Lorentz claim*).

This locality result is a statement about the *free* effective operator, and it should not be conflated with the separate — and still open — question of whether the interacting, condensate-dressed coarse-graining map realises a sufficient ultraviolet *smearing* to protect the rotational sector at loop level (§3.1 (*Scope of the emergent-Lorentz claim*, point (iii))). The two questions share the object G_{RR} but are different computations: the chirality question asks whether the free kernel is finite-range (answered here: yes), while the Lorentz-radiative-stability question asks whether the dressed induced links inherit the matter-field smearing (an interacting order-parameter question, addressed in §3.1 (*A present-day constraint, and the rotational sector's distinct status*) and [Ch. 19 §19.3.7], where the current evidence leans toward the un-smeared regime). A local free chiral operator and an un-smeared dressed condensate are consistent: the former secures the chirality derivation of this section, while the latter leaves the radiative-stability protection of §3.1 (*Scope of the emergent-Lorentz claim*) as the open item it is stated to be there.

4.9 Anomaly cancellation and hypercharges

Theorem 14 (U(1)_Y automaticity). *The existence of a U(1) gauge factor is automatic in the multi-component framework.*

Proof. Each factor of the commutant decomposes as $\mathrm{U}(n_i) = \mathrm{SU}(n_i) \times \mathrm{U}(1)_i$. For K = 6 with multiplicities (3, 2, 1): $G = [\mathrm{SU}(3) \times \mathrm{U}(1)_B] \times [\mathrm{SU}(2) \times \mathrm{U}(1)_L] \times \mathrm{U}(1)_S$. Three U(1) factors exist automatically. A general abelian charge is $Q = \alpha B + \beta L + \gamma S$. Given SU(3) $\times$ SU(2) and the minimal fermion content (5 representations per generation), the six anomaly conditions impose 4 independent constraints on the 5 hypercharge unknowns, leaving exactly one free parameter (overall normalization). There is therefore exactly one anomaly-free U(1). □

Remark (the two non-gauge U(1) combinations). The two combinations of (B, L, S) orthogonal to hypercharge cannot be gauged: gauging them would introduce non-cancelling chiral anomalies. They survive as accidental global symmetries of the emergent theory — the same status that baryon number B and lepton number L have in the standard SM construction, broken only by the Weinberg operator and non-perturbative effects. No new gauge bosons are introduced, and the framework's prediction of B and L as accidental global symmetries matches observation (proton stability; lepton number conservation up to the small Majorana mass).

Theorem 15 (Unique hypercharges). *Given* SU(3)×SU(2)×U(1) *with fermions in fundamental or singlet representations, the six anomaly conditions [18] determine the hypercharges uniquely up to the overall normalization and the interchange $Y_u \leftrightarrow Y_d$ — the latter a relabeling of which quark species is called up-type, since the anomaly conditions are symmetric in (Y_u, Y_d); the labeling is fixed by the electric-charge identification* $Q_{\text{em}} = T_3 + Y$ [16]:

$$Y_Q = \tfrac{1}{6}, \quad Y_u = \tfrac{2}{3}, \quad Y_d = -\tfrac{1}{3}, \quad Y_L = -\tfrac{1}{2}, \quad Y_e = -1$$

Corollary. |q_p| = |q_e| is a theorem, not a coincidence.

4.10 Renormalizability

Theorem 16. *The Yang-Mills coupling has mass dimension [g] = (3−d)/2; dimensionless iff d = 3.*

Proof. The Yang-Mills action $S = \frac{1}{4g^2}\int d^{d+1}x\, F_{\mu\nu}^2$ requires $[g^{-2}][F^2] = [x]^{-(d+1)}$. With $[F] = [x]^{-2}$: $[g]^{-2} = [x]^{d-3}$, so $[g] = (3-d)/2$. □

Since d = 3 is derived (§3.2), the emergent QFT supports renormalizable gauge interactions with asymptotic freedom.

4.11 Spontaneous symmetry breaking and the hierarchy

Chiral symmetry (Theorem 3) combined with the center-independent gauge sector forbids explicit masses. The unique renormalizable unitarity-preserving mechanism is the Higgs [13, 14]. The minimal scalar breaking $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \to \mathrm{U}(1)_{\text{em}}$ while preserving $\mathrm{SU}(3)_c$ is $H = (\mathbf{1}, \mathbf{2}, +\frac{1}{2})$.

The hierarchy problem is dissolved: in the OI framework, the QFT is emergent. The Higgs mass is set by eigenvalues of M (properties of φ), not by radiative corrections.

4.12 The filter chain

The complete derivation from (S, φ) to the Standard Model is laid out across §§3–5. The primary chain is proved end-to-end: wave equation → KG → staggered fermions → chiral symmetry → K = 2d = 6 → cubic decomposition (3,2,1) → local gauge invariance → anomaly cancellation → chirality → three generations → one Higgs. Theorem 21 carries the construction to $\bar{\theta} = 0$ within the emergent description it produces; §5.5 states the four conditions standing between that and the strong CP problem of the Standard Model, none of them discharged. Redundant second-route confirmations (asymptotic freedom, minimal N_c, minimal chiral group) are at proposition level and provide independent checks.

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5. The Strong CP Problem

5.1 T-invariance of the wave equation

Theorem 17. *The discrete wave equation is invariant under time reversal T: φ(n, t) → φ(n, −t).*

Proof. Under T, define ψ(n, t) = φ(n, −t). Substitution gives ψ(n, t+1) = ψ(n−1, t) + ψ(n+1, t) − ψ(n, t−1): the same wave equation. □

The checkerboard partition marginalizes over spatial sites, not temporal data. It preserves T while breaking P (§4.8).

5.2 $\theta = 0$

Theorem 18. *The QCD θ-parameter vanishes: θ = 0.*

Proof. The θ-term $\mathcal{L}_\theta = (\theta/32\pi^2)\mathrm{Tr}(F\tilde{F})$ is T-odd: $\mathrm{Tr}(F\tilde{F}) = 2\,\mathbf{E}\cdot\mathbf{B}$, and under T, $\mathbf{E} \to \mathbf{E}$, $\mathbf{B} \to -\mathbf{B}$. A T-invariant Lagrangian cannot contain a T-odd term. The emergent Lagrangian inherits T-invariance because the partition is spatial, so time reversal commutes with the visible/hidden projection. □

5.3 Detailed balance and $\bar{\theta} = 0$

Theorem 19 (Detailed balance). *For a spatial partition (one preserved by time reversal: $\pi_V \circ \mathcal{T} = \pi_V$), the visible-sector transition probabilities satisfy $T_{ij} = T_{ji}$ for all i, j.*

Proof. The full dynamics φ is a bijection, and T-invariance (Theorem 17) gives $\varphi^{-1} = \mathcal{T} \circ \varphi \circ \mathcal{T}$. For each hidden state h with $\pi_V(\varphi(i,h)) = j$, let $h' = \pi_H(\varphi(i,h))$. Then $\varphi(i,h) = (j,h')$, so $\varphi^{-1}(j,h') = (i,h)$. Since the partition is spatial, $\mathcal{T}$ preserves the visible/hidden classification. The map $h \mapsto h'$ is an injection from $\{h : \pi_V(\varphi(i,h)) = j\}$ into $\{h' : \pi_V(\varphi(j,h')) = i\}$. Since φ is a bijection on a finite set, the injection is a bijection between sets of equal cardinality. Dividing by $|\mathcal{C}_H|$: $T_{ij} = T_{ji}$. □

Theorem 20 (Detailed balance implies T-invariant Hamiltonian). *If $T_{ij} = T_{ji}$ for all i, j, then the emergent Hamiltonian satisfies $[\hat{H}_{\mathit{eff}}, \hat{\Theta}] = 0$ where $\hat{\Theta}$ is the anti-unitary time-reversal operator.*

Proof. The Born rule gives $T_{ij}(t) = |U_{ij}(t)|^2$ (Prerequisites; Barandes correspondence [28, 29]). Combined with $T_{ij} = T_{ji}$: $|U_{ij}|^2 = |U_{ji}|^2$ for all i, j, t. This symmetry of the transition amplitudes, by Wigner's theorem, implies $U(t) = \hat{\Theta}\,U(t)^*\,\hat{\Theta}^{-1}$ for a suitable anti-unitary $\hat{\Theta}$, which gives $[\hat{H}_{\mathrm{eff}}, \hat{\Theta}] = 0$. □

Theorem 21 ($\bar{\theta} = 0$ at the kinematic level of the construction). *In the lattice construction of §2–§4, with the substratum bijection exactly T-invariant and the emergent Hamiltonian satisfying detailed balance, the physical parameter $\bar{\theta} = \theta + \arg\det(Y_u Y_d) = 0$, and the vanishing propagates to every scale of the emergent description so constructed.* The theorem is conditional on the construction's hypotheses and does not by itself dissolve the strong CP problem of the physical Standard Model; see §5.5.

Proof. **Step 1:** Since φ is a bijection, φ^n is a bijection for every n. Applying the counting argument of Theorem 19 to φ^n: $T_{ij}(n) = T_{ji}(n)$ for all n. (Verified numerically: T-violation is exactly zero for L = 4, 6, q = 2, 3, 5, and all n = 1, ..., 12.)

Step 2: By Theorem 20 applied to $T_{ij}(n)$ at each scale: $[\hat{H}_{\text{eff}}(n), \hat{\Theta}] = 0$. The effective Hamiltonian is T-invariant at every time scale.

Step 3: T-invariant Yukawa couplings at each scale are simultaneously real in an appropriate basis [16, §6.3]. Real Yukawa matrices have $\arg\det(Y_u Y_d) = 0$. Combined with $\theta = 0$ (Theorem 18): $\bar{\theta} = 0$ at every scale of the constructed emergent description. ☐

[Status note, 2026-08-04: Step 3 requires ***simultaneous*** *reality — that Y_u and Y_d are real in one common basis — and that is what a T-invariant H_{eff} supplies. The weaker statement that each Yukawa is real in* its own *mass basis is vacuous: every complex matrix is diagonal and positive in its own mass basis by singular value decomposition, so it would give* $\arg\det(Y_u Y_d) = 0$ *for any Yukawa sector whatever. The common-basis reading that carries this step also forces $V_{CKM} = O_u^{\mathrm{T}} O_d$ to be real orthogonal and the Jarlskog invariant to vanish identically; see the scope discussion in §5.5, condition (i).]*

Remark. This proof bypasses the instanton question. T-invariance of the transition probabilities is an exact consequence of the bijection structure, holding at every time scale without perturbative or non-perturbative approximation. The RG flow cannot generate T-violation because the underlying bijection structure forbids it at every scale.

Remark (scope of Theorem 21). The vanishing of $\bar{\theta}$ does not extend to the flavor-sector CP-violating phases observed in CKM and PMNS mixing. These phases arise from a distinct structural source and are treated as solution-specific inputs; see §5.4 for the scope distinction and its pre-registration.

Conditional prediction: $\bar{\theta} = 0$ with no axion required, *within the construction*. The framework does not yet deliver a quantitative neutron-EDM prediction: an exactly vanishing d_n follows only if $\bar{\theta} = 0$ survives the conditions of §5.5, and until radiative stability against a CP-violating weak sector is established the framework's commitment is the weaker one that d_n lies below the current bound, $|d_n| < 1.8 \times 10^{-26}\ e \cdot \mathrm{cm}$ [19].

5.4 Scope: flavor-sector CP violation as solution-specific input

Theorem 21 establishes $\bar{\theta} = 0$, the *static* CP-odd combination $\theta + \arg\det(Y_u Y_d)$ — the quantity that couples to QCD instantons and would generate a neutron electric dipole moment. The theorem does *not* forbid the CKM and PMNS CP-violating phases observed in flavor-changing processes. This subsection delineates the scope of the $\bar{\theta} = 0$ result and makes explicit that the framework treats flavor-sector CP phases as solution-specific inputs rather than framework-level predictions.

The distinction. $\bar{\theta}$ is a property of $\det(Y_u Y_d)$: it depends only on the products of Yukawa eigenvalues and vanishes whenever each Yukawa matrix is individually diagonalizable to real form. CKM CP violation, by contrast, is encoded in the Jarlskog invariant $J \approx 3 \times 10^{-5}$ and arises from the *basis mismatch* between the up-type and down-type Yukawa mass eigenstates. In the Standard Model the Yukawas are not required to be real in any common basis: each is real in its own mass basis, the up-type and down-type mass bases generically differ by a unitary transformation, and when that transformation has complex entries that cannot be absorbed by field redefinitions the resulting CKM matrix contains an irreducible CP-violating phase [16, §6.3]. That route is closed in the present construction, and the difference is the premise. The detailed balance of Theorem 19, holding at every time scale as Theorem 21 Step 1 requires, is equivalent to H_{eff} being rephasing-equivalent to a real symmetric matrix — the obstruction enters at third order in t and is the rephasing-invariant triangle phase, which must vanish. Two matrices real in a common basis have real orthogonal singular value decompositions, so the mismatch unitary is real orthogonal and $J = 0$ exactly. The consequence for the scope of Theorem 21 is stated in §5.5, condition (i).

What the framework does and does not derive. T-invariance at the substrate level (Theorem 17) plus detailed balance (Theorem 19) forces each Yukawa matrix to be individually diagonalizable to real form, fixing $\bar{\theta} = 0$ at every scale of the construction (Theorem 21, conditional per §5.5). It does *not* force the up-type and down-type mass bases to coincide. The basis-alignment unitary is a solution-specific property of the particular bijection φ, analogous to the absolute scale of fermion masses (§7.6) and the specific eigenvalues of the Yukawa matrices: the framework derives the structural form of the Yukawa sector but does not predict the basis-alignment parameters. What the construction does force is that both Yukawas are real in one common *interaction* basis, and that alone makes the mismatch unitary real orthogonal, hence $J = 0$, whatever the mass bases are. The CKM and PMNS CP-violating phases are therefore not accommodated by the framework's structural commitments as they stand; §5.5, condition (i), states what remains open and what does not.

[Status note, 2026-08-04: earlier revisions of this paragraph presented those phases as compatible with — though not derived from — the framework's structural commitments; that argument is withdrawn.]

The reducible phases are exactly the field redefinitions (nullity). The $\bar{\theta} = 0$ result relies on a supporting fact worth stating precisely: among the CP phases carried by a T-invariant Yukawa sector, those that are physically vacuous — removable by field redefinition — are *exactly* the diagonal rephasings, and nothing else is spuriously counted as removable. Cast in the natural language: a phase assignment θ_{ab} on generation pairs is field-redefinition-removable iff it is a coboundary $\theta_{ab} = \alpha_a - \alpha_b$ of a rephasing vector α. The claim that the removable phases *exhaust* the closed ones — those consistent around every triangle $\theta_{ab} + \theta_{bc} + \theta_{ca} = 0$ — is the vanishing of the first cohomology of the generation-coupling complex. For the physical sector, where all three generations couple pairwise, that complex is the filled triangle K_3, which is simply connected, so $H^1 = 0$: every closed phase configuration is a coboundary, and the only irreducible phase is the single Jarlskog invariant that survives H^1 of the *mismatch* between the up- and down-sectors. The result generalizes to any number of generations whose coupling graph is chordal (every cycle triangulated); it can fail only on a coupling graph with an unfilled cycle, which the all-pairs-coupled generation structure excludes. This closes the nullity step numerically observed across generation counts, replacing the case check with the topological identity $H^1(K_n) = 0$.

Pre-registration. Flavor-sector CP phases are listed as solution-specific inputs in §8.3 and in the pre-registered falsification conditions ([Substratum, §6.4, "what does not falsify"]). They join the absolute fermion mass scale m_s and the specific neutrino Yukawa eigenvalues in the class of quantities the framework treats as properties of the particular φ. The framework makes no commitment to any specific value of δ_{CKM}, δ_{PMNS}, the Jarlskog J, or the Wolfenstein parameters (ρ, η), and the measurement of these quantities does not bear on the framework's structural status.

5.5 Scope of Theorem 21: what is established and what is not

Theorem 21 is a statement about the construction of §2–§4, not yet about the physical Standard Model, and this subsection states the gap precisely so the result is not read as more than it is.

What is established. Given an exactly T-invariant substratum bijection and the detailed balance that follows from it, the emergent Hamiltonian is T-invariant, the θ-term is excluded, and T-invariant Yukawas are simultaneously real, so $\bar{\theta} = 0$ at every scale of the emergent description the construction produces. This is a kinematic consequence of the symmetry assumed at the substratum; nothing in it depends on instanton dynamics.

What is not established. Four conditions stand between that statement and a resolution of the strong CP problem, and none is discharged here.

(i) **A consistent CP-violating weak sector.** §5.4 places the observed CKM and PMNS phases outside the theorem's scope by attributing them to solution-specific basis alignment. That division does not survive. Detailed balance at **every** time scale — the premise Theorem 21 Step 1 asserts — is equivalent to the emergent Hamiltonian being rephasing-equivalent to a real symmetric matrix: expanding $U = e^{-iHt}$, the anti-symmetric part of $|U_{ij}|^2$ enters at $O(t^3)$ with coefficient $-2\,\mathrm{Im}[H_{ij}(H^2)^*_{ij}]$, which for a three-level system reduces to the rephasing-invariant triangle phase $\mathrm{Im}[H_{12}H_{23}H_{31}]$; a Hermitian matrix is rephasing-equivalent to a real one exactly when that vanishes. A real H_{eff} places Y_u and Y_d in a common real basis, and two real matrices have real orthogonal singular value decompositions, so V_{CKM} is real orthogonal and the Jarlskog invariant vanishes identically — a rephasing-invariant statement, so no field redefinition recovers it.

A quantitative computation on the emergent operator of the §4.8 reconstruction settles both horns. If the operator is octahedral-equivariant, the Jarlskog invariant vanishes identically. If the equivariance is broken, the Jarlskog and the forbidden E_g rotational anisotropy turn on together — measured exponents $J \sim \varepsilon^3$ and $E_g \sim \varepsilon$ — so the breaking strength that reproduces the observed $J = 3.0 \times 10^{-5}$ produces a fractional E_g anisotropy of order 10^{-3}, some twelve orders of magnitude above dimension-four rotational-Lorentz-violation bounds, in the most favourable of the sampled breaking directions and sector assignments. Within the construction, rotational isotropy and CP conservation stand or fall together, and observation requires one of each. The supports of this conclusion are the identification of the §4.8 staggered reconstruction as the coarse-graining map, the all-order detailed balance asserted in Theorem 21 Step 1, and the single-hypercube operator family on which the exponents were measured.

[Status note, 2026-08-04: earlier revisions of this condition described it as awaiting an example of a T-invariant bijection with nonzero Jarlskog invariant; the results above supersede that reading.]

(ii) **An explicit $\bar{\theta}$-fixing mechanism at the emergent level.** The argument fixes $\bar{\theta}$ by symmetry of the substratum. A mechanism operating on the emergent effective Lagrangian — the level at which $\bar{\theta}$ is measured — is not supplied, and the transfer from the one level to the other is the step the construction assumes rather than proves.

(iii) **Radiative stability.** In the Standard Model a CP-violating weak sector regenerates $\bar{\theta}$ radiatively, at high loop order and far below the experimental bound but not at zero. Once condition (i) is met, the framework owes an argument that its $\bar{\theta} = 0$ survives that regeneration. "Exactly zero at all energy scales" is not sustainable as a claim about the physical theory before that argument exists.

(iv) **A quantitative neutron-EDM prediction.** The falsification statement attached to this result should be a predicted d_n with an error, or a bound the framework commits to. An exactly vanishing d_n is a prediction of the construction, not yet of the framework applied to the observed world.

Accordingly the framework's commitment on strong CP is: $\bar{\theta} = 0$ at the kinematic level of the lattice construction, a conditional result. The four conditions above are the content of the open item, and are tracked as part of the matter-sector programme of §8.9.

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6. Gauge Coupling Prediction

The derivation chain of §§3–5 determines the gauge group $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$, the matter content, and the discrete symmetries. This section extends the chain to the gauge coupling strengths — the values of α_1, α_2, α_3 at the Z boson mass.

6.1 The induced coupling

The gauge field is not an independent degree of freedom — it is the state-dependent coupling matrix $M(\mathbf{n}, \hat{e}_j)$ of §3.1, determined by the matter field $\phi \in (\mathbb{Z}/q\mathbb{Z})^{K\times N}$. The gauge kinetic term is therefore *induced* by the fermion determinant: the effective action is $S_{\text{eff}}[U] = -N_f \operatorname{Tr}\ln(D^\dagger D[U])$, and the gauge propagator arises from expanding S_{eff} to second order in A_μ.

The induced coupling is computed from the one-loop staggered vacuum polarization $\Pi_s(0)$ — the lattice momentum integral over the fermion bubble with the OI vertex $V_j = ig\mu T^\alpha \cos(k_j + p_j/2)$. Since the coupling matrix is scalar, $M = \mu \cdot I_6$ (Theorems 6–7 jointly; the gauge group derives from the condensate, Theorem 5 revised), the Dynkin index $T(R) = 1/2$ per Dirac fermion is the same for all three gauge factors. With $N_f = 6$ (one per internal component), the induced coupling is universal:

$$\frac{1}{\alpha_0} = 6 \times \Pi_s(0) \times 4\pi = 23.25$$

This is a *structural* prediction — it depends on N_f and $T(R)$, both determined by the lattice structure (§§4.5–4.6), not on the eigenvalues μ_c, μ_w of M. The universality follows from the Dynkin index equality $T_3(R) = T_2(R) = T_1(R)$, which holds because all six components transform in the fundamental representation of their respective gauge group.

Note on the perturbative form. The integrand above is hybrid in dimensional structure: the loop integration $\int d^4q$ and the lattice propagator $D(q) = \sum_\nu \sin^2(q_\nu) + m^2$ run over four directions (consistent with the substratum's 3+1D Lorentzian structure), while the vertex factors $\cos^2(q_\mu)$ are spatial-only ($\mu \in \{1, 2, 3\}$, consistent with gauge fields living on spatial links only — there is no time-direction gauge connection in the substratum). The result is a perturbative form that is neither standard 4D staggered VP (which has $\cos^2$ on all four directions) nor standard 3D PT (which would have a 3D loop integral). At one loop, this form gives a finite, well-defined integral converging to $\Pi_s \to 0.3084$ on a robust plateau. At two loops, the strict 2-loop result for the universal threshold δ_0 vanishes at large N (the dominant scalar-bubble integrand falls as $1/N^{3.09}$), which is why $\delta_0 \approx 10$ is treated as empirical input rather than derived (§6.2). A proof that this hybrid form is a consistent perturbative scheme — closing at 1-loop and giving the observed asymptotic behavior at higher loops — is open work.

6.2 The threshold and logarithmic resummation

The universal induced coupling α_0 is defined at the lattice cutoff $\mu = M_{\rm Pl}$. At this scale, gauge self-interactions — which are proportional to the quadratic Casimir C_2 — generate a group-dependent threshold. A universal (C_2-independent) threshold δ_0 also arises from higher-loop corrections to the vacuum polarization.

The expansion parameter for the C_2-dependent threshold is $C_2 \cdot g_0^2 = C_2 \times 4\pi\alpha_0 = 1.08$ (SU(2)) and 1.62 (SU(3)) — both $\mathcal{O}(1)$. Finite-order perturbation theory converges too slowly: using known Wilson-action plaquette coefficients, 3-loop PT explains only $\sim 26\%$ of the threshold at SU(3). The all-orders result is well parameterized by a geometric-series resummation of the gauge self-energy:

$$\frac{1}{\alpha_i(M_Z)} = \underbrace{23.25}_{1/\alpha_0} + \underbrace{10.0}_{\delta_0} + \underbrace{8.3 \cdot \ln(1 + 5.59 \cdot C_2 g_0^2)}_{\text{resummed gauge self-energy}} + \underbrace{\frac{b_i}{2\pi} \ln \frac{M_{\rm Pl}}{M_Z}}_{\text{SM RG running}}$$

Above $M_{\rm Pl}$, the OI induced theory has no tree-level gauge action — gauge dynamics arise entirely from the fermion determinant — so its β-function differs at 2-loop and beyond from the standard SM running below the matching scale. The threshold function structurally absorbs this discontinuity at $M_{\rm Pl}$, which is why the matching formula has the form shown rather than continuous running across the scale. This reflects the substratum-vs-emergent split (group structural at the substratum; couplings emergent through the fermion determinant).

The four components:

- $1/\alpha_0 = 23.25$: computed structurally (one-loop staggered vacuum polarization, §6.1). Analytically verified from the lattice integral $\Pi_s(0) = \int d^4q\, [\cos^2(q_\mu)/D(q) - \cos^2(q_\mu)\sin^2(q_\mu)/D(q)^2]$ with $D(q) = \sum_\nu \sin^2(q_\nu) + m^2$, spatial average over $\mu = 1, 2, 3$; converges to $\Pi_s \to 0.3084$ as the grid refines, giving $1/\alpha_0 = 6 \cdot 0.3084 \cdot 4\pi = 23.25$ on a robust plateau $m \in [0.001, 0.05]$.
- $\delta_0 = 10.0$: the C_2-independent threshold. In the current formulation, δ_0 is determined empirically by the U(1) row (see below). A 2-loop staggered VP computation was implemented as a consistency check, and subsequently rebuilt from first principles (scalar and vector self-energy insertions, vertex correction, sails/crossed-gluon, and ghost 2-loop) using the Π_s convention above, with the conversion $\delta_0 = (1/\alpha_0) \cdot (\delta\Pi_s^{(2L)}/\Pi_s)$ tied to $1/\alpha_0 = 24\pi \cdot \Pi_s$. Large-$N$ scans of the dominant scalar-bubble SE piece across $N = 6, 8, 10, 12, 14, 16$ at $m = 0.05$ show its raw integrand falls as $1/N^{3.09}$ with per-point residuals below 6%, consistent with continuum-limit value zero; the apparent "plateau" at intermediate N is a finite-N coincidence. The assembled 2-loop sum at $N = 8$ of approximately $+72$ in δ_0 units is therefore a finite-N artifact, and strict 2-loop lattice PT

in OI does not reproduce the empirical $\delta_0 \approx 10$. We therefore treat δ_0 as a universal-row matching parameter determined by the U(1) threshold (below), analogous to how Standard Model input parameters are fixed by observation rather than derived. A natural 3-loop bound provides the structural plausibility check:

3-loop bound. The non-perturbative coupling at the staggered VP scale gives an effective expansion parameter $\sim g_0^2/\pi \approx 0.17$. Standard lattice perturbation theory in this regime has successive-order ratios of 0.20–0.30. Taking the structural input $1/\alpha_0 = 23.25$ as the leading scale and applying one power of the 2-loop perturbative ratio gives an expected threshold of order $0.2\text{–}0.3 \times 23.25 \approx 4.6\text{–}7.0$, and an expected 3-loop correction of order $0.2\text{–}0.3 \times 4.6\text{–}7.0 \approx 1\text{–}2$. The empirical value $\delta_0 \approx 10$ exceeds this natural 2-loop order by a factor consistent with the accumulation through third order — not tight enough to serve as a derivation, but consistent with typical non-asymptotic lattice-PT behavior.

Empirical determination via the U(1) row. The U(1) row (where $C_2 = 0$, eliminating the gauge self-energy contribution entirely) determines $\delta_0 = 10.02$ from the observed $1/\alpha_1(M_Z) = 59.00$, given the structural input $1/\alpha_0 = 23.25$ and SM RG running. This empirical determination is consistent with the natural 3-loop expectation (order of magnitude).

Numerical observation. The empirically-fixed value $\delta_0 = 10.02$ coincides to 0.2% with the hypercharge-squared trace $\mathrm{Tr}[Y^2] = 10$ of three generations of SM fermions in standard convention ($Q = T_3 + Y$): Q_L contributes $6 \cdot (1/6)^2 = 1/6$, u_R contributes $3 \cdot (2/3)^2 = 4/3$, d_R contributes $3 \cdot (1/3)^2 = 1/3$, L_L contributes $2 \cdot (1/2)^2 = 1/2$, e_R contributes $1 \cdot 1^2 = 1$, summing to 10/3 per generation, or 10 for three generations. The analogous group-theoretic sums for the other two factors are $\sum_R T(R) = 6$ for both SU(2) and SU(3) (with matter in fundamentals, three generations), coinciding with $N_f = 6$ (the internal-component count from $K = 2d$). A naive group-specific interpretation $\delta_0^{U(1)} = \mathrm{Tr}[Y^2]$, $\delta_0^{SU(N)} = \sum_R T(R)$ is ruled out within the paper's framework: it would require (A, B) yielding $A \cdot B \approx 90$, inconsistent with the independently-derived $A \cdot B = 48 \pm 1.5$ (§6.2.1). An anomaly-coefficient identification also fails structurally: $\mathrm{Tr}[Y^2]$ is the fermion-only contribution to b_Y via the (2/3) factor in the one-loop β-function formula, not a member of the anomaly polynomial — the four standard SM gauge-anomaly invariants ($\sum Y^3$, $\sum Y$, $\sum_{\text{doublets}} Y$, $\sum_{\text{triplets}} T \cdot Y$) all cancel for SM matter content. A β-function matching-scale offset $\delta_0 = (b_1/2\pi)\ln(M_{\text{Pl}}/\mu_{\text{match}})$ reproduces $\delta_0 = 10.02$ at $\mu_{\text{match}} \approx 2.6 \times 10^{12}$ GeV in the U(1) row, but the same mechanism applied at the same μ_{match} to the SU(2) and SU(3) rows yields $\delta_0 = -7.74$ and -17.11 respectively, contradicting the framework's universality requirement on the C_2-independent threshold. The universal $\delta_0 = 10.02$ is therefore required by self-consistency of the framework;

whether the precise $\mathrm{Tr}[Y^2] = 10$ coincidence for the U(1) row is a numerical accident or reflects higher-order structural physics — substratum-level mode-counting on the cubic lattice, non-perturbative effects in the discrete substratum measure (§6.5), or lattice topological invariants beyond the framework's currently-developed scope — remains open and is treated as numerical pending such structural mechanisms.

- $A \cdot \ln(1 + B \cdot C_2 g_0^2)$ with $(A, B) = (8.3, 5.59)$: the resummed gauge self-energy contribution. The logarithmic form is motivated by geometric-series resummation of the gauge self-energy $1/(1-\Sigma)$ in the induced gauge theory; the effective exponent $p \approx 0.42$ at the OI coupling is the local slope of the logarithm at $C_2 g_0^2 \sim 1.3$, not a fundamental constant. In the current formulation, A and B are determined by matching to the observed SU(2) and SU(3) couplings at M_Z, given the structurally computed $1/\alpha_0$ and the empirically fixed δ_0. The leading coefficient $A \cdot B$ is independently derived from 2-loop lattice perturbation theory in the induced gauge theory (§6.2.1), giving $A \cdot B \approx 48$ under the $1/N^2$ leading-power assumption, against the fitted 46.4; the extrapolation is leading-power-dependent and the comparison inconclusive (§6.2.1).
- SM RG running: the one-loop β-function coefficients $b_1 = 41/10$, $b_2 = -19/6$, $b_3 = -7$ over $\ln(M_{\mathrm{Pl}}/M_Z) = 39.4$.

6.2.1 First-principles derivation of $A \cdot B$

The leading coefficient of the C_2-dependent threshold in the expansion of $A \ln(1+B \cdot C_2 g_0^2)$ — namely the product $A \cdot B$ multiplying $C_2 g_0^2$ at first order — is extracted from the 1-loop gauge self-energy $\Sigma_{\mu\nu}(p)$ in the induced gauge theory. The induced propagator $D_{\mathrm{ind}}(k) = 1/(k_{\mathrm{lat}}^2 \cdot \Pi_s)$ and the Wilson-plaquette 3-gluon vertex (the leading continuum-limit piece of the full induced Γ_3) enter the bubble diagram; a Faddeev–Popov ghost loop with the induced propagator restores transversality. The transverse self-energy is extracted via

$$\Pi_T(p^2) = \frac{\Sigma^{11}(p) - \Sigma^{00}(p)}{\hat{p}^2} \qquad \text{for } p \text{ along the 0-direction,}$$

with linear extrapolation $p^2 \to 0$ at each lattice size N (three external momenta $\lambda = 1, 2, 3$ at the smallest accessible lattice values).

Conversion formula. The relation between Π_T and the coupling shift is

$$\delta(1/\alpha) = \frac{4\pi \cdot |\Pi_T|}{b_0^{\mathrm{capture}}} \qquad \text{(per } C_2\text{, coupling-stripped),}$$

where b_0^{capture} is the fraction of the textbook pure-YM 1-loop β-function coefficient $b_0 = 11/3$ that is reproduced by the finite-lattice computation. This conversion is validated by a standard-QCD cross-check (replacing $D_{\text{ind}} \to 1/\hat{k}^2$, no Π_s factor), which reproduces the textbook Π_T to 100% at $N = 28, m = 0.05$ via the formula.

Lattice data. Computing at each lattice size N using each N's own $\Pi_s(N)$ and $b_0^{\text{capture}}(N)$ (measured from a standard-QCD cross-check at the same N and m), the resulting per-N values of $A \cdot B$ at $m = 0.05$ are:

N	$\Pi_s(N)$	$b_0^{\text{capture}}(N)$	$A \cdot B(N)$
20	0.3451	0.523	22.14
24	0.3255	0.460	28.55
28	0.3174	0.427	33.31
32	0.3135	0.408	36.92
36	0.3115	0.398	39.61
40	0.3104	0.392	41.70

Extrapolation to $N \to \infty$. Three independent estimators applied to the $A \cdot B(N)$ sequence give consistent results:

Estimator	$A \cdot B(\infty)$
Linear $1/N^2$ fit on $(N = 32, 36, 40)$	50.1 (residual std 0.05)
Shanks acceleration, level 2 (last term)	48.0
Shanks acceleration, level 3 (last term)	47.9
Shanks acceleration, level 4 (last term)	47.9

The extrapolated value is model-dependent beyond the estimator spread above, and the comparison to the fit is conditional on the assumed leading finite-size power. Fitting the six data points with a leading $1/N$ correction gives $A \cdot B(\infty) \approx 61$ and fits the sequence an order of magnitude better (rms residual 0.06 against 0.67 for the pure $1/N^2$ model over all six points); a $1/N^2 + 1/N^3$ model fits best of all (rms 0.02) and gives ≈ 52. Only the pure-$1/N^2$ extrapolation and the Shanks family (which presupposes near-geometric convergence) land near 48; the largest computed value is 41.7, so every candidate sits 15–45% above the data's endpoint. The physically motivated default for staggered actions is $O(a^2) = O(1/N^2)$ artifacts, which is the ground for preferring the 48-family — but the induced-propagator normalization $\Pi_s(N)$ and the $b_0^{\text{capture}}(N)$ division can each introduce $O(1/N)$ pieces, and the data in this range in fact prefer a leading $1/N$ term. The comparison against the fitted 46.4 is therefore *conditional on the* $1/N^2$ *leading-power assumption*: under it, agreement at the few-percent level; under the better-fitting $1/N$ model, a $\sim 30\%$ disagreement. The cross-check is **inconclusive pending either substantially larger N or a derivation of the leading finite-size power of the induced theory**.

Uncertainty budget.

- $A \cdot B$ extrapolation to $N \to \infty$ (Shanks vs $1/N^2$ fit spread): $\pm 3\%$
- Subleading contributions. The F^3 correction has been propagated through the bubble: the sensitivity is $|\Delta(A\cdot B)/(A\cdot B)| \approx 8.4\cdot|c_{F^3}|$, extracted from a c_{F^3} sweep at $N = 8, 12, 16, 20$ with $1/N^2$ extrapolation. Under the continuum 1-loop fermion-induced value $c_{F^3} \approx -1/(360\pi^2) \approx -2.8 \times 10^{-4}$, the shift is $\approx 0.24\%$ — negligible within the extrapolation uncertainty. Under a conservative upper bound (naive $N_f = 6$ flavor multiplication, $c_{F^3} \approx -1.7 \times 10^{-3}$), the shift is $\approx 1.4\%$. A raw lattice value $c_{F^3} \approx -0.034$ extracted separately is in a different normalization (a $\sim 120\times$ discrepancy vs the continuum Wilson coefficient was noted in that extraction) and is not directly applicable in the bubble integration without a normalization-matching step. The dominant remaining subleading contribution is lattice V_{4g} cos-factor corrections to the tadpole's p^2-coefficient (absent in the continuum V_{4g} used here), bracketed at $\sim$ 3–12% (see Open work in §6.2.1).
- Convention and gauge-fixing prescription in the induced theory: $\pm 2\%$, bounded by the standard-QCD cross-check's 100% agreement.

The combined uncertainty is dominated by the extrapolation (statistical-like, $\pm 3\%$). Including estimated subleading contributions gives a total uncertainty $\lesssim 6\%$ at the lower end of the V_{4g} bracket (under which the computed value and the fitted value agree); the upper end of the bracket would widen the combined uncertainty correspondingly. The central extrapolated value $A \cdot B \approx 48$ carries the extrapolation-only uncertainty of the $1/N^2$ model, and is conditional on that leading power (§6.2.1).

Methodological notes. The Wilson-plaquette 3-gluon vertex is the leading piece of the full induced Γ_3; the full induced vertex is transverse by Ward identity (verified numerically at $N = 20$ to 1%), while the Wilson-plaquette piece alone requires the ghost contribution to restore transversality. The tadpole with continuum V_{4g} contributes only to the gauge-violating mass term and does not enter Π_T at leading order; lattice V_{4g} cos-factor corrections contribute to Π_T at the few-percent level and are currently subsumed into the systematic uncertainty (see Open work below). The OI framework has no formal Lagrangian gauge-fixing (the gauge field is induced rather than independent), but the induced propagator being a scalar in color and Lorentz is consistent with effective Feynman gauge, which the ghost prescription assumes.

Reproducibility. The full calculation runs in ~ 40 minutes on a single CPU core at $N \leq 40$. The implementation (`reproduce_AB.py` and supporting files) is archived with the code repository referenced in §8.

Open work. Separating A from B individually requires the next-order coefficient of $(C_2 g_0^2)^2$ in the expansion of the gauge self-energy. In the OI induced theory, this coefficient is structurally suppressed at strict 2-loop across all color channels. The C_A^2 channel receives only negligible Faddeev-Popov ghost contributions, verified numerically at approximately 0.2% of $A \cdot B$ via a direct 2-loop ghost self-energy computation. The mixed $C_A \cdot C_F \cdot T_F \cdot N_f$ channel, arising from crossed-topology fermion-line diagrams (vertex correction, sails), also approaches zero at continuum limit: large-N scans of the vertex-correction integrand show values decreasing as $1/N^{\sim 5}$ (raw: $9.96, 2.59, 0.64, 0.046$ at $N = 6, 8, 10, 12$), consistent with Ward-identity cancellation against the bubble/seagull self-energy pieces. Induced 3- and 4-gluon vertices first appear at 3-loop order (no tree-level gauge self-interactions exist in the induced theory). A clean separation of A from B therefore requires 3-loop lattice perturbation theory or non-perturbative methods, both beyond the current scope. Under the one-parameter constraint $A \cdot B = 48$ alone (§6.2.1), fitting the remaining freedom to the observed SU(2) and SU(3) couplings gives $(A, B) \approx (8.07, 5.95)$, numerically close to the factorization $(A, B) = (I_{\text{loop}}, N_f) = (8, 6)$, where $N_f = 6$ is the structural fermion count from $K = 2d$ (Theorem 6) and $I_{\text{loop}} = 8$ is the bare 1-loop lattice integral factor in the OI Π_s-normalized scheme (recovered from $A \cdot B = N_f \cdot 2T(R) \cdot I_{\text{loop}}$ with $2T(R) = 1$ for fundamentals). This factorization is compatible with the framework's universality requirement on A: I_{loop} is a property of the lattice scheme (geometry plus Π_s normalization), independent of which gauge factor the threshold is being evaluated for. The identification $A = \dim \operatorname{adj} \mathrm{SU}(3)$ — paired with the numerical match $\dim \operatorname{adj} \mathrm{SU}(3) = 8$ — is incompatible with universality (it would make A group-specific) and is a numerical coincidence rather than a structural identification. Both factorizations are consistent with the unconstrained fit $(8.3, 5.59)$ within the $\pm 3\%$ systematic uncertainty on $A \cdot B$. No rigorous derivation of individual A and B is yet available: the structural origin of $I_{\text{loop}} = 8$ on the $d = 3$ cubic lattice (candidates include $2^d = 8$ BZ corners, 4 staggered tastes $\times$ 2 sublattices, or another lattice-geometric factor) is an open follow-up; identifying it would upgrade $A \cdot B = 48$ to a structural derivation of (A, B) individually. The F^3 correction has been propagated (see uncertainty bullet above); the remaining systematic component is dominated by lattice V_{4g} cos-factor corrections. Two independent derivations of the lattice V_{4g} tadpole contribution have been carried out — one based on Wilson-extension of the continuum pair-topology decomposition, the other on per-topology Lorentz structure following Peskin Eq. 16.6 with Wilson cosines per Capitani Eq. 5.28. They differ structurally in the placement of cosine factors and yield Π_T contributions whose magnitudes differ by a factor ~ 14 at the integrand level (traced analytically to the $2\cos(p_\mu) \sum_\gamma \cos(k_\gamma)$ piece present in the per-topology form but absent from the simpler pair-topology form). The corresponding shift on $A \cdot B$ is bracketed at $\sim$ 3–12% depending on which form is normalized correctly; both are honest derivations and neither matches the textbook continuum tadpole formula directly at $p = 0$. Closing this ambiguity requires either a verified reference Wilson V_{4g} implementation or a fully-worked textbook 1-loop lattice Yang-Mills gluon self-energy computation as an external anchor; both are beyond current scope. The $\pm$4–5% residual systematic quoted above accommodates the lower end of this bracket; the upper end ($\sim$ 12%) would widen the systematic correspondingly. The central extrapolated value $A \cdot B \approx 48$, derived from $1/N^2$ extrapolation of the two-loop gauge self-energy and conditional on that leading power (§6.2.1), is independent of this V_{4g} subleading correction.

Resolution of the cosine-placement ambiguity (per-topology confirmed). The structural ambiguity above has been resolved at the level of cosine placement by direct comparison against the explicit Wilson four-gluon vertex of lattice perturbation theory (Capitani, *Lattice Perturbation Theory*, Phys. Rept. **384** (2003) 71, Eq. (5.28), attributed therein to Rothe, *Lattice Gauge Theories: An Introduction*, 1997). The vertex's cosine factors carry difference arguments spanning all four legs (of the form $\cos(a(q-s)_\mu/2)$ etc.); on the tadpole contraction, where two legs carry the loop momentum k and two the external momentum p, such an argument becomes $\cos(a(p \mp k)_\mu/2) = \cos(ap_\mu/2)\cos(ak_\mu/2) + \sin(ap_\mu/2)\sin(ak_\mu/2)$, which produces external×loop cosine cross-terms at the vertex. The lattice four-gluon vertex thus carries loop-momentum cosine dependence at the vertex, unlike the fermion seagull (whose tadpole vertex depends only on external momentum), so the $2\cos(p_\mu)\sum_\gamma \cos(k_\gamma)$ cross-term is a feature of the gauge tadpole rather than a retained artifact. The per-topology form retains this cross-term; the pair-topology form omits it. The comparison therefore selects the per-topology form at the level of cosine placement (not magnitude): on this resolution the residual V_{4g} systematic sits at the upper ($\sim 12\%$) end of the bracket above. The exact $A \cdot B$ shift remains to be computed by inserting Eq. (5.28) into the tadpole integral in the present Π_s-normalized scheme (a finite, well-defined task), but does not affect the central extrapolated value $A \cdot B \approx 48$, which is independent of the V_{4g} correction.

6.3 The prediction

Group	$1/\alpha_0$	δ_0	$A \cdot \ln(1+Bx)$	RG	Predicted	Observed
U(1)	23.25	10.02	0.00	+25.73	**59.00**	59.00
SU(2)	23.25	10.02	16.17	−19.88	**29.57**	29.57
SU(3)	23.25	10.02	19.13	−43.93	**8.47**	8.48

Parameter count. The chain contains one structurally computed input ($1/\alpha_0 = 23.25$) and three fitted parameters of the threshold function (δ_0, A, B) matched to the three observed couplings. Three parameters against three observables produces zero residual by construction; the zeroes in the right column are a consequence of the counting, not evidence of predictive power in themselves. The predictive content of §6 consists of the one computed value, two structural constraints on the threshold function, and a consistency check.

(i) The computed value $1/\alpha_0 = 23.25$. The one-loop staggered vacuum polarization, evaluated analytically in §6.1. No free parameters. This is the foundational structural input and would falsify the framework if observation required a significantly different value.

(ii) Structural constraint — universality of δ_0. The C_2-independent part of the threshold is by definition group-independent: a correction with no C_2 dependence cannot distinguish the three gauge factors. This is a structural constraint on the functional form of the threshold, not a fit. Its force is visible in the parameter count: without the universality constraint, the threshold function would carry five free parameters (three independent $\delta_0^{U(1)}$, $\delta_0^{SU(2)}$, $\delta_0^{SU(3)}$ plus A, B) matched to three observables — underdetermined, with the apparent match of the right column becoming empty. The universality reduces the count from five to three and makes the three-observable fit non-trivial: the fit could have been incompatible with a single δ_0 across rows.

(iii) Structural constraint — the log-resum form. The C_2-dependent threshold is parameterized as $A \ln(1 + B \cdot C_2 g_0^2)$, motivated by a geometric-series resummation of the gauge self-energy $1/(1-\Sigma)$ with $\Sigma \propto C_2 g_0^2$. This constrains the C_2 dependence to a two-parameter family rather than an unrestricted function. The step from $1/(1-\Sigma)$ to the logarithm is not derived in the current work; the log form should be read as a constrained parameterization consistent with the resummation structure, not as a derived result. The leading coefficient $A \cdot B$ (the $(C_2 g_0^2)^1$ term in the expansion) is computed from 2-loop lattice PT in the induced gauge theory (§6.2.1), giving $A \cdot B \approx 48$ under the $1/N^2$ leading-power assumption, against the fitted 46.4; the extrapolation is leading-power-dependent and the comparison inconclusive (§6.2.1). Separating A from B requires computing the next-order $(C_2 g_0^2)^2$ coefficient and is left for future work.

(iv) Consistency check with the 2-loop VP. The U(1) row fixes $\delta_0 = 10.02$ from observation, given (i). A first-principles 2-loop staggered VP computation was implemented as a cross-check, but on examination uses a scheme-locked Π_s convention. The consistency of a correctly-implemented 2-loop value with empirical δ_0 is an open question; the U(1)-row determination is the primary source of δ_0 in the current framework.

Status of the rows. U(1) uses one fitted parameter (δ_0) against one observable; the structural content is (i) together with the consistency check of (iv). SU(2) and SU(3) use two additional fitted parameters (A, B) against two additional observables; their structural content is carried by the universality constraint (ii) and the log-form constraint (iii). The product $A \cdot B$ (the leading coefficient in the $C_2 g_0^2$ expansion) is independently derived in §6.2.1 from 2-loop lattice PT in the induced gauge theory, giving $A \cdot B \approx 48$ under the $1/N^2$ leading-power assumption (inconclusive across finite-size models, §6.2.1) against the fit 46.4.

6.4 What is structural and what is solution-specific

§8.3 distinguishes structural predictions (properties of the equivalence class) from solution-specific properties (determined by the particular φ). The coupling calculation clarifies this distinction.

Structural: The universal induced coupling $1/\alpha_0 = 23.25$ is structural — it depends on $N_f = 6$ (from $K = 2d$, Theorem 6) and $T(R) = 1/2$ (fundamental representation), not on the eigenvalues of M. The logarithmic form of the C_2 dependence is structural — it follows from the perturbative structure of the gauge self-energy at any coupling. The SM β-functions are structural.

Solution-specific: The eigenvalues μ_c, μ_w of M determine which modes are massless (Theorem 4: $\cos\omega = \mu\cos k$), hence the mass spectrum, but they do NOT affect the gauge coupling at the cutoff — the induced coupling depends on the Dynkin index, which is eigenvalue-independent. The gauge couplings at M_Z therefore depend on the eigenvalues only indirectly, through the RG running (which involves particle masses that set decoupling thresholds). For the leading-order prediction above, this indirect dependence is subleading.

Non-unification. The couplings do not unify at M_{Pl}. The common value $1/\alpha_0 = 23.25$ is the induced coupling before gauge self-interactions; the C_2-dependent threshold splits the couplings at the cutoff itself. This is not a failure of unification — it is the correct structure of an induced gauge theory, where all factors share the same fermion-induced coupling but have different gauge self-energies.

6.5 The confinement question

A natural question is whether the purely induced theory — with no fundamental gauge action — is in the deconfined phase required for the perturbative matching to be valid. Direct Monte Carlo simulation of $Z = \int DU_{\mathrm{Haar}}\,|\det D(U)|^{2N_f}$ reveals that it confines: $\langle P\rangle \approx 0$ for all gauge groups (SU(3), SU(2), compact U(1)), all lattice volumes ($L = 2$–8), all fermion masses ($m = 0.01$–1.0), and all anisotropy parameters ($\mu = 0.5$–2.0). Cold-start simulations confirm monotonic decline of $\langle P\rangle$ from 1.0 toward 0. The fermion determinant cannot sustain weak coupling against the gauge field entropy.

This confinement is an artifact of the Haar measure. In the OI framework, the gauge field $M(\mathbf{n}, \hat{e}_j)$ is not drawn from the Haar measure on SU(N) — it is determined by the matter field $\phi \in (\mathbb{Z}/q\mathbb{Z})^{K\times N}$ (§3.1). The induced measure is the pushforward of the uniform measure on $(\mathbb{Z}/q\mathbb{Z})^{K\times N}$, which concentrates gauge configurations near the identity at large q. The rate depends on which reading of "uniform" is taken. Counting non-trivial link values, whose fraction scales as $1/q$, gives $\beta_{\mathrm{eff}} \sim \ln q$; but that counting presumes the site-independent field, whereas the on-shell measure is the smooth $\Box^{-1}$ field of §4.2, for which the exponential map's $2\pi/q$ amplitude gives $1 - \langle P\rangle \sim q^{-4}$ and hence $\beta_{\mathrm{eff}} \sim q^4$ (verified numerically for SU(3) over $q = 20$–200, and consistent with the plaquette-matching of §6.6). *[Status note, 2026-07-24: the code that produced the archived numerical checks in this section and in §7.5 is no longer extant. The construction itself is documented — the measure is uniform on the substratum field, pushed through the on-shell exponential map — and its q^{-4} plaquette-deficit scaling has been independently reproduced (fitted exponent -4.11 over $q = 15$–127). The specific archived values have not been reproduced and are unverified* *pending a pre-registered re-derivation. Update, 2026-07-25: that re-derivation has now been carried out on the directly sampled construction. At the archived point (4^4, $mL = 3$, matched $\langle P \rangle$) the directly sampled ensemble differs from the Wilson comparator in Z_S by $+2.6\% \pm 0.15\%$, beyond the pre-registered 2% tolerance, reproducibly across independent streams. The universality reported in this section is thereby established only for ensembles generated by Wilson dynamics on the image; for the directly sampled measure it is empirically violated at this point.]* Since q is gauge (§2.7) and the physical coupling is q-independent after renormalization, neither form is load-bearing — $\beta = 11.1$ below is fixed by α_0, not by q — but the two differ by orders of magnitude in where they place a given β_{eff}, and only the second is consistent with §4.2. Direct simulation of the 1D state-dependent coupling (§3.1, explicit construction) confirms that the wave equation creates no gauge correlations beyond random matter, but the $(\mathbb{Z}/q\mathbb{Z})$ constraint itself acts as an effective gauge kinetic term.

Since q is gauge (§2.7), the physical coupling is q-independent after renormalization. The perturbative calculation (§6.1) works in the q-independent sector and correctly determines the physical coupling. The Haar-measure Monte Carlo explored a confined phase that does not correspond to the OI framework's gauge measure.

Status of the induced-measure identification. The step this argument turns on — that the substratum's uniform measure pushes forward to a near-identity (rather than Haar) gauge measure — is a load-bearing commitment whose status should be stated at the rigor level of §A.3. The free-sector correspondence (the matter/pseudofermion measure equivalence, §4.2 Remark) is established at Level 2; the interacting statement used here, that the pushforward under the explicit map $\phi \mapsto M(\mathbf{n}, \hat{e}_j) \in \mathrm{SU}(N)$ concentrates near the identity, is not at that level, because the explicit map is under-determined at several enumerable joints (the real-to-complex step, the unitarization prescription) and the natural candidate map $M = \mathrm{polar}(\phi(\mathbf{n})^\dagger \phi(\mathbf{n} + \hat{e}_j))$ has been found numerically to produce a *Haar* (hence confining) pushforward, not the near-identity one this section requires. The near-identity claim is therefore a conjecture about the correct map, not a derived property of the natural candidate; the concentration "at large q" holds for an un-centered variant whose canonicity is not established. This has a direct consequence for the falsification-capable lattice work: the A^2 Monte Carlo and any production gauge-sector computation sample a specific induced measure, so their verdict is conditional on the measure identification being the near-identity one rather than the candidate's Haar — a dependency that the pre-registration of those computations should carry explicitly. Resolving which map is correct, and whether its pushforward is near-identity or Haar, is the open structural task on which the coupling sector's derived-versus-fitted status ultimately rests.

A candidate resolution. The under-determination isolated above lives at two enumerable joints — the amplitude of the matter measure and the map prescription — and both admit a natural resolution, set down here for scrutiny as a candidate rather than asserted as closed. First, the measure feeding the pushforward is the uniform measure *on the energy shell*, not the independent-uniform measure on $(\mathbb{Z}/q\mathbb{Z})^{K\times N}$: by the sphere-Gaussian projection of §4.2 the on-shell measure is the smooth Gaussian with covariance $[\Box_{\text{lat}}^{-1}]_{xy}$ — the same $O(1)$-amplitude field the pseudofermion equivalence already requires — so the two readings of "uniform" give opposite physics, and only the on-shell/smooth one is consistent with §4.2. Second, the canonical map from the $\mathbb{Z}/q\mathbb{Z}$-valued matter data to the group is the exponential $M(\mathbf{n},\hat{e}_j) = \exp(\frac{2\pi i}{q}\,\mathrm{Lie}\,[\phi(\mathbf{n}+\hat{e}_j)-\phi(\mathbf{n})])$ — the composite of the canonical roots-of-unity homomorphism $\mathbb{Z}/q\mathbb{Z}\to\mathrm{U}(1)$ with the Lie exponential $\mathfrak{g}\to G$ — which respects the $2\pi/q$ scale that the scale-invariant polar decomposition ($\mathrm{polar}(cA)=\mathrm{polar}(A)$) discards; this is why the polar candidate returns Haar irrespective of amplitude. On the smooth $\Box^{-1}$ field the exponential map yields a *near-identity* pushforward carrying weak, non-zero, octahedrally-equivariant non-abelian field strength — the $\beta_{\text{eff}}\sim\ln q$ regime this section requires — while the polar candidate's Haar result is recovered only on the (inconsistent) independent-uniform field. This has been checked numerically for SU(2) and SU(3): $\langle P\rangle\to 1$ as $q\to\infty$, with the deviation from unity vanishing as the field strength while remaining strictly positive. The identification's *kinematic* half — which measure, and which map — is thereby raised to theorem level, conditional on the additive-automorphism (amplitude-scale) gauge principle it invokes, which [Substratum §4, Theorem 24, *Status*] classifies as an adjoined operational input rather than a theorem (§2.7); the map-uniqueness therefore inherits that principle's open status. The measure is fixed by the sphere-Gaussian projection of §4.2. The map is fixed by a uniqueness statement grounded in the framework's own gauge principle. The physical gauge is the alphabet's additive-automorphism group — not arbitrary relabellings of $\mathbb{Z}/q\mathbb{Z}$, since a nonlinear relabelling changes the fluctuation dispersion and so is not a symmetry of the observables — so the induced connection must be a homomorphism of the matter field's additive ($\mathbb{Z}/q\mathbb{Z}$) structure, the structure the wave equation evolves. Under that requirement the exponential is the *unique* map up to the choice of generator (the homomorphisms $\mathbb{Z}/q\mathbb{Z}\to\mathrm{U}(1)$ being exactly the q-th roots of unity), and the polar candidate, a function of the multiplicative bilinear $\phi^\dagger\phi$ rather than the additive difference, is excluded because it is not such a homomorphism. The same map is selected independently by the continuum limit: q is a resolution parameter, so a sensible $q\to\infty$ limit is required, and the exponential's rescaled connection $(q/2\pi)\log M$ tends to the continuum gauge field q-independently, whereas the polar/Haar map is confining at every q with no such limit. The substantive residual is now the *dynamical* statement, that the interacting measure — the smooth pushforward reweighted by $\det(D^\dagger D)^{N_f}$ — retains this near-identity concentration. Because that reweighting is extensive, it does not follow from the classical pushforward result and must be argued directly. It does follow from a free-energy comparison. The links are unitary, so the staggered spectrum is bounded and $|\ln\det(D^\dagger D)|$ is bounded by cN with $c=O(1)$ *independent of* q: the determinant's total influence on the measure is extensive but q-blind. The prior, by contrast, is not: disordering the links requires field differences of order q, which the on-shell Gaussian penalizes by $\sim Nq^2/\pi^2$. The determinant therefore cannot pull the field out of the near-identity region once q exceeds an $O(1)$ threshold — numerically $q \gtrsim 9$ at $N = 4^4$, $m = 0.1$, $N_f = 6$ — and at the physical alphabet size the prior wins by a factor $\sim q^2$. The determinant moreover does not oppose the identity in the first place: at the same parameters $\ln\det(D^\dagger D)$ is $+260$ at $U = I$ against -85 on Haar-random links, so it *reinforces* the near-identity concentration rather than competing with it, which is why the Haar-measure confinement of the induced theory reported above is a statement about the gauge entropy of the Haar measure, absent here, and not about the determinant. The residual is the rigor of the comparison rather than its direction: the constants are measured at one small volume and mass, and a first-order transition at intermediate q is not excluded, though none appears. Whether the near-identity measure additionally carries the chiral condensate the A^2 amplitude requires (§7.5) is a separate question, not settled here. The abelian row is not a further gap: its classical field strength is identically zero (the gradient connection is pure gauge), but this is immaterial, since in the induced theory every coupling is the fermion vacuum polarization — nonzero for U(1) as for the non-abelian factors — not a classical field strength. The one residual that remains separate is whether the near-identity measure additionally carries the chiral condensate the A^2 amplitude requires (§7.5), which is not settled here. For audit purposes the bridge's status under the [appendix-A §A.3] scheme is recorded explicitly: the kinematic half (measure and map) is derived-conditional (solid modulo the amplitude-scale operational input of [Substratum §4, Theorem 24, *Status*]); the classical near-identity pushforward and the determinant free-energy comparison are numerically supported at one volume and mass (sketch-grade); the condensate transport is a documented gap. The induced measure is therefore a derived-conditional commitment, not a theorem, and any verdict of the A^2 falsification computation is a verdict on the conjunction of the amplitude closure with this bridge; any successor run order accordingly carries plaquette $\langle P\rangle$ and chiral-condensate measurements as first-class observables, so that a wrong-ensemble failure is empirically separable from a wrong-amplitude failure. The 2026-07-14 reconnaissance halt makes this concrete: the $\beta = 0$ induced-measure ensembles proved near-identity-leaning ($\langle P\rangle \approx 0.62$–$0.65$, ordering as m falls) but chirally symmetric (anti-GMOR trend, cutoff-null vector channel) — empirical support for the concentration half of this bridge and an empirical caution on the condensate half. Whether the physical induced measure carries the chiral condensate is therefore the live gating question for both this bridge and the A^2 observable itself.

Note on the dynamical-fermion MC. A separate set of dynamical-fermion HMC simulations is used to extract empirical inputs to the §7.5 mass relations (notably the chiral condensate $\Sigma \approx 0.20$ and the matching-scale verification $m^{\mathrm{MC}}_{\mathrm{match}} = 0.1234 \pm 0.0002$). These simulations use the standard Haar measure on $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ together with the standard staggered pseudofermion Berezin measure, run in the weak-coupling regime $\beta \in \{11.10, 7.40, 3.70\}$. Whether these standard-measure observables agree with the corresponding observables in the framework's induced $(\mathbb{Z}/q\mathbb{Z})$-pushforward measure is open work. At the level of Σ specifically, both measures concentrate near the identity at weak coupling — Haar via $\beta \gg 1$ giving $U \approx I + O(1/\sqrt{\beta})$, and the $(\mathbb{Z}/q\mathbb{Z})$-pushforward via $q \gg 1$ giving the framework's "fraction of nontrivial link values scales as $1/q$" — so the leading-order Σ values plausibly agree, with the empirical agreement of $m_{\text{match}}^{\text{MC}}$ with λg_0^2 at 1.4% consistent with this scenario. The agreement can be established directly, and a check at accessible parameters supports it. Matching the plaquette between the two ensembles fixes the $q \leftrightarrow \beta$ correspondence empirically, and Z_S may then be compared at matched $\langle P \rangle$ — a non-trivial test, since two measures agreeing on one observable need not agree on another. On a 4^4 lattice at $mL = 3$, the induced $(\mathbb{Z}/q\mathbb{Z})$-pushforward and the standard Haar–Wilson ensemble trace the same Z_S-versus-$\langle P \rangle$ curve to within 0.4–0.9% across the full range from strong coupling ($\langle P \rangle \approx 0.14$) to $\langle P \rangle \approx 0.82$, the latter being the plaquette of $\beta = 11.1$ itself: the induced action is Wilson-universal for this observable at the coupling the mass relations use, so the induced measure inherits the standard-measure Z_S. The same matching locates $\beta = 11.1$ at $q \approx 11$, rather than at the exponentially large value a $\beta_{\text{eff}} \sim \ln q$ reading would place it; since q is gauge (§2.7) this fixes where the correspondence sits in the parameterization, not a physical alphabet size. The two families of residual correction separate cleanly under this test. The $O(1/q)$ corrections are the discreteness of the $(\mathbb{Z}/q\mathbb{Z})$-valued field, isolated by comparing the integer-valued field against its continuum idealization: at the §4.2 amplitude the field occupies only some eight letters whatever q may be — q setting the exponential map's $2\pi/q$ scale rather than the field's range, as its gauge status requires — and rounding renormalizes the $q \leftrightarrow \beta$ map by an $O(1)$ factor, inflating $1 - \langle P \rangle$ by some ten to twenty percent, while leaving the Z_S-versus-$\langle P \rangle$ relation itself intact to 0.01–0.14%, the deviation shrinking toward weak coupling. Discreteness is thus absorbed into the correspondence and does not touch the universal relation. The $O(1/\beta)$ corrections — the induced action's departure from the Wilson form — are what remains, and they are the larger of the two by roughly an order of magnitude, at the half-percent level in Z_S at matched $\langle P \rangle$. Three caveats bound all of this: the check is at one volume, one mass, and modest statistics without error bars, so the half-percent figure is an upper bound that may be partly statistical, and universality at the production parameters ($m \approx 0.12$, $L = 32$–64) is inferred rather than demonstrated; and the located q is contingent on the field-amplitude normalization of §4.2.

6.6 Status of the calculation

The chain in §6.3 reproduces all three SM gauge couplings at M_Z through the structural input $1/\alpha_0 = 23.25$ (computed analytically from the one-loop staggered VP, §6.1), the universal threshold $\delta_0 = 10.02$ (empirically fixed by the U(1) row; a 2-loop VP rebuild indicates δ_0 is not reproduced by strict 2-loop PT and is accordingly treated as empirical input), and the C_2-dependent threshold parameters $(A, B) = (8.3, 5.59)$ (matched to the observed SU(2) and SU(3) couplings at M_Z, with the product $A \cdot B = 46.4$ independently computed at ≈ 48 under the $1/N^2$ leading-power assumption, inconclusive across finite-size models; see §6.2.1). The structural predictions in this sector are: (i) the specific value $1/\alpha_0 = 23.25$, (ii) the universality of δ_0 across gauge groups, and (iii) the independent derivation of $A \cdot B$, consistent with the fitted value to $\sim 3\%$ under the $1/N^2$ leading-power assumption but inconclusive across finite-size models (§6.2.1). Separating A from B individually is open.

6.7 No-GUT structural prediction: a three-layer synthesis

§§4.6 and 6.4 individually establish components of a unified structural prediction that the framework makes against grand-unified scenarios. This subsection brings these components into one place and articulates them as a falsifiable forward-prediction set.

Layer A (group-theoretic). The Standard Model gauge group SU(3) $\times$ SU(2) $\times$ U(1) emerges directly from the cubic-commutant decomposition of the 6 nearest-neighbor link directions (§4.6, Theorem 7), with no intermediate large gauge group at any scale. The decomposition $6 = T_1(3) \oplus E(2) \oplus A_1(1)$ is single-step: three eigenspaces of the coupling matrix M, three commutant factors. The framework therefore has *no GUT stage* — no SU(5), SO(10), E_6, or Pati-Salam intermediate gauge structure at any energy. This contrasts with all standard string-phenomenology routes to the SM, each of which traverses some intermediate large gauge group: heterotic chains $E_8 \to E_6 \to \mathrm{SO}(10) \to \mathrm{SU}(5) \to \mathrm{SM}$; F-theory SU(5) $\to$ SM via flux; type IIA Pati-Salam $\mathrm{SU}(4) \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R \to$ SM.

Layer B (coupling-level). The gauge couplings do not unify at any scale. §6.4 establishes this as a structural feature of induced gauge theory: the universal $1/\alpha_0 = 23.25$ at the cutoff is shared across all three gauge factors before gauge self-interactions, but the C_2-dependent threshold $A \ln(1 + B \cdot C_2 g_0^2)$ splits the couplings *at the cutoff itself*, with C_2 different across SU(3), SU(2), U(1). Below M_{Pl}, pure SM β-functions run the (already-split) couplings to M_Z, with no intermediate gauge structure modifying the running. Above M_{Pl}, the induced theory has no tree-level gauge action and the running differs from standard SM. The framework therefore predicts that the three SM gauge couplings remain non-meeting under any RG running consistent with SM matter content.

Layer C (mechanistic). The gauge couplings emerge from the fermion determinant via $S_{\mathrm{eff}}[U] = -N_f \ln \det(D^\dagger D)$, with no fundamental gauge action at the substratum level (§6.1). This induced-gauge-theory structure is structurally distinct from: (a) standard QFT (gauge action is fundamental input, couplings are renormalizable parameters); (b) string-theoretic compactifications (gauge couplings depend on moduli — compactification volumes, dilaton vacuum expectation values, brane positions — with values determined by flux stabilization). In OI, the gauge couplings are forced by $N_f = 6$ (from $K = 2d$, Theorem 6) and $T(R) = 1/2$ (fundamental representation), with no moduli parameters connecting predicted values to other values. This is a stronger statement than "no GUT": it asserts that the framework's gauge couplings exist as quantities through a specific mechanism (fermion determinant) that string compactifications do not share.

Forward-prediction consequences. The three layers jointly predict:

(i) *No proton decay through GUT-mediated channels.* Standard SU(5) GUT proton decay through dimension-6 operators with suppression scale $M_{\text{GUT}} \sim 10^{16}$ GeV gives $\tau_p \sim 10^{34}$–10^{36} years. In OI's framework, the only available cutoff is $M_{\text{Pl}} \sim 10^{19}$ GeV; dimension-6 baryon-number-violating operators are suppressed by M_{Pl}^2, giving $\tau_p \sim M_{\text{Pl}}^4/m_p^5 \sim 10^{45}$–$10^{46}$ years. This is roughly 10 orders of magnitude beyond Hyper-Kamiokande's projected $\sim 10^{35}$ year sensitivity.

(ii) *No GUT-mechanism magnetic monopoles.* The 't Hooft-Polyakov monopole production at GUT phase transitions does not occur in OI's framework. The cosmological monopole problem (which inflation was historically invoked to dissolve) does not arise. Electroweak-scale monopoles remain possible at $\sim M_W/\alpha_W \sim 10^4$ GeV, with production rates set by the (crossover) electroweak phase transition; OI inherits the standard SM treatment of these.

(iii) *No GUT-chain cosmic strings.* Topological defects from SO(10) $\to$ SM chains with non-trivial π_1 of intermediate vacuum manifolds do not occur in OI. Standard cosmic-string searches non-detection is consistent with OI's prediction.

(iv) *Discriminating prediction: shared origin of the three gauge couplings.* The three SM gauge couplings emerge from a single fermion-determinant computation with $N_f = 6$ and $T(R) = 1/2$; they are not independent moduli. This is the OI-specific structural feature distinguishing the framework from string-with-no-GUT scenarios, in which gauge couplings depend on independent compactification data. Empirically, the framework's matching of α_3, α_2, α_1 at M_Z to the observed values via the §6.3 chain confirms this structural prediction at the few-percent level.

Empirical status. The framework's predictions are jointly tested by: (a) continued non-observation of proton decay at Super-Kamiokande and forthcoming Hyper-Kamiokande (proton lifetime bounds tightening past 10^{35} years would support OI; detection would falsify it); (b) continued non-observation of magnetic monopoles in cosmic-ray and condensed-matter searches (consistent with OI); (c) absence of cosmic string signatures in CMB and gravitational-wave observations (consistent with OI); (d) the precision matching of α_3, α_2, α_1 at M_Z via the universal-induced-coupling chain (already empirically supported at few-percent level).

What this synthesis does not do. The three-layer prediction does not uniquely identify OI among no-GUT frameworks. Any framework with no GUT and Planck-scale-suppressed dimension-6 operators makes prediction (i); any framework without intermediate symmetry breaking makes (ii) and (iii). What is OI-specific is prediction (iv): the shared-origin structural relation among the three gauge couplings. The full no-GUT prediction set is therefore: confirmation of (i)-(iii) supports OI but is also consistent with other no-GUT frameworks; (iv) plus the universal $1/\alpha_0 = 23.25$ value is the OI-distinguishing content. Refutation of any of (i)-(iv) would be evidence against OI; (i)-(iii) would also be evidence against the broader no-GUT class.

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7. Quantitative Predictions

The derivation chain of §§4–6 fixes the structural content of the Standard Model — gauge group, matter content, discrete symmetries, and gauge couplings. This section presents the quantitative predictions for the remaining SM observables: the CKM matrix elements, the charged-lepton mass relations, the fermion mass chain (six masses from one input), the PMNS angles, the Higgs mass, and the bottom-to-tau mass ratio. All predictions follow from the cubic-lattice geometry. The structural-vs-bijection-level distinction (developed further in §8.3) applies throughout: the framework predicts the *angular* and *ratio* structure of fermion mass relations from cubic-group representation theory, but the overall mass scale of each chain is set by the specific bijection φ and is taken as one empirical input per chain. The predictions in this section therefore have the form "structural relations among observables, with a small number of empirical inputs setting the overall scales." Subsection 7.6 collects the full prediction table.

7.1 The CKM matrix

The Cabibbo angle.

The three generations occupy T_1 BZ corners: $X_j = \pi\, e_j$. The distance between adjacent corners is $|X_i - X_j| = \pi\sqrt{2}$, a geometric constant of the cubic lattice. The Cabibbo angle equals the inverse of this distance:

$$\boxed{\lambda = \frac{1}{\pi\sqrt{2}} = 0.22508}$$

Observed [20]: $\lambda = 0.22500 \pm 0.00067$. Match: 0.12σ (0.04%).

The mixing matrix element between generations i and j is the continuum fermion propagator at the inter-generation momentum: $|M_{ij}| = |S(X_j - X_i)| = 1/|X_j - X_i| = 1/(\pi\sqrt{2})$. The $1/|q|$ form (not $1/|q|^2$) arises because the taste-changing transition preserves chirality.

Chirality preservation of the taste-changing vertex. This is established in the staggered fermion literature. Mason et al. [32] show explicitly that taste-changing transitions in staggered quarks have spinor structure given by a combination of γ_μ and $\gamma_{5\mu} = \gamma_5\gamma_\mu$ (their Section 2.1: "Each quark line has an odd number of gamma-matrices and therefore the spinor structure must be a combination of γ_μ, $\gamma_{5\mu}$"). Both γ_μ (vector current) and $\gamma_5\gamma_\mu$ (axial current) commute with the projection operators $P_{L,R} = (1 \mp \gamma_5)/2$ in the standard sense — they map $L \to L$ and $R \to R$ rather than $L \leftrightarrow R$ — so any combination of them is chirality-preserving. This is also forced by the unbroken $U(1)_\epsilon$ chiral symmetry of the staggered formulation: a chirality-flipping inter-generation vertex would require an explicit mass insertion violating $U(1)_\epsilon$, which is forbidden in the massless limit.

In the massless limit, the chirality-preserving propagator structure $\gamma \cdot q/q^2$ contracted with the chirality-preserving vertex gives $\mathrm{Tr}[\gamma \cdot S(q)] \propto 1/|q|$. Hence $|M_{ij}| \propto 1/|X_j - X_i|$ rather than $1/|X_j - X_i|^2$. The $1/|q|^2$ form (chirality-flipping with mass insertion) gives $\lambda^2 = m_d/m_s$ — the GST relation [21] follows as a corollary (self-energy requires two propagator insertions, so the down-strange mass ratio inherits the squared scaling). The observer's continuum theory has no BZ periodicity, so the propagator at $|q| = \pi\sqrt{2}$ is smooth.

Status of the identification (named hypothesis). The chain above contains three links of different strength, and it is important to grade them separately. The corner geometry ($|X_i - X_j| = \pi\sqrt{2}$ for adjacent generations) and the chirality-forced $1/|q|$ power (established via the staggered-fermion literature) are solid. The load-bearing link is the identification itself — that the dimensionless mixing-matrix element *equals* the taste-changing propagator magnitude at the inter-corner momentum, in lattice units, with prefactor exactly unity. This is a hypothesis, not a derivation: a mixing angle is a property of the relative misalignment of two mass matrices, and no derivation from the substrate Yukawa diagonalization to "mixing element = free propagator magnitude, coefficient one" is currently given; the unit prefactor (no coupling, wavefunction, or normalization factor) and the selection of the adjacent-corner distance (rather than another invariant of the same corner set) are commitments of the hypothesis rather than consequences of the geometry. The resolving computation is the taste-changing two-point function between T_1 corner states with explicit external-leg normalization, evaluated on the $d = 3$ staggered lattice — the same machinery class as the $1/\alpha_0$ integral of §6.1 and the $\mathcal{N}$ integral of §7.2 — whose output is the dimensionless coefficient c_λ in $\lambda = c_\lambda/(\pi\sqrt{2})$. The hypothesis commits to $c_\lambda = 1$; the empirical match at 0.04% confirms the commitment but does not derive it, and a computed c_λ away from unity by more than the experimental precision falsifies the identification. Taken by itself, this link would rest at "geometry and power forced, identification hypothesized and empirically confirmed." The *Structural narrowing* below discharges the load-bearing part of the resolving computation: the unit-prefactor condition is shown to be an algebraic identity of the staggered spin-taste map, so the structural level of the identification is closed and the only residual is the emergent-theory dressing. The prediction's status is therefore "geometry and power forced, unit prefactor a structural identity, emergent dressing pending" — no longer merely hypothesized at the structural level, and a complementary open link to (rather than the same footing as) the Koide amplitude's dynamical-closure caveat (§7.2).

Structural narrowing of the hypothesis. Three exact results reduce what the resolving computation must decide. First, the free-level taste-changing amplitude vanishes identically: the free staggered spectrum is exactly eight-fold taste-degenerate at every momentum and taste-diagonal in its eigenbasis, so there is no free-level object for the identification to refer to — c_λ is an interaction-order quantity. Second, the generation states are taste *eigenstates* — superpositions across Brillouin-zone corners — rather than single-corner modes; the $\pi\sqrt{2}$ geometry survives at the operator level (the chirality-preserving taste-changing operators are exactly the two-flip class), but the external legs of the resolving computation must be taste eigenprojections. Third, the $1/|q|$ power arises as the energy denominator of second-order transition perturbation theory, and the candidate dispersions at the transfer momentum discriminate sharply: the substrate boson kernel gives $\lambda = 1/(2\sqrt{2}) = 0.3536$, excluded by the measured Cabibbo angle at $\approx 190\sigma$; the substrate fermion kernel is singular there (the transfer momentum is itself a Brillouin-zone corner); the emergent continuum dispersion gives $\lambda = 1/(\pi\sqrt{2})$ exactly. The hypothesis $c_\lambda = 1$ is therefore equivalent to the statement that the intermediate states of the mixing perturbation theory are the emergent continuum fermions — an emergent-EFT statement by necessity rather than by choice — and the remaining content of the resolving computation is the one-loop coefficient: whether the cubic-group Yukawa vertex and the taste-eigenprojected, residue-normalized external legs multiply to exactly unity. A scoping computation of the branch-restricted transfer operator found it to be $(1/\sqrt{3})$ times a unitary — all four singular values equal $1/\sqrt{3}$, the $(1 - A^2, A^2)$ projection split appearing in the transfer's branch structure — sharpening the gate question to whether the on-shell residue normalization absorbs this factor exactly. The full on-shell computation answers it: between physical legs — the $(1 + \gamma_0)/2$-projected spin$\otimes$taste states of the staggered-to-Dirac reconstruction, at the massless point the free fermion sector occupies — every spin-resolved taste-changing element of the site-local Yukawa transfer has unit modulus exactly; the vertex acts as a taste permutation with unit weight in each physical channel, and the $1/\sqrt{3}$ of the scoping pass is identified as a branch/spin conflation that the physical projection removes. The gate as defined — vertex and residue-normalized legs multiplying to exactly unity — is therefore answered affirmatively at its structural level, and the answer is algebraic rather than numerical: under the spin-taste reconstruction the two-flip transfer maps to a unitary monomial ($\gamma_0\gamma_1\gamma_3$ in the spin slot for the axes-$(1, 2)$ transfer, times a taste monomial) — a generalized permutation whose nonzero matrix elements have unit modulus identically — and this monomial commutes with γ_0, so the physical energy projection preserves the structure. The unity condition is thus an algebraic identity of the staggered spin-taste map, with the finite computation serving as its witness. What remains beyond this structural level is the emergent-theory dressing.

The Wolfenstein A parameter.

The A_1 (Higgs) taste sits along the democratic direction $\hat{h} = (1,1,1)/\sqrt{3}$. Each generation axis e_j makes an angle θ with $\hat{h}$ where $\cos\theta = 1/\sqrt{3}$. The CKM mixing is driven by the perpendicular component:

$$\boxed{A = \sin\theta = \sqrt{2/3} = 0.8165}$$

Observed [20]: $A = 0.826 \pm 0.012$. Match: 0.8σ (1.2%).

Combined: $|V_{cb}| = A\lambda^2 = \sqrt{2/3}/(2\pi^2) = 0.04136$ (observed: 0.0408 ± 0.0014, 0.4σ).

Structural identity. $A^2 = (d-1)/d = C_2(T_1)/d$ is a representation-theoretic identity for vector representations, not a numerical coincidence. The off-democratic projection-squared $1 - 1/d$ (from $\hat{e}_i \cdot \hat{h}$ geometry) and the cubic-group quadratic Casimir $C_2 = d - 1$ for the vector representation are the same quantity. For $d = 3$ this gives $A^2 = 2/3 = C_2/d$ directly. This identity underlies the structural status of the natural cubic-lattice perturbation amplitude $\sqrt{C_2}/d = \sqrt{2}/3$ and the consequent Layer 1 prediction of $\sin^2\theta_{13}$ (§7.3).

Where A is protected and where it is not. The geometric value $A = \sqrt{2/3}$ is exact — it is the sine of a fixed lattice angle, independent of any bijection or dynamics. Its two downstream uses, however, carry different correction status, and the distinction determines which comparisons test the geometry alone. In the reactor angle, A enters *protected*: $\sin\theta_{13} = A^2\lambda$ is exact at $O(\lambda^2)$ because the second-order correction Δ_{13} is independent of the A_2 generator (§7.3), so that prediction tests the geometric A with no Layer-2 contamination. In $|V_{cb}| = A\lambda^2$, by contrast, A enters *unprotected*: the $(2,3)$ element's leading contribution arises at $O(\lambda^2)$ through the second-order coefficient b_{23}, so identifying the measured Wolfenstein A with the geometric value is a leading-order statement whose correction is the same open Layer 2(a)-tractable object as the Condition-2 ratio (§7.3, §8.3) — sharpening the $|V_{cb}|$-side prediction beyond $\sqrt{2/3}$ is that derivation, not a separate task. The observed deviation, $(0.826 - 0.8165)/0.8165 \approx 1.2\%$, is smaller than the natural correction scale $\lambda^2 \approx 5\%$, consistent with a suppressed b_{23} correction; the suppression is observed, not derived. A global-fit A drifting outside $\sqrt{2/3}\,(1 \pm \lambda^2)$ would strain the leading-order identification, while the protected $\sin\theta_{13}$ comparison would remain a clean test of the geometry regardless.

The down-to-strange mass ratio.

The Gatto–Sartori–Tonin relation [21] $\sin\theta_C \approx \sqrt{m_d/m_s}$ gives:

The GST relation [21] is an external empirical relation in flavor physics with known higher-order corrections. The framework's prediction is structural in λ (the cubic-lattice Cabibbo derivation above); the conversion to m_d/m_s inherits GST's empirical status. To this approximation:

$$\boxed{\frac{m_d}{m_s} = \lambda^2 = \frac{1}{2\pi^2} = 0.05066}$$

Observed [20]: $m_d/m_s = 0.0503 \pm 0.0007$ (FLAG 2024 Nf=2+1+1). Match: 0.56σ.

The Jarlskog invariant.

The Jarlskog invariant $J = \mathrm{Im}(V_{us}V_{cb}V_{ub}^*V_{cs}^*)$ measures CP violation in the quark sector. In the Wolfenstein parametrization, $J \approx A^2\lambda^6\eta$. Using $\lambda = 1/(\pi\sqrt{2})$ and $A = \sqrt{2/3}$:

$$J = \frac{\eta}{12\pi^6}$$

where $\eta = 0.348$ [20] is the solution-specific CP-violating parameter. This gives $J = 3.02\times10^{-5}$ (observed: $(3.08\pm0.13)\times10^{-5}$, 0.5σ). The structural suppression factor $12\pi^6 \approx 11{,}537$ is purely geometric.

CP-violating parameters.

The Wolfenstein parameters ρ and η require complex CKM entries. On the OI lattice, complex phases arise from the specific bijection φ and are likely solution-specific rather than structural.

7.2 Charged lepton masses and the fermion mass chain

Structural and bijection-level content. The three lepton generations come from the three triplet (T_1) staggered tastes, related at the structural level by the cubic symmetry O. The framework predicts the *angular position* $\theta_0 = C_2/d^2 = 2/9$ in the $\mathbb{Z}_3$-symmetric mass parametrization (a property of the cubic-group representation theory of T_1), but not the overall splitting magnitude (set by the bijection-level φ, taken as one empirical mass input per chain). Given the structural angular prediction and one mass input, the Koide relation determines the remaining masses; the same structural-vs-bijection-level distinction applies to the down-quark chain (below) and the cross-sector relations (below).

The Koide parameter.

The Koide relation [22] $Q = (m_e+m_\mu+m_\tau)/(\sqrt{m_e}+\sqrt{m_\mu}+\sqrt{m_\tau})^2 = 2/3$ holds to 0.001%. On the OI lattice the masses take the form $\sqrt{m_k} = \mu(1 + A\cos(\theta_0 + 2k\pi/3))$. The $\mathbb{Z}_3$ symmetry of the T_1 representation — cyclic permutation of the three BZ corners — fixes the *phases* to equal angular spacing $2k\pi/3$; it does not by itself fix the amplitude A, and the Koide value $Q = \frac{1}{3} + \frac{A^2}{6} = 2/3$ is the specific statement $A^2 = 2$. The amplitude shares the angle's structural origin: $A^2 = C_2(T_1) = l(l+1) = 2$, the quadratic Casimir of the generation representation, entering at the eigenvalue level as the amplitude-squared while the same Casimir fixes the phase $\theta_0 = C_2/d^2$ at the angle level (next paragraph) — the amplitude-versus-eigenvalue correspondence used for m_u/m_d below. Both Koide degrees of freedom thus descend from the single invariant $C_2(T_1) = 2$. The status of the amplitude is that of the angle's normalization (below), not stronger: the Casimir form is structural and $A^2 = 2$ matches observation exactly, but it is not yet derived from the dynamics — one-loop perturbative taste-breaking gives $A^2 \approx 0.34$, undershooting $C_2 = 2$, so establishing that the substrate dynamics produces $A^2 = C_2$ non-perturbatively is an open computation ([Ch. 19 §19.3.1]); its observable class is constrained in advance by an exact shift-symmetry identity — the ensemble single-quark two-point function is exactly eight-fold corner-degenerate at every coupling (driver `oi_lattice_code/fermion/shift_identity_check.py`), so quark-level taste splitting vanishes identically and any dynamical measurement is necessarily composite (multi-fermion channels). The relation is therefore "structural form forced, amplitude argued from the Casimir and empirically confirmed" — the same footing as θ_0, weaker than a full derivation and stronger than a fit.

The Koide angle.

The T_1 representation of SO(3) has angular momentum $l = 1$ and quadratic Casimir $C_2 = l(l+1) = 2$. The mass splitting within the T_1 triplet is an O-invariant quadratic form on the generation space. The unique such invariant is the quadratic Casimir C_2, which measures the strength of the anisotropy at each BZ corner. Normalized by d^2 (the lattice bandwidth in generation space):

$$\boxed{\theta_0 = \frac{C_2}{d^2} = \frac{2}{9} = 0.22222}$$

Observed: $\theta_0 = 0.22227$. Match: 0.02%. For the A_1 (Higgs) taste, $l = 0$ gives $\theta_0 = 0$ — no generational splitting, consistent with a single Higgs field.

Uniqueness of the normalization. The Koide angle is dimensionless, so it must be a pure-number ratio of structural constants of the $(d = 3, T_1)$ data: angular momentum l, Casimir $C_2 = l(l+1)$, dimension $\dim(T_1) = 2l+1$, lattice dimension d, and the dimensions of related representations and tensor products. Enumerating the natural ratios with the right order of magnitude:

Ratio	Value	Match to $\theta_0 = 0.22227$
$C_2/d^2 = 2/9$	0.22222	**0.02%** ✓
$\dim(E)/\dim(T_1 \otimes T_1) = 2/9$	0.22222	**0.02%** ✓
$C_2/(d(d+1)) = 2/12$	0.16667	25% off
$l/d^2 = 1/9$	0.11111	50% off
$\dim(T_2)/\dim(T_1 \otimes T_1) = 3/9$	0.33333	50% off
$(2l+1)/(d(d+1)) = 3/12$	0.25000	12% off
$1/d^2 = 1/9$	0.11111	50% off

Ratio	Value	Match to $\theta_0 = 0.22227$
$l/(2d) = 1/6$	0.16667	25% off
$C_2/(2d) = 2/6$	0.33333	50% off
$\dim(E)/\dim(\mathrm{Sym}^2(T_1)) = 2/6$	0.33333	50% off

Only 2/9 matches. The Casimir construction $C_2(T_1)/d^2$ is the physically correct derivation: $C_2(T_1) = 2$ measures the anisotropy strength of the T_1 generation representation (it is the unique $O(3)$-invariant quadratic scalar on the generation space), and $d^2 = 9$ is the natural geometric normalization from the cubic lattice bandwidth squared. The numerical coincidence with $\dim(E)/\dim(T_1 \otimes T_1) = 2/9$ arises because $\dim(E) = C_2(T_1) = 2$ for O acting on its vector representation — a genuine structural feature of this group. However, the operator space for the Hermitian mass-splitting matrix δM is $\mathrm{Sym}^2(T_1)$ of dimension 6 (not $T_1 \otimes T_1$ of dimension 9, which includes antisymmetric operators irrelevant to symmetric mass splittings), and $\dim(E)/\dim(\mathrm{Sym}^2(T_1)) = 1/3$, not 2/9. The dim-ratio interpretation using the full $T_1 \otimes T_1$ space is therefore an arithmetic artifact; the Casimir construction is the structural derivation, with the factor $d^2 = 9$ understood as the geometric bandwidth rather than an operator-space dimension.

Among dimensionally consistent $(d = 3, T_1)$ structural ratios enumerated above, $\theta_0 = C_2/d^2 = 2/9$ is the unique match to observation, and it has a clean physical interpretation: the anisotropy strength of the generation representation, normalized by the lattice bandwidth.

Connection to the §7.1 structural identity. The natural cubic-lattice perturbation amplitude on T_1 is $\sqrt{C_2}/d = \sqrt{2}/3$ (Casimir scaling over bandwidth). Second-order perturbation theory on the eigenvalues then gives $\theta_0 \sim (\sqrt{C_2}/d)^2 = C_2/d^2 = A^2/d$, where $A^2 = (d-1)/d = C_2/d$ is the structural identity of §7.1 for vector representations. The bandwidth-squared normalization $(1/d^2)$ reflects this PT scaling: amplitudes scale as $\sqrt{C_2}/d$, eigenvalue shifts as the square of the amplitude.

A first-principles derivation of the specific $1/d$ amplitude scaling (vs. alternatives like $1/(2d)$) from the cubic-lattice Yukawa structure remains an open task. It is important to be precise about what is forced and what is open here, because the distinction bears directly on whether the cluster of predictions sharing this normalization (Wolfenstein A, the Koide angle θ_0, $\sin^2\theta_{13}$, and m_u/m_d) is structurally derived or empirically fitted. Three things are forced and one is open. *Forced:* (a) that the relevant invariant is the quadratic Casimir $C_2(T_1) = 2$ — this is representation theory of the cubic group on its vector representation, not a choice; (b) that the eigenvalue shift scales as the square of the amplitude — this is the standard order-counting of second-order perturbation theory, not a choice; (c) that the bandwidth normalization is a power of d — this is dimensional, fixed by the lattice geometry. *Open:* whether the amplitude normalization is $1/d$ (giving $C_2/d^2 = 2/9$) or an alternative such as $1/(2d)$ (giving $C_2/(4d^2) = 1/18$). The open element is therefore a single discrete choice among a short enumerable list of geometrically admissible normalizations, not a continuous fit parameter. The framework commits to $1/d$ on the structural ground that the amplitude is the Casimir scaled by the single-axis bandwidth d (not $2d$), and this commitment is falsifiable: the $1/(2d)$ alternative predicts $\theta_0 = 1/18 = 0.0556$, which misses the observed 0.2227 by a factor of four and is excluded by data. The empirical match at 0.02% does not *select* the normalization from a fit; it *confirms* a structural commitment that was constrained, before any fit, to a discrete choice the framework resolves on geometric grounds. The geometric ground for $1/d$ over $1/(2d)$ is argued rather than proved from the Yukawa structure; closing this is the Layer 2(a)-tractable computation referenced in §7.3. One structural component of that closure can be stated now. The per-axis weight that controls the taste-off-diagonal normalization — the Brillouin-zone average $w = \langle \sin^2 q_1 / \sum_k \sin^2 q_k \rangle$ that appears in the second-order self-energy — evaluates to exactly $1/d$, and does so as a consequence of cubic symmetry alone: the d axis-fractions are equal by the cubic point group and sum to unity by construction, so each equals $1/d$ independently of the propagator's detailed form (verified across the massless staggered weight, the propagator-squared self-energy weight, and the Klein-Gordon $\sin^2(q/2)$ form, all giving $1/d$ to numerical precision). This forces the *per-axis* $1/d$ scaling structurally rather than leaving it to geometric analogy, and it is the mechanism behind why $1/d$ and not $1/(2d)$: the loop's axis-resolved weight is the symmetric fraction $1/d$, with no factor of two available from the axis decomposition. The overall coefficient relating this per-axis weight to the full perturbation-theory amplitude — the $m \to 0$ limit of $\mathcal{N}$ below and its prefactor, which the symmetry argument does not fix — has now been computed (driver `oi_lattice_code/fermion/n_ratio_pt.py`, anchored end-to-end: the 8×8 corner operator is derived numerically from the position-space staggered operator rather than assumed, the derived corner matrices satisfy the Clifford algebra to machine precision, and the sum-rule identity $d\,w + m^2\langle 1/D \rangle = 1$ holds to 10^{-16}). The computation resolves a genuine underdetermination in the specification en route: the matrix-element readings of the taste-off-diagonal vertex vanish identically — the free staggered propagator has no two-flip corner entries, the same structural zero as the free-level taste-changing amplitude of §7.1 — leaving the substrate-eigenvalue reading of the following subsection as the unique non-vanishing one-loop construction; under it the per-axis object is $\hat{V}_a(\hat{D}^\dagger \hat{D})^{-1}\hat{V}_a = (\sin^2 k_a / D)\,\mathbf{1}$ exactly, Schur-diagonal, with prefactor exactly one. The residual is thereby discharged: the cluster's status is "structural form forced, per-axis normalization symmetry-forced, amplitude prefactor computed ($= 1$)" — the magnitude normalization is derived, conditional only on the substrate criterion of the following subsection. This closure promotes the *magnitude* normalization only; the bijection-specific directional content of the flavor structure remains Layer 2(b) by construction (it transforms under substratum state-relabeling and is not fixed by any substrate-only computation), so closure sharpens the magnitude rather than promoting the cluster as a whole.

Specification of the resolving computation. So that the open task is precise rather than gestured, the calculation that would settle $1/d$ vs $1/(2d)$ is the following, statable as a self-contained lattice perturbation-theory problem independent of the rest of the framework. Take the staggered fermion action on the $d = 3$ simple-cubic lattice with the OI taste structure (the T_1 vector representation on the three Cartesian link directions), and introduce the flavor-breaking perturbation as the second-order taste-mixing vertex induced by the cubic-group-symmetric Yukawa coupling. The quantity to compute is the ratio of the off-diagonal second-order self-energy on the T_1 representation to the perturbation amplitude squared:

$$\mathcal{N} \;=\; \lim_{m\to 0} \; \frac{\Sigma^{(2)}_{T_1,\text{off-diag}}(0)}{\left(g\,\sqrt{C_2}\right)^2},$$

where $\Sigma^{(2)}$ is the one-loop (second-order in the Yukawa) taste-off-diagonal self-energy evaluated at zero external momentum, computed as the lattice momentum integral over the staggered propagator $D(q) = \sum_\nu \sin^2 q_\nu + m^2$ with the cubic-symmetric taste vertices on the T_1 corners. The prediction is that this ratio evaluates to $\mathcal{N} = 1/d^2 = 1/9$ (the framework's $1/d$ amplitude normalization), as against $1/(2d)^2 = 1/36$ (the $1/(2d)$ alternative). (The following subsection refines this comparison: the decisive alternative is the *time-direction* value $\mathcal{N} = 1/16$ — i.e. whether the taste loop carries a time direction — not $1/36$; the $1/(2d)$ framing here is superseded by that analysis.) The two candidates differ by a factor of four and are not close, so the computation is decisive at modest numerical precision: a lattice evaluation of $\Sigma^{(2)}$ converging to within, say, 20% of either candidate settles the normalization. This is the same class of computation as the $1/\alpha_0 = 23.25$ staggered vacuum-polarization integral of §6.1 (which converges on a robust plateau), differing only in the vertex structure (taste-off-diagonal rather than gauge-current), so it is tractable with the same lattice-PT machinery already validated in §6.1. The framework predicts $\mathcal{N} = 1/9$; a specialist running this integral either confirms it — promoting the *magnitude* normalization $C_2/d^2 = 2/9$ from argued to derived — or refutes it (falsifying the $1/d$ normalization and with it the $\theta_0 = 2/9$ magnitude). This computation has now been run to closure (same driver as above): under the construction the specification's own refinement selects, $\mathcal{N} = w^2 = 1/d^2 = 1/9$ exactly at $m \to 0$, with the finite-mass plateau approaching from below as the sum-rule identity requires ($w^2 = 0.10906$ at $m = 0.1$, 0.11056 at $m = 0.05$) — inside the 20% band called decisive above, and exact rather than numerical by the identity. The computation requires no input not already fixed by the lattice geometry, which is what makes the magnitude question Layer 2(a)-tractable rather than open-ended. It is important to delimit what such a closure would and would not establish: the quantity $\mathcal{N}$ is a magnitude with a gauge-invariant interpretation, and its closure promotes the magnitude of θ_0 only. It does not, and cannot, fix the bijection-specific directional content of the broader flavor structure (which pair-ratio carries a given coefficient, the generation-label assignment, the CP phase), which remains Layer 2(b): these quantities transform non-trivially under substratum state-relabeling and are therefore not determined by any substrate-only computation, gauge-invariant by construction. The lattice computation specified here is thus a magnitude-closure target, not a route to deriving the full cluster; the directional/label content stays bijection-conditional by design.

Structure of $\mathcal{N}$: the dimensionality of the taste loop is the load-bearing choice. The forced second-order scaling makes $\mathcal{N}$ the square of a per-axis weight $w = \langle \sin^2 q_i / D \rangle$, and the staggered sum rule $\sum_\nu \sin^2 q_\nu = D - m^2$ makes w an exact rational, so $\mathcal{N} = w^2$ is fixed entirely by which directions the loop runs over. The decisive feature is that there is no intermediate value. The time direction carries exactly the same share of the sum rule as each spatial one, $\langle \sin^2 q_0 / D_4 \rangle = 1/4$, so a four-dimensional loop gives $w = 1/4$ and $\mathcal{N} = 1/16$ — and this holds *even when the taste vertex is purely spatial*, because the time direction still claims its quarter of the sum rule regardless of the vertex. Only a loop with no time direction at all gives $w = 1/d = 1/3$ and $\mathcal{N} = 1/d^2 = 1/9$. The two outcomes are far apart in observable terms: $1/9$ gives $\theta_0 = C_2/d^2 = 2/9 = 0.2222$ (observed 0.2227), whereas $1/16$ gives $\theta_0 = 1/8 = 0.125$, off by nearly a factor of two. The genuine open question is therefore not "$1/d$ versus $1/(2d)$" but whether the splitting is computed *with or without a time direction*; and the comparison to the four-dimensional $1/\alpha_0$ integral misleads on exactly this point, since $1/\alpha_0$ is a genuinely four-dimensional running-coupling quantity while θ_0 is not.

The framework's own architecture selects the time-free computation. The generation masses are eigenvalues of the substrate coupling structure — properties of the semisimple part of the deterministic bijection (the μ_k of the mass-generation mechanism) — fixed at the $d = 3$ spatial substrate, before the emergent time direction exists. The Koide angle is the splitting pattern of those substrate eigenvalues, so its second-order shift is an instantaneous $d = 3$ quantity carrying no time loop, giving $w = 1/3$ and $\mathcal{N} = 1/9$. On this reading the eigenvalue-perturbation description above is the correct one; the "$\Sigma^{(2)}$ self-energy" phrasing of the resolving computation must be understood as this substrate eigenvalue calculation and *not* as a four-dimensional emergent self-energy, which would carry the time direction and yield $1/16$. The two phrasings are otherwise in tension precisely on the dimensionality, and it is the substrate-eigenvalue origin of the masses that resolves it. The same criterion, stated once for both sectors: quantities that are *spectral* — eigenvalues of the substrate coupling structure, involving no intermediate-state propagation — are computed at the $d = 3$ substrate (the mass ratios and the Koide angle, as here); quantities that are *transitional* — matrix elements between distinct eigenstates, mediated by intermediate-state propagation — carry the emergent continuum kinematics on the propagating line (the mixing elements, §7.1, where the substrate alternatives are respectively vanishing, excluded, and singular). Spectrum is substrate; transition is emergent.

The open element is correspondingly sharper, and part of it can now be settled directly. The three generations form the irreducible vector representation T_1, so by Schur's lemma any cubic-invariant (taste-blind) emergent dressing acts on them as a multiple of the identity — an affine map on the mass eigenvalues — which rescales the democratic mass component but leaves the splitting phase θ_0 exactly invariant. The generic four-dimensional Higgs-loop mass renormalization is of this taste-blind kind (the tree-level Yukawa is generation-independent, Theorem 10, and the Higgs occupies the singlet taste), so it provably cannot shift θ_0; only cubic-symmetry-*breaking* structure can. That breaking is the substrate coupling-matrix eigenvalue structure (the μ_k of M, Theorem 4) — the same substrate-eigenvalue origin the mass mechanism uses elsewhere — so $\theta_0 = C_2/d^2$ introduces no assumption beyond that mechanism. What Schur's lemma does *not* exclude is a loop-level cubic-symmetry-breaking emergent operator: cubic symmetry alone does not forbid the radiative generation of such an operator (the same operator-mixing phenomenon that governs the radiative-stability question for emergent Lorentz invariance, §4.5), though none arises at leading order.

A second class escapes the Schur step without being cubic-symmetry-breaking: a dressing that is cubic-invariant and acts on the mass matrix as a superoperator rather than as a multiple of the identity on T_1. Cubic invariance forces such a map to be a scalar on each irreducible component of $\mathrm{Sym}^2(T_1) = A_1 \oplus E \oplus T_2$ and permits those scalars to differ; a correction proportional to the tree mass matrix is the special case in which they coincide. For the one-loop staggered taste kernel, $\lambda_{T_2} = 0$ exactly and $\lambda_E/\lambda_{A_1} \approx 0.22$–$0.24$ in three and four dimensions. On the $\sqrt{m}$ pattern such a dressing leaves θ_0 invariant: the angle is a direction within a two-dimensional irreducible component, on which an equivariant map acts as a scalar. On the mass pattern the $\sqrt{\cdot}$ nonlinearity breaks that invariance, with $d\theta_0/\theta_0 = \kappa\,(1-\lambda_E/\lambda_{A_1})\,(\delta m/m)$ and $\kappa \approx -35$ at the physical Koide point, the magnitude reflecting the near-degeneracy $\sqrt{m_e} \ll \sqrt{m_\mu}$. The 0.2% agreement of $\theta_0 = 2/9$ bounds the taste-structured part of an emergent mass-level dressing by $\delta m/m \lesssim 8 \times 10^{-5}$; a second-order lepton-Yukawa loop supplies $\sim 10^{-6}$. The open element is a loop-level cubic-*breaking* operator. The status is thus "$\mathcal{N} = 1/9$ protected by representation theory against all taste-blind emergent dressing and fixed by the substrate eigenvalue structure, with the sole residual a possible loop-level cubic-breaking contribution that is shared with the emergent-Lorentz radiative-stability problem rather than specific to the flavor sector" — closer to "derived" than "argued geometrically."

Mass predictions.

With $Q = 2/3$, $\theta_0 = 2/9$, and $m_\tau = 1776.86$ MeV as input:

Mass	Predicted	Observed	Deviation
m_e	0.51096 MeV	0.51100 MeV	0.007%
m_μ	105.652 MeV	105.658 MeV	0.006%

The quark sector does not satisfy $Q = 2/3$: $Q_{\text{down}} = 0.731$, $Q_{\text{up}} = 0.849$. The down-quark deviation is numerically described by $Q_{\text{down}} \approx (2/3)(1 + \alpha_s(2\,\text{GeV})/\pi)$ to 0.15%, a suggestive regularity that matches the leading-order form of a color correction but for which a first-principles derivation from framework primitives is not currently available. Two observations argue against treating this form as structural: (i) the generation $\mathbb{Z}_3$ symmetry that produces $Q_{\text{lepton}} = 2/3$ acts on the staggered-taste space (BZ corners), while SU(3) color acts on the internal T_1 (K-component) space — a direct breaking of the former by the latter would require a cross-coupling between the two T_1 representations beyond what §4.7.1.2 establishes; and (ii) the 0.15% match holds only in the PDG mixed scheme (light quarks at $\mu = 2$ GeV, b quark at $\mu = m_b$); in physically consistent single-scale schemes $Q_{\text{down}} \approx 0.755$–$0.760$ and the $(1 + \alpha_s/\pi)$ form misses by 4–5%. The observed numerical match is reported below as empirical input to the m_b determination, not as a derived structural relation. A full derivation would require either the link-carrier cross-coupling mentioned above, or a taste-decomposed Coleman–Weinberg analysis yet to be carried out.

The bottom quark mass from the down-sector Koide.

Taking $Q_{\text{down}} = (2/3)(1 + \alpha_s(2\,\text{GeV})/\pi)$ as a phenomenological input (above), the Koide formula determines m_b from m_d and m_s. Using $m_d/m_s = 1/(2\pi^2)$ (§7.1) and $m_s = 93.4$ MeV as the single input:

$$\boxed{m_b = 4144 \text{ MeV} \quad (\text{obs: } 4180 \pm 30,\ 0.9\%)}$$

The up-to-down mass ratio.

The Koide angle $\theta_0 = C_2/d^2 = 2/9$ determines the inter-generation hierarchy. Its square root determines the intra-doublet splitting:

$$\boxed{\frac{m_u}{m_d} = \sqrt{\theta_0} = \sqrt{2/9} = 0.4714}$$

The PDG value [20] is $m_u/m_d = 0.474 \pm 0.056$. Match: 0.05σ. Both arise from the same quadratic Casimir $C_2 = 2$ of the T_1 representation, acting in different channels: the inter-generation hierarchy enters at the *eigenvalue* level (Koide angle parametrizing the three masses in the T_1 representation, where $\theta_0 = C_2/d^2$ enters the eigenvalue formula directly through second-order energy-level structure), while the intra-doublet m_u/m_d ratio enters at the *amplitude* level (mass matrix element between the up and down components within a doublet). The square-root relation reflects the standard QM scaling: amplitudes $\sim V$, eigenvalues $\sim V^2$, so ratios at the amplitude level scale as the square root of ratios at the eigenvalue level.

Status: solution-specific (Layer 2(b)). The same-perturbation hypothesis underlying this derivation — that *one* perturbation V controls both the cubic-group T_1 Casimir splitting and the SU(2)-doublet up-down splitting — requires alignment between the SU(2) Yukawa structure and the cubic-group T_1 perturbation that is not forced by representation theory. Operator counting on the first-generation Yukawas Y_u^{11}, Y_d^{11} shows that each receives contributions from A_1 (singlet) and E (generation-dependent diagonal) channels of $T_1 \otimes T_1$ with four independent Wilson coefficients $(a_u^{(1)}, a_u^{(E)}, a_d^{(1)}, a_d^{(E)})$ in total. The ratio Y_u^{11}/Y_d^{11} depends on these four coefficients; no representation-theory argument forces it to equal $\sqrt{C_2}/d$. Per the §8.3 four-layer classification, this places $m_u/m_d = \sqrt{\theta_0}$ at Layer 2(b): the empirical match at 0.05σ indicates the framework's bijection has the relevant alignment property, but identifying that alignment as forced by symmetry (Layer 2(a)) requires explicit substrate-level Yukawa derivation that has not been performed. A 1-loop OI-vertex computation, while useful, would calculate the alignment for the specific bijection rather than show it is forced.

Mechanism specificity. Within the §8.3 sub-classification of Layer 2(b), m_u/m_d is *single-parameter direction selection*: the up-down asymmetry in OI's structure comes from the Higgs's choice of internal V_2-axis to condense along (which by gauge invariance is equivalent at the Lagrangian level for any direction, but is fixed once chosen by the bijection). The specific value $\sqrt{2/9}$ then follows from how the cubic structure projects this single-axis choice into the up vs down components — a one-parameter family of bijection-dependent values, not a multi-parameter Wilson coefficient ambiguity. The 0.05σ empirical match therefore constrains the bijection's V_2-axis selection more tightly than generic Layer 2(b) classification suggests.

The mass chain: six masses from one input.

The structural relations link all light fermion masses through a single input (m_s):

$$m_s \xrightarrow{\lambda^2} m_d \xrightarrow{\sqrt{\theta_0}} m_u \xrightarrow{Q_{\text{down}}} m_b \xrightarrow{Z_S} m_\tau \xrightarrow{\theta_0} m_e,\, m_\mu$$

Mass	Formula	Predicted	Observed	Match
m_d	$m_s/(2\pi^2)$	4.73 MeV	4.67 ± 0.48	1.3%
m_u	$m_d \times \sqrt{2/9}$	2.20 MeV	2.16 ± 0.49	0.1σ
m_b	Koide(Q_{down})	4144 MeV	4180 ± 30	0.9%
m_τ	$m_b \times Z_S/4.28$	1755 MeV	1776.9	1.2%
m_e	Koide(θ_0, m_τ)	0.505 MeV	0.511	1.2%
m_μ	Koide(θ_0, m_τ)	104.4 MeV	105.7	1.2%

Six masses from one input, all within $\sim$ 1–1.3%. The remaining input m_s sets the overall mass scale.

The top quark mass.

The tree-level Yukawa couplings are taste-independent within each generation (Theorem 10 with standard staggered-lattice Yukawa structure [12]) and are normalized to unity at the compositeness scale M_{Pl} as the lattice-to-continuum matching convention: in the normalized wave equation (coefficient $+1$ on nearest-neighbor terms, §4.1), the staggered fermion-to-scalar overlap integral at a single site evaluates to an O-invariant Clebsch-Gordan coefficient normalized to unity. For y_t, the relation between this convention and the observed top mass is open rather than closed. Under standard one-loop SM running, the IR quasi-fixed point sits at $y_t(m_t) \approx 1.25$ ($m_t \approx 215$–220 GeV), not at unity; the normalization $y_t(M_{\mathrm{Pl}}) = 1$ flows down to $y_t(m_t) \approx 1.19$, i.e. $m_t \approx 207$ GeV — a 20% excess over observation — and the attractor basin $y_{\mathrm{UV}} \in [0.5, 5]$ focuses only to $m_t \approx 177$–221 GeV, a 25% spread. The observed top corresponds to $y_t(M_{\mathrm{Pl}}) \approx 0.4$–$0.5$, not unity. What is empirically exact to 0.9% is the *weak-scale* statement $y_t(m_t) \approx 1$; the compositeness-scale normalization plus IR attraction does not deliver it. A derivation of $y_t = 1$ at the electroweak matching scale specifically — or of $y_t(M_{\mathrm{Pl}}) \approx 1/2$, which runs to ≈ 177 GeV at one loop — would close the gap; neither is currently available. The relation $m_t = v/\sqrt{2}$ is accordingly classified as a retrodiction of the known weak-scale $y_t \approx 1$ coincidence with the structural mechanism open (§7.6), not as a parameter-free structural prediction. Since all other fermion-mass predictions use either the taste-independence equality $y_b = y_\tau$ (m_b/m_τ) or Koide relations referencing observed masses, the tree-level Yukawa convention does not enter any other quantitative prediction:

$$\boxed{m_t = \frac{v}{\sqrt{2}} = 174.1 \text{ GeV} \quad (\text{obs: } 172.5 \pm 0.3,\ 0.9\%)}$$

7.3 PMNS mixing angles

Tribimaximal (TBM) mixing — $\sin^2\theta_{12} = 1/3$, $\sin^2\theta_{23} = 1/2$, $\sin^2\theta_{13} = 0$ — arises from $A_4 \subset O$ [23]. The deviations from TBM are controlled by three structural ingredients: (i) the overall scale $\lambda^2 = 1/(2\pi^2)$, the squared Cabibbo angle; (ii) the sum rule $2\Delta_{12} + \Delta_{23} = 0$ associated with a U-parity grading (μ-τ exchange antisymmetry) of the charged-lepton perturbation and motivated by the projection geometry below; and (iii) the ratio $\Delta_{13}/\Delta_{23} = A^4 = 4/9$ following from the Wolfenstein $A = \sqrt{2/3}$ of §7.1 via a second-order projection onto the Higgs democratic direction. These three ingredients determine the PMNS corrections:

$$\sin^2\theta_{12} = \frac{1}{3} - \frac{1}{4\pi^2} = 0.3080$$

$$\sin^2\theta_{23} = \frac{1}{2} + \frac{1}{2\pi^2} = 0.5507$$

$$\sin^2\theta_{13} = \frac{4}{9}\cdot\frac{1}{2\pi^2} = \frac{4}{18\pi^2} = 0.02252$$

Angle	Predicted	Observed	Match
$\sin^2\theta_{12}$	0.3080	0.3085 ± 0.0073 (post-JUNO global fit [27])	0.07σ
$\sin^2\theta_{23}$	0.5507	$0.561^{+0.012}_{-0.015}$ (NuFIT 6.0, NO)	0.74σ
$\sin^2\theta_{13}$	0.02252	0.02195 ± 0.00056 (NuFIT 6.0, NO)	1.02σ

The three predictions have different structural status, made precise in the derivation below and in §8.3's four-layer classification. The reactor angle prediction $\sin^2\theta_{13}$ depends only on A_1 at second order — the A_2 coefficients cancel from $|U_{e3}|^2$ — and is therefore independent of Condition 2 (the structural relation among A_2 coefficients introduced below). The solar angle and atmospheric angle predictions $\sin^2\theta_{12}$ and $\sin^2\theta_{23}$ depend on Cond 2 jointly: they are equivalent (via the sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$) to the relation $b_{23} = (4/3)(b_{12}+b_{13})$. After the structural decomposition of "Status of Cond 2" below, this relation has an A_1 component closed at Layer 1 (sum rule preserved by A_1 alone via the $\cos^2\theta_{13}$ dressing) and an A_2 component whose substrate ratio $b_{23}/(b_{12} + b_{13}) = 4/3 = 2A^2$ was Layer 2(a)-tractable until its defining substrate calculation was performed (§7.3, "Status of Cond 2") and excluded the second-order route — demoting it to Layer 2(b), with the 7/6 sum rule as its direct experimental test. Within the four-layer structural framework of §8.3, $\sin^2\theta_{13}$ is a Layer 1 unconditional structural prediction, while $\sin^2\theta_{12}$ and $\sin^2\theta_{23}$ are layered predictions: Layer 1 form combined with the Layer 2(b) Cond 2 coefficient.

Experimental confirmation (JUNO). The Jiangmen Underground Neutrino Observatory reported its first measurement of reactor neutrino oscillations in November 2025 [24], achieving the world's most precise determination of $\sin^2\theta_{12}$: 0.3092 ± 0.0087 (a factor of 1.6 improvement over all previous measurements combined). The OI prediction $1/3-1/(4\pi^2) = 0.3080$ matches this measurement at 0.14σ. The updated global fit including JUNO data gives $\sin^2\theta_{12} = 0.3085 \pm 0.0073$ [27], matching the prediction at 0.07σ. As JUNO accumulates data over its 30-year design lifetime, the error bar is projected to reach ± 0.0014, testing the prediction at sub-percent precision.

Structural relations among the prefactors. The reactor angle has the clean primitive form

$$\sin\theta_{13} = A^2\lambda$$

where $A = \sqrt{2/3}$ is the Wolfenstein A of §7.1 and $\lambda = 1/(\pi\sqrt{2})$ is the Cabibbo angle. The A^2 factor arises from the path-summed calculation: for the A_1-determined first-row entries of $V_\ell^\dagger$ $(-\lambda\alpha, +\lambda\alpha$ with $\alpha = \sqrt{C_2}/d = \sqrt{2}/3$, the natural cubic-lattice perturbation amplitude on T_1) combined with the TBM third column $(0, -1/\sqrt{2}, +1/\sqrt{2})$, the two non-trivial paths contribute $\lambda\alpha/\sqrt{2}$ each, summing to $\sin\theta_{13} = \sqrt{2}\alpha\lambda = (C_2/d)\lambda = A^2\lambda$ via the structural identity $A^2 = C_2/d = (d-1)/d$ for vector representations (§7.1). The result $\sin^2\theta_{13} = A^4\lambda^2 = (4/9)\lambda^2$ is Layer 1 structural; the ratio $\Delta_{13}/\Delta_{23} = A^4$ follows immediately with $\Delta_{23} = \lambda^2$ by convention.

Determination of A_1 and second-order structure. The charged-lepton rotation in the AF basis admits the expansion $V_\ell = \exp(\lambda A_1 + \lambda^2 A_2 + O(\lambda^3))$ with A_n antisymmetric. Three framework constraints — vanishing $O(\lambda)$ contributions to both Δ_{12} and Δ_{23} (pushing these deviations to $O(\lambda^2)$), plus the correct $\sin^2\theta_{13} = 4\lambda^2/9$ coefficient — uniquely determine

$$A_1 = \frac{\sqrt{2}}{3}(E_{12} - E_{13}),$$

where $(E_{ij})_{kl} = \delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}$. Geometrically, A_1 is a rotation by angle $A^2 = 2/3$ around the axis $(e_2 + e_3)/\sqrt{2}$, reproducing $\sin\theta_{13} = A^2\lambda$ as above. The structural property is that A_1 is purely *U-odd* under the μ-τ exchange generator of $S_4 \supset A_4$: $UA_1U^{-1} = -A_1$.

Decomposing the most general antisymmetric $A_2 = b_{12}E_{12} + b_{13}E_{13} + b_{23}E_{23}$ and computing $|U_{\text{PMNS},ij}|^2$ to $O(\lambda^2)$ (with $\cos^2\theta_{13}$ correction included), the deviations are:

$$\Delta_{13} = \tfrac{4}{9}\lambda^2 = A^4\lambda^2, \qquad \Delta_{12} = -\tfrac{2}{3}(b_{12} + b_{13})\lambda^2, \qquad \Delta_{23} = b_{23}\lambda^2.$$

Three points: (i) Δ_{13} is independent of A_2 — the structural identity $\sin\theta_{13} = A^2\lambda$ is exact at $O(\lambda^2)$; (ii) $b_{12} - b_{13}$ does not enter any of the three angles, and is identified with δ_{CP}; (iii) under U-parity, $(b_{12} + b_{13})$ is the U-even and b_{23} the dominant U-odd coefficient of A_2, with $(b_{12} - b_{13})$ the sub-dominant U-odd combination.

The absolute normalization is fixed by the cubic-lattice realization of the charged-lepton mass matrix. The natural-normalization choice $b_{23} = 1$ — corresponding to A_2 entering the perturbation series at unit strength relative to λ^2 — combined with the structural relation

$$b_{23} = \tfrac{4}{3}(b_{12} + b_{13})$$

This condition has a direct experiment-facing form: it is equivalent to the exact sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$, since $2\Delta_{12} + \Delta_{23} = \lambda^2\left[b_{23} - \tfrac{4}{3}(b_{12} + b_{13})\right]$ vanishes precisely when the condition holds — the $O(\lambda^2)$ deviations cancel in that combination and in no simpler one. The sum rule is therefore the observable statement of the Condition-2 ratio (companion sum-rule paper, §3): a measured violation of 7/6 beyond $O(\lambda^4)$ falsifies the ratio directly, independently of how the individual angles are attributed.

between the dominant U-odd coefficient and the U-even combination yields

$$b_{12} + b_{13} = \tfrac{3}{4}, \qquad b_{23} = 1,$$

which translates into the angle-level sum rule

$$2\Delta_{12} + \Delta_{23} = 0.$$

Status of Cond 2: structural decomposition. The relation $b_{23} = (4/3)(b_{12} + b_{13})$ admits a decomposition into two structurally distinct components, with closure achieved at different layers.

Equivalence with TBM sum rule preservation. Combining the explicit results above:

$$2\Delta_{12} + \Delta_{23} = \lambda^2 \left[b_{23} - \tfrac{4}{3}(b_{12} + b_{13})\right]$$

so Cond 2 is *equivalent* to the statement that the TBM sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ — which holds at zeroth order from cubic geometry as $2/d + 1/(d-1) = (3d-2)/(d(d-1)) = 7/6$ for $d = 3$ — continues to hold at $O(\lambda^2)$. The Wilson-coefficient relation and the angle-level sum rule encode the same content under different normalizations.

Layer 1 closure of the A_1 component. With $A_2 = 0$, direct calculation gives $2|U_{e2}|^2 + |U_{\mu 3}|^2 = 7/6 - (14/27)\lambda^2 + (\cos^2\theta_{13}$ corrections). Dividing by $\cos^2\theta_{13} = 1 - (4/9)\lambda^2$ — which itself follows from $\sin^2\theta_{13} = A^4\lambda^2$, a Layer 1 quantity — exactly cancels the $-14/27$ offset:

$$(2\sin^2\theta_{12} + \sin^2\theta_{23})\big|_{A_2=0} = \tfrac{7}{6} + O(\lambda^4).$$

The TBM sum rule is therefore *structurally protected* against the A_1 perturbation by cubic geometry alone — no substrate input required for this component. The cancellation depends on A_1's Layer 1-forced form: the magnitude $\sqrt{2}/3$ is fixed by $\sin\theta_{13} = A^2\lambda$, and the U-odd character is fixed by the requirement $\Delta_{12}, \Delta_{23} = O(\lambda^2)$ rather than $O(\lambda)$. With those two Layer 1 constraints, the A_1^2-induced $-14/27$ contribution and the $\cos^2\theta_{13}$-induced $+14/27$ contribution cancel automatically.

Layer 2(a)-tractable residual question for the A_2 component. What remains is the question of whether A_2 preserves the sum rule (which A_1 alone has structurally protected) or breaks it. Cond 2 reduces to: *why does* $b_{23}/(b_{12} + b_{13}) =$

$4/3 = 2A^2 = 4(d-1)/(2d)$*?* This is a single numerical ratio with a structural reading from the squared off-democratic projection for the vector representation, sharply specified at the level of substrate Wilson coefficients. The question admits definite outcomes from substrate calculations targeting this specific ratio (e.g., one-loop staggered-fermion calculation on the cubic lattice testing whether the $C_2(T_1)$ Casimir structure produces the $4(d-1)/(2d)$ coefficient). This is Layer 2(a)-tractable rather than the prior "arbitrary T_2-direction alignment" formulation.

The geometric reading of $4/3$. The relation admits the interpretation

$$\tfrac{4}{3} = 2A^2 = 2\sin^2(\angle(e_i, \hat{h}))$$

decomposing as a *partner-count* $\times$ *per-partner projection*: the U-even combination $E_{12}+E_{13}$ couples generation e_1 to its two partners $\{e_2, e_3\}$, each notionally contributing factor A^2 from the $\hat{h}$-perpendicular projection geometry — total $2 \times A^2 = 4/3$. The dominant U-odd combination E_{23} couples e_2 to its single partner e_3, with no analogous multiplicity, giving relative weight 1. The structural identity $A^2 = (d-1)/d = C_2(T_1)/d$ for vector representations is what makes $4/3$ a representation-theoretic quantity rather than a numerical coincidence. However, the load-bearing claim of the geometric reading is the *operator alignment* — that the substrate's T_2-direction in $\Lambda^2(T_1)$ space is aligned such that the partner-count $\times$ projection picture holds. Operator counting on $T_1 \otimes T_1 = A_1 \oplus E \oplus T_1 \oplus T_2$ shows that the T_2 direction of the substrate's second-order corrections is *not forced by representation theory alone* — the natural Higgs-democratic spurion $\xi_h \propto (1,1,1)/\sqrt{3}$ does not produce the relation; the relation requires a specific cubic-lattice Yukawa structure that has not been independently derived.

The empirical match $2\sin^2\theta_{12} + \sin^2\theta_{23} = 1.178 \pm 0.020$ (using post-JUNO global fit and NuFIT 6.0 atmospheric) tests the bijection's specific operator structure at the few-percent level on the structural coefficient, conditional on the atmospheric octant. Promoting the residual question to Layer 2(a) requires demonstrating that the cubic-lattice substrate Yukawa structure produces the specific $4(d-1)/(2d)$ ratio in the T_2-channel; the closure path is the one-loop staggered calculation referenced above.

Positive support: explicit substrate calculation. An explicit calculation of the natural cubic-symmetric staggered Yukawa structure on T_1 BZ corners shows that both the leading A_h-mediated and subleading T_2-corner-mediated channels produce equal off-diagonal $Y^{(2)}_{ab}$ entries (by cubic symmetry of BZ corner separations and uniform staggered phase prefactors). Equivalently, the natural-form substrate Yukawa correction satisfies $\delta M^{\text{nat}}_{ij} = \delta M_0$ (generation-democratic) for all off-diagonal pairs (i,j).

This produces b^{sub} ratios $b_{12} : b_{13} : b_{23} = m_\tau/m_\mu : 1 : 1$ exactly in the heavy-tau limit. The hierarchy follows from a Layer 1 mass-matrix algebraic identity: at order λ^2 the off-diagonal entries of $M_\ell M_\ell^\dagger$ are

$$m_{ij}^{(2)} = (m_i + m_j)\,\delta M_{ij}$$

(from $M_{ik}^{(0)}\delta M_{jk}^* + \delta M_{ik} M_{jk}^{(0)*}$ with $M^{(0)} = \mathrm{diag}(m_a)$ at LO and symmetric δM). Combined with the natural form $\delta M_{ij} = \delta M_0$ and the b normalization $b_{ij} = m_{ij}^{(2)}/m_{\max(i,j)}^2$, this gives $b_{ij} \propto (m_i + m_j)/m_{\max(i,j)}^2 \approx 1/m_{\max(i,j)}$ — the linear inverse-mass scaling that produces the $m_\tau/m_\mu \approx 17$ hierarchy in b_{12}/b_{13}. The factor 17 is therefore not a numerical happenstance but a derived consequence of (i) the natural form's generation-democratic substrate Yukawa and (ii) the linear $(m_i + m_j)$ algebraic factor in the second-order off-diagonal mass-squared.

The framework's bijection therefore has structure beyond the natural cubic-symmetric form. Cond 2's symmetric U-even split $b_{12}^{\mathrm{sub}} = b_{13}^{\mathrm{sub}}$ requires the substrate amplitudes to obey a different scaling:

$$\delta M_{ij}^{\mathrm{sub}} \propto m_{\max(i,j)}$$

— linear in the heavier external mass (rather than democratic). This is the substrate-level structural pattern Cond 2 imposes: not "alignment" in some abstract operator space, but a definite scaling law for the off-diagonal Yukawa correction. The required substrate matrix-element hierarchy (from Cond 2 plus the symmetric U-even split $b_{12}^{\mathrm{sub}} = b_{13}^{\mathrm{sub}} = 3/8$) is

$$m_{12}^{(2)} : m_{13}^{(2)} : m_{23}^{(2)} = \tfrac{3}{8} m_\mu^2 : \tfrac{3}{8} m_\tau^2 : \tfrac{10}{9}(m_\tau^2 - m_\mu^2)$$

— sharply specified, quantitative, and falsifiable. Any first-principles substrate calculation of $m_{ij}^{(2)}$ produces a definite outcome on whether this hierarchy emerges.

Status under emergent quantum mechanics. In standard effective field theory, where quantum mechanics is taken as fundamental, the U-even and dominant U-odd second-order coefficients $b_{12} + b_{13}$ and b_{23} are independent Wilson coefficients of distinct cubic-invariant operators. No EFT-level symmetry relates them. In the OI framework, where quantum mechanics is itself emergent — a tool for describing the marginal statistics of an embedded observer over a deterministic substrate (companion paper [Main]) — EFT Wilson coefficients are not independent inputs but derived quantities arising from substrate-level mode counting, and constraints among them are *expected* whenever multiple EFT operators share a common substrate-level origin. This places OI in the broader research program of emergent QM developed by 't Hooft [30] (cellular automaton interpretation), Adler [31] (trace dynamics from matrix models), and Barandes [28] (stochastic-quantum correspondence). The relation $b_{23} = (4/3)(b_{12} + b_{13})$ is consistent with such a substrate-level origin; the empirical match indicates this is realized in the framework's bijection. The current Layer 2(a)-tractable status — sharply specified as the $4(d-1)/(2d)$ ratio with structural numerology — reflects that the substrate-level mechanism producing this specific coefficient has not yet been derived, while the question of whether the cubic-lattice substrate Yukawa produces it is concretely tractable rather than open at the level of arbitrary operator alignment.

Sharpening the open task: the cross-term contribution. The substrate-level question admits a concrete formulation. In the AF basis where the charged-lepton mass matrix $M_\ell M_\ell^\dagger$ is diagonal at LO, second-order perturbative diagonalization gives the A_2 coefficients in terms of the substrate's direct second-order off-diagonal corrections $m_{ij}^{(2)}$ *plus* cross-terms involving products of first-order amplitudes. Direct computation in the framework's $A_1 = (\sqrt{2}/3)(E_{12} - E_{13})$ ansatz yields, in the $m_e^2 \to 0$ limit:

$$b_{12} = \frac{m_{12}^{(2)}}{m_\mu^2}, \quad b_{13} = \frac{m_{13}^{(2)}}{m_\tau^2}, \quad b_{23} = -\frac{1}{9} - \frac{m_{23}^{(2)}}{m_\tau^2 - m_\mu^2}.$$

The $-1/9$ contribution to b_{23} is *hierarchy-independent* and is forced by A_1's cubic-group structure ($a_{12} \cdot a_{13} = -2/9$ from the explicit A_1 form). It arises with no substrate input at second order. The relation Cond.~2 at the observable level, $b_{23} = (4/3)(b_{12} + b_{13})$, therefore translates to the following constraint on the substrate's direct second-order corrections:

$$b_{23}^{\text{sub}} = \tfrac{4}{3}(b_{12}^{\text{sub}} + b_{13}^{\text{sub}}) + \tfrac{1}{9}.$$

The $+1/9$ augmentation compensates for the $-1/9$ cross-term contribution from A_1. The Layer 2(a) closure question reduces to: *does the cubic-lattice Yukawa structure produce direct second-order corrections $m_{ij}^{(2)}$ that satisfy this augmented relation?* With $b_{23}^{\text{sub}} = 10/9$ and $b_{12}^{\text{sub}} + b_{13}^{\text{sub}} = 3/4$ in the framework's natural normalization, the required substrate matrix-element hierarchy is $m_{12}^{(2)} : m_{13}^{(2)} : m_{23}^{(2)} = (3/8)m_\mu^2 : (3/8)m_\tau^2 : (10/9)(m_\tau^2 - m_\mu^2)$ (assuming symmetric U-even split).

Progress on the sharpened task (scoping-level, inputs declared). Three results from the resolving computation's scoping phases bear on this formulation. First, the symmetric U-even split assumed above now has structural support: the geometric second-order taste channel (two single-flip hops through the heavy non-T_1 corners) is exactly democratic — equal coefficients for all three pairs — as a consequence of cubic symmetry, computed over the full range of heavy-corner gap assignments. Second, the dressing rule is forced rather than chosen: reproducing A_1's mass-independent angles requires the mass-matrix off-diagonals to carry the column (heavier) mass, $M_{ij} = T_{ij}\, m_j$ — the taste transition living in the left sector — under which the heavier-mass-squared scaling of the required $m_{ij}^{(2)}$ hierarchy is automatic. Third, with these two results the augmented relation reduces to a single condition on the substrate's heavy-corner spectrum: the democratic two-hop amplitude must equal $-1/15$ (in the $m_\mu/m_\tau \to 0$ limit), equivalently $|1/\Delta_S - 1/\Delta_{T_2}| = 1/(30\rho)$, where Δ_S, Δ_{T_2} are the denominators of the singlet- and T_2-class exchange channels — the wave-sector corner frequencies of Theorem 8b, since the taste-changing messenger at a corner-difference momentum propagates at that corner's frequency — and ρ the spin-to-chirality projection. The gaps are not free: Theorem 8b fixes the corner-class couplings at $\mu = (1, 1/3, -1/3, -1)$ — universality-class numbers — so the rest frequencies $\arccos \mu$ determine Δ_S and Δ_{T_2} outright (with opposite signs: the singlet sits below the generation corners, the T_2 class above). The condition then fixes the required projection: $\rho \approx -0.015$ in frequency units or -0.030 in mass-squared units — small rather than order unity, satisfiable but not yet derived. The remaining content of Condition 2 is thereby a single number: the spin-to-chirality projection with proper on-shell normalization — the same machinery as the c_λ computation of §7.1. That machinery resolves the question, and the resolution is negative — algebraically, in the same identity class as the §7.1 result. In the complete $(\gamma_S \otimes \xi_T)$ channel decomposition, the two-hop operator is a *single pure monomial* ($\gamma_0\gamma_1\gamma_3$ in the spin slot for the axes-$(1,2)$ transfer, times a taste monomial), and a monomial is orthogonal to every other channel: its weight in each mass-channel spin structure — scalar, γ_0, and both γ_5-graded partners — vanishes identically. The projection ρ is therefore not small but exactly zero, against the required -0.015 to -0.03: the substrate dressing chain is retired. Condition 2 accordingly stands as what the construction always declared it to be — a stated condition, with the 7/6 sum rule as its direct experimental test — and the derivation-mechanism space is now mapped: first-order-squared effects are excluded by the spectrum-dominance computation, and the geometric second-order route is excluded by monomial orthogonality. One scalar route survives the mapping and is recorded without being pursued here: the $O(\lambda^3)$ direct–two-hop interference is spin-scalar (the square of the transfer monomial is proportional to the identity — explicitly $(\gamma_0\gamma_1\gamma_3)^2 = -1$ in the derived corner algebra), a different construction that would require its own chain and carries an additional factor of λ. Specifying that chain requires two elements the present hop model does not fix: the operator-level rule by which the direct and two-hop insertions compose in the mass-squared matrix, and the on-shell normalization of the composite. Declared inputs of this reduction: the hop model of the taste geometry, the projection ρ, and the neglect of the $O(\lambda^3)$ interference; one convergence caveat is recorded — the $(1,2)$ channel carries a hierarchy-enhanced correction $\lambda^2 m_\tau^2/m_\mu^2 \approx 0.7$ at physical λ, so its strict-λ^2 extraction is the least robust of the three. The limitation extends beyond that channel. The value of $m_{23}^{(2)}$ at which the sum rule holds exactly converges, as $\lambda \to 0$, to $b_{23}^{\rm sub} = \frac{4}{3}(b_{12}^{\rm sub} + b_{13}^{\rm sub}) + \frac{1}{9}$, and in the $m_\mu/m_\tau \to 0$ limit the required democratic two-hop amplitude converges to $-1/15$; the deviation from these grows linearly in λ with a large coefficient, reaching order unity relative to the target at physical λ. The substrate-level form of Condition 2 holds as a $\lambda \to 0$ statement, and a test at physical λ requires exact diagonalization rather than the strict second-order extraction. The Layer 2(a) question is thereby reduced, not closed: its remaining content is the heavy-corner gap determination.This is concrete, quantitative, and falsifiable — any first-principles substrate calculation of $m_{ij}^{(2)}$ can be tested against this hierarchy.

The observed $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ (observed: 1.178 ± 0.020 using post-JUNO global fit and NuFIT 6.0 atmospheric) therefore follows from the cubic-lattice structure conditional on the substrate-level relation above. The empirical match distinguishes this prediction from TM1 or TM2 partial-TBM patterns (which do not produce this combination as an exact sum rule); the match indicates the framework's bijection has the alignment property required by Cond 2 — equivalently, that the substrate-level mechanism producing the $4(d-1)/(2d)$ amplitude ratio is realized in the bijection.

The quoted value uses the NuFIT 6.0 normal-ordering best fit $\sin^2\theta_{23} \approx 0.561$, the upper octant. Under the lower-octant solution ($\sin^2\theta_{23} \approx 0.455$) the same combination is ≈ 1.07, roughly 5σ below 7/6, and the natural-form alternative discussed below fails in the opposite direction. A lower-octant determination excludes the prediction.

7.4 The Higgs mass

The Higgs is the A_1 taste — a composite scalar. Its quartic self-coupling at the compositeness scale has no tree-level contribution: $\lambda(M_{\rm Pl}) = 0$. The SM quartic is generated entirely by RGE running. The observed $\lambda(M_{\rm Pl}) \approx -0.013 \pm 0.020$ [25] is consistent with zero at 0.6σ. The structural prediction is $\lambda(M_{\rm Pl}) = 0$; the numerical value of m_H in GeV is derived from this structural boundary condition combined with the measured m_t as a one-input parameter.

Using the 3-loop SM RGE [25], $\lambda(M_{\rm Pl}) = 0$ gives $m_H \approx 129$–132 GeV for $m_t = 172$–173 GeV. The observed $m_H = 125.10 \pm 0.14$ GeV is 4–7 GeV below; the gap is sensitive to m_t ($\partial m_H/\partial m_t \approx -1.1$ GeV/GeV). The CMS measurement $m_t = 170.5 \pm 0.8$ GeV [26] would bring the prediction to $\sim 128 \pm 2$ GeV. The structural claim $\lambda(M_{\rm Pl}) = 0$ is therefore confirmed; the specific m_H value in GeV is consistent with observation within the current m_t uncertainty rather than predicted at sub-GeV precision.

Higgs near-criticality. The observed $\lambda(M_{\rm Pl}) \approx 0$ is part of a broader empirical pattern: the SM Higgs sector sits at the boundary between vacuum stability and metastability, with $\lambda \approx 0$ and $\beta_\lambda \approx 0$ both occurring near the Planck scale (Buttazzo et al. [25]). This "near-criticality" is treated as a remarkable coincidence in standard SM analyses requiring beyond-the-Standard-Model explanation. The Multiple Point Principle (Bennett-Froggatt-Nielsen [34], Froggatt-Nielsen [35]) postulated $\lambda(M_{\rm Pl}) = 0$ as a fine-tuning condition; this prediction matched the observed Higgs mass before its discovery. Subsequent BSM model-building (singlet scalar dark matter, vector-like leptons, strong dynamics extensions, 2HDM realizations) [36, 37, 38] has produced extensions enforcing $\lambda(M_{\rm Pl}) = 0$ as a boundary condition postulated from outside the SM. The framework's prediction $\lambda(M_{\rm Pl}) = 0$ is therefore the structural realization within a derived framework of a boundary condition that an active BSM literature has been postulating for three decades.

The asymptotic-safety convergence. Wetterich (asymptotic-safety prediction [39]) showed that if the SM is valid up to the Planck mass and gravity contributes an irrelevant fixed-point structure to the running, the observed $M_H \approx 125$ GeV predicts $M_t \approx 171$ GeV. This is structurally the same prediction as the framework's, viewed from the opposite direction: from observed m_H to predicted m_t. The convergence between two structurally distinct frameworks (asymptotic safety vs OI) predicting the same near-criticality boundary condition strengthens the empirical content of the prediction.

Empirical confirmation status of independent lattice composite-Higgs work. Aoki et al. [40] observed via lattice simulations of $N_f = 8$ QCD that the singlet-taste channel produces a light composite scalar (technidilaton candidate) consistent with the observed 125 GeV Higgs mass. The framework's specific mechanism — Higgs as composite scalar in the singlet staggered taste with chiral protection — has independent empirical support from lattice gauge theory studies of strongly-coupled gauge theories.

Joint boundary-condition structure: $y_t(M_{\rm Pl}) = 1$ and $\lambda(M_{\rm Pl}) = 0$. The framework fixes two boundary conditions at the compositeness scale — the top Yukawa $y_t(M_{\rm Pl}) = 1$ from the staggered fermion-to-scalar overlap normalization (§7), and the Higgs quartic $\lambda(M_{\rm Pl}) = 0$ from the composite-Higgs structure (this section) — but their empirical statuses differ and must not be conflated. The quartic condition $\lambda(M_{\rm Pl}) = 0$, combined with the observed m_t and the Higgs vev $v = 246$ GeV (Tier 3 input), predicts $m_H \approx 128$–132 GeV via SM 3-loop RGE running [25] — consistent with the observed 125.10 ± 0.14 GeV to ~ 2–3σ (reduced to $\lesssim 1\sigma$ for the CMS $m_t = 170.5 \pm 0.8$ GeV [26]). The Yukawa condition $y_t(M_{\rm Pl}) = 1$, by contrast, does *not* reproduce the observed top mass: one-loop running carries it toward the IR quasi-fixed point at $y_t(m_t) \approx 1.19$, i.e. $m_t \approx 207$ GeV — a $\sim 20\%$ excess (§7.2) — while the observed $m_t = 172.7$ GeV corresponds to $y_t(M_{\rm Pl}) \approx 0.4$–$0.5$, not unity ($\lambda$ does not enter the one-loop y_t β-function, so the quartic condition cannot rescue it). The weak-scale coincidence $m_t = v/\sqrt{2}$ ($y_t(m_t) \approx 1$) is therefore a retrodiction with the fixed-point mechanism open (§7.2, §7.6), not a forward prediction from the compositeness-scale normalization. The structural content that survives is the single boundary condition $\lambda(M_{\rm Pl}) = 0$ — the same condition the MPP literature *postulates*, here *derived* from the composite-Higgs (singlet-taste) construction.

Unified matching at the compositeness scale. The framework's substratum produces a unified matching picture at the single scale $M_{\rm Pl}$: gauge coupling matching ($1/\alpha_0 = 23.25$ and the universal threshold $\delta_0 = 10$, both independent of gauge group; §6), Higgs quartic matching ($\lambda(M_{\rm Pl}) = 0$; this section), top Yukawa matching ($y_t(M_{\rm Pl}) = 1$; this section), and bare Higgs mass matching (Veltman condition $m_H^{\rm bare}(M_{\rm Pl}) = 0$; §8.8). All four matching conditions hold at the same scale — the substratum's intrinsic UV cutoff — because they all arise from the same compositeness structure: emergent SM couplings inherit their UV boundary conditions from the substratum's structural features (group-theoretic, taste, chiral) at the compositeness scale. This is structurally analogous to (and significantly more constrained than) conventional composite-Higgs scenarios at $f \sim$ TeV, where multiple matching conditions are imposed at the compositeness scale but typically require additional assumptions (custodial symmetry, partial compositeness ansatz, top partners). The framework's single-scale unified matching with no additional assumptions is a substantive structural feature: the same compositeness scale yields the gauge couplings (§6), the Higgs quartic and mass (§7.4 and §8.8), the top Yukawa (§7), and the bare Higgs mass (§8.8) without independent matching parameters for each. The Buttazzo near-criticality coincidence — $\lambda(M_{\rm Pl}) \approx 0$, $\beta_\lambda(M_{\rm Pl}) \approx 0$, $y_t(M_{\rm Pl})$ at minimum, Veltman condition near-satisfied — is therefore not four separate coincidences but four consequences of a unified compositeness-scale matching structure.

7.5 The bottom-to-tau mass ratio

Tree-level result.

The tree-level Yukawa coupling is taste-independent [12]: $y_b = y_\tau$. With two-loop SM RGE [25]: $m_b/m_\tau|_{\rm tree} = 4.276$ (with $\sim 1\%$ residual systematic from higher-loop corrections and EW-threshold matching), indicating a substantial non-perturbative correction.

Non-perturbative correction.

The correction is the scalar density renormalization $Z_S(m) = \langle\bar\psi\psi\rangle_{\rm int}/\langle\bar\psi\psi\rangle_{\rm free}$, computed on SU(3) gauge backgrounds at $\beta = 11.1$ as a function of bare mass m. The denominator $\langle\bar\psi\psi\rangle_{\rm free}$ is evaluated analytically on the same lattice with the eight BZ-corner modes excluded from the momentum sum (`scripts/analyze_zs.py`); this finite-L regulator differs from the standard staggered convention (corner modes included) by $\sim 2\%$ at $L = 32$, $\sim 0.3\%$ at $L = 64$, and converges to it at $L \to \infty$. The prediction: $m_b/m_\tau = 4.28/Z_S(m_{\rm match})$.

Monte Carlo simulations at $L = 16$ (30 configs) and $L = 32$ (50 configs), scanning 30 masses from $m = 0.005$ to 0.50, reveal that $Z_S(m)$ is monotonically decreasing in the volume-converged region ($mL \gtrsim 3$). At $m = 0.10$: Z_S converges from 1.70 (L=16) to 1.92 (L=32) to 1.94 (L=64).

Chiral condensate formation.

Comparison of $L = 16$ and $L = 32$ confirms spontaneous chiral symmetry breaking: Z_S at small m grows 8× between volumes, and the apparent peak shifts from $m = 0.087$ (L=16) to $m = 0.024$ (L=32, $Z_S = 2.828 \pm 0.008$), tracking the finite-volume boundary $mL \sim 1$. The chiral condensate $\Sigma \approx 0.20$ (from linear extrapolation in the converged region at $L = 32$).

The matching mass.

The matching scale where lattice dynamics connects to the perturbative Yukawa description is set by the product of the two relevant dimensionless parameters:

the taste-breaking amplitude $\lambda = 1/(\pi\sqrt{2})$ and the unified gauge coupling $g_0^2 = 4\pi\alpha_0 = 4\pi/23.25$:

$$m_{\text{match}} = \lambda \times g_0^2 = \frac{1}{\pi\sqrt{2}} \times \frac{4\pi}{23.25} = \frac{4}{23.25\sqrt{2}} = 0.1217$$

Physically, λ controls inter-generation mixing (taste-breaking) while g_0^2 controls the non-perturbative condensate (confinement). The matching scale is derived in the taste-decomposed Coleman–Weinberg potential by tracing three ingredients through the framework's existing structure:

(i) The b-quark is identified with the democratic eigenstate $\hat{h} = (|e_1\rangle + |e_2\rangle + |e_3\rangle)/\sqrt{3}$ of the cubic-symmetric taste-breaking operator (this follows from the §7.1 Wolfenstein $A = \sqrt{2/3}$ identification and the §7.2 Koide derivation, both of which place the heaviest fermion along $\hat{h}$).

(ii) The taste-breaking operator in the X-corner basis has the cubic-symmetric form $M_{\text{taste}} = m_{\text{off}} \cdot (J - I)$ (after trace removal), giving the diagonal element in the b-quark eigenstate $c_{bb} = \langle b|M_{\text{taste}}|b\rangle = 2m_{\text{off}}$.

(iii) The off-diagonal element m_{off} is identified with the BZ-distance geometric primitive of §7.1: $m_{\text{off}} = 1/|X_i - X_j| = \lambda$.

Combining these gives $c_{bb} = 2\lambda$. The scalar density self-energy takes the form $\Pi_{\text{taste}}(m) = K \cdot c_{bb} \cdot g_0^2 \cdot \Sigma/m$ for some loop kinematic coefficient K, and matching at the natural condensate-scale threshold $T = \Sigma$ gives $m_{\text{match}} = 2K\lambda \cdot g_0^2$.

Structural form: $K = 1/(n_{\mathbf{gen}} - 1)$. The factor 2 in $c_{bb} = 2\lambda$ has a representation-theoretic origin parallel to the Wolfenstein $A^2 = (d-1)/d$ identity of §7.1. For the democratic eigenstate $\hat{h} = (e_1 + ... + e_{n_{\text{gen}}})/\sqrt{n_{\text{gen}}}$ in n_{gen} generations:

$$J\hat{h} = n_{\text{gen}} \cdot \hat{h}, \quad \langle \hat{h}|J|\hat{h}\rangle = n_{\text{gen}}, \quad \langle \hat{h}|I|\hat{h}\rangle = 1$$

giving

$$c_{bb} = \langle \hat{h}|M_{\text{taste}}|\hat{h}\rangle = (n_{\text{gen}} - 1) \cdot m_{\text{off}} = (n_{\text{gen}} - 1) \cdot \lambda.$$

The $(n_{\text{gen}} - 1)$ factor counts the off-democratic "partners" of $\hat{h}$ in the generation rep — the inverse of the Wolfenstein A structural quantity $\sqrt{(d-1)/d}$ applied to the same off-democratic projection.

Substituting into the matching condition gives $m_{\text{match}} = K \cdot (n_{\text{gen}} - 1) \cdot \lambda \cdot g_0^2$. The framework's prediction $m_{\text{match}} = \lambda \cdot g_0^2$ (with unit prefactor: the proportional form $m_{\text{match}} \propto \lambda g_0^2$ is fixed by power-counting in the taste-decomposed CW potential with the natural threshold convention $T = \Sigma$, while the specific dimensionless coefficient is currently MC-verified to 1.4% rather than derived in closed form — see below) therefore requires:

$$\boxed{K = \frac{1}{n_{\text{gen}} - 1}}$$

For $n_{\text{gen}} = 3$ — itself forced by the cubic-group decomposition of the 6-link representation $T_1 \oplus E \oplus A_1$ giving 3 fermion generations on T_1 (Theorem 7) — this gives $K = 1/2$.

This is a representation-theoretic structural identity, not a numerical happenstance. $K = 1/2$ is the $n_{\text{gen}} = 3$ value of a generation-count-dependent identity $K = 1/(n_{\text{gen}}-1)$, paralleling the family of $(d-1)$-based identities the framework already employs:

Identity	Formula	$d = 3$ / $n_{\text{gen}} = 3$ value	Source
Wolfenstein A^2	$(d-1)/d$	2/3	Off-democratic projection (§7.1)
Cond 2 ratio	$4(d-1)/(2d) = 2A^2$	4/3	Squared off-democratic projection (§7.3)
K **(this work)**	$1/(n_{\mathbf{gen}} - 1)$	1/2	**Inverse off-democratic partner count**

Unifying identity. Wolfenstein $A^2 = (d-1)/d$ and $K = 1/(d-1)$ are dual aspects of the same off-democratic structure on a d-dimensional vector representation: A^2 is the projection magnitude per unit dimension, K is inverse partner count. Their product is the per-component weight:

$$K \cdot A^2 = \frac{1}{d-1} \cdot \frac{d-1}{d} = \frac{1}{d}.$$

For $d = 3$: $A^2 = 2/3$, $K = 1/2$, $K \cdot A^2 = 1/3$. The framework's $(d-1)$-based identity family is internally consistent and unified, not a list of independent coincidences.

Consistency check on $n_{\mathbf{gen}} = 3$. Given the matching condition $K \cdot (n_{\text{gen}}-1) = 1$ (forced by $m_{\text{match}} = \lambda g_0^2$ with prefactor unity from power counting and confirmed by MC at 1.4%) and the value $K = 1/2$ inherited from the standard staggered Z_S calculation (Argument 1; plausibility evidence pending the explicit one-loop OI derivation), $n_{\text{gen}} - 1 = 2$ — consistent with the framework's three-generation structure. This is a *consistency check* rather than an independent over-determination: $n_{\text{gen}} = 3$ is itself fixed independently by the cubic decomposition $T_1 \oplus E \oplus A_1$ (Theorem 7), and the matching consistency uses an inherited (not independently derived) value of K. The framework's commitment to three generations rests primarily on the cubic decomposition; the matching-condition consistency adds a self-consistency confirmation rather than an independent forcing.

Layer stratification of $K = 1/2$. The structural identity admits the following decomposition (analogous to §7.3's Cond 2 stratification):

Component	Required value	Layer status
Democratic eigenstate $\hat{h}$ as O-singlet	from cubic-group rep theory	**Layer 1** (rep theory)
$c_{bb} = (n_{\text{gen}} - 1) \cdot \lambda$	from $\langle \hat{h} \Vert J - I \Vert \hat{h} \rangle = n_{\text{gen}} - 1$ algebraic identity	**Layer 1** (algebraic)
Power counting forces $m_{\text{match}} \propto \lambda \cdot g_0^2$	from CW dimensional analysis	**Layer 1** (already noted in §7.5)
$n_{\text{gen}} = 3$	from cubic decomposition $T_1 \oplus E \oplus A_1$	**Layer 0/1** (Theorem 7)
Loop calculation gives $K = 1/(n_{\text{gen}} - 1)$ explicitly	from OI-specific taste-decomposed CW one-loop	**Layer 2(a)-tractable** (residual gap)

The first four components are Layer 1 closed: they together fix $K = 1/(n_{\text{gen}} - 1) = 1/2$ as the value the loop calculation *must* yield for the framework's matching prediction to hold. The Layer 2(a)-tractable residual is whether the OI's specific taste-decomposed CW one-loop calculation actually produces this kinematic prefactor — addressed by Argument 1 below (inheritance from standard staggered) and the Residual gap subsection.

The framework's prediction $m_{\text{match}} = \lambda \cdot g_0^2$ requires $K = 1/2$. Given the algebraic identity $c_{bb} = (n_{\text{gen}} - 1)\lambda$ (Layer 1, clean), the matching condition $m_{\text{match}} = K \cdot c_{bb} \cdot g_0^2$ forces $K = 1/(n_{\text{gen}} - 1) = 1/2$ as a structural identity whenever the matching condition itself holds. The matching condition $m_{\text{match}} = \lambda g_0^2$ is supported by the following arguments, MC-verified to 1.4% precision:

Argument 1 (Staggered convention). The standard staggered fermion action uses the symmetric kinetic term $S_{\text{kin}} = \frac{1}{2} \sum_{x,\mu} \eta_\mu(x)\bar{\chi}(x)[U_\mu(x)\chi(x+\hat{\mu}) - U_\mu^\dagger(x-\hat{\mu})\chi(x-\hat{\mu})]$, with the gauge vertex $V_\mu = (g_0/2)\eta_\mu(x)[U_\mu + U_\mu^\dagger]$. Patel & Sharpe [33] derive the standard one-gluon vertex $V_\mu \propto -ig\cos(q'_\mu + k_\mu/2)(\gamma_\mu \otimes 1)$ (their Eq. 17) — the same form as the OI vertex $V_j = ig_0\mu T^\alpha \cos(k_j + p_j/2)$ — and the two-gluon (tadpole) vertex with explicit factor $-i\frac{1}{2}g^2$ (their Eq. 18) tracing directly to the symmetrization. The full one-loop renormalization of staggered scalar density bilinears computed there and in subsequent work [Constantinou-Costa-Panagopoulos and others] is a specific known number consistent with $K = 1/2$ in the framework's decomposition.

Scope of the inheritance argument. The algebraic identity $\cos^2(k_\mu/2) = (1 + \cos(k_\mu))/2$ is dimension-independent per direction, but its "1/2" content propagates into the lattice integral only when the $\cos(k_\mu)$ residual vanishes by symmetry. The relevant symmetry is the π-shift $k_\mu \to \pi - k_\mu$: integrands depending on k only through $\sin^2(k_\mu)$ (i.e., the staggered fermion propagator) are π-shift symmetric and the residual integrates to zero exactly, leaving the 1/2 factor;

integrands involving the lattice gluon propagator $\hat{q}^2 = 4\sum \sin^2(k_\mu/2)$ are *not* π-shift symmetric (under the shift, $\sin^2(k_\mu/2) \to \cos^2(k_\mu/2)$), and the residual does not vanish in isolation. The actual one-loop Z_S integrand involves both fermion and gluon propagators, so the simple $\cos^2 \to 1/2$ algebra does not by itself constitute the analytical derivation of $K = 1/2$; the residual $\cos(k_\mu)$ pieces from the vertex-correction diagram combine with analogous pieces from the fermion self-energy and tadpole diagrams to produce the published staggered Z_S. In the analogous condensate-ratio construction the residual does not cancel: in four dimensions the residual fraction is finite and convergent at ≈ 0.11 (tadpole-normalization bracket 0.11–0.23), and the three-dimensional analogue falls toward the chiral limit without converging at accessible volumes. $Z_S^{(1)}$ is a well-defined finite number with that residual included, and the $\cos^2 \to 1/2$ symmetrization is a numerical approximation at the ten-percent level rather than an identity. The inheritance argument requires only the published value. The object evaluated is this framework's condensate-ratio Z_S; a scheme difference from the published-literature object is not excluded. The inheritance argument therefore reduces to: the published staggered $Z_S^{(1)}$ is a specific number consistent with $K = 1/2$, and OI uses the same vertex, action, and gauge fixing as the published calculations, so OI's Z_S should reproduce the same number. This is plausibility evidence, not analytical derivation; the matching mass μ enters the OI calculation as an overall factor (it sets the scale at which the matching is performed, not the dimensionless loop coefficient), so it does not modify K by inspection.

Consistency check 2 (Symmetric averaging — definitional). The taste-breaking insertion can appear before or after the gauge exchange in the fermion loop. By Hermitian conjugation symmetry these orderings contribute equally; if the full amplitude is *defined* as the average $\Pi^{\text{full}}_{\text{taste}} = \frac{1}{2}[\Pi^{(L)} + \Pi^{(R)}]$, the prefactor 1/2 appears explicitly and equals $\Pi^{(L)}$. Defining Π^{full} as the sum (rather than the average) gives no factor; the 1/2 here is therefore a convention that aligns with Argument 1's normalization, not an additional first-principles derivation.

Consistency check 3 (Empirical exclusion of natural fractions). Among the natural fractions $K \in \{1/4, 1/2, 1, 2\}$, the observed m_b/m_τ excludes all but $K = 1/2$:

K	m_{match}	m_b/m_τ	Discrepancy
1/2	0.122	2.36	0.4% ✓
1	0.243	≈ 3.17	35% ✗
1/4	0.061	≈ 1.78	24% ✗
2	0.487	$\gg 4$	excluded ✗

Intermediate fractions ($K \in \{1/3, 2/3, 3/4\}$) give m_{match} between 0.08 and 0.18, with corresponding m_b/m_τ in the 1.8–3.2 range; no fraction outside $\{1/2\}$ falls within $\sim 5\%$ of observation. This is empirical fitting on a moderately-restricted set rather than a first-principles derivation, but the resulting selection is sharp.

$K = 1/2$ is therefore *consistent with* the algebraic identity $c_{bb} = (n_{\text{gen}} - 1)\lambda$ and the matching condition $m_{\text{match}} = \lambda g_0^2$, with the latter MC-verified to 1.4% precision. The Layer 1 content is the algebraic identity for c_{bb}; the matching condition has the *form* of a Layer 1 structural relation (dimensional analysis plus the "natural threshold" convention $T = \Sigma$) but the specific dimensionless coefficient is currently established by MC rather than by closed-form analytical derivation. Argument 1 (inheritance from standard staggered convention), Consistency check 2 (definitional symmetric averaging), and Consistency check 3 (empirical exclusion of natural-fraction alternatives) provide plausibility evidence for the matching condition rather than analytical closure.

Within the four-layer framing (§8.3), $K = 1/2$ is MC-verified at 1.4% precision; the Layer 2(a)-tractable residual is an explicit one-loop calculation on the OI cubic lattice deriving $m_{\text{match}} = \lambda g_0^2$ from the taste-decomposed Coleman–Weinberg potential without appealing to MC verification of the matching scale. Given the matching condition, $K = 1/(n_{\text{gen}} - 1)$ follows by structural identity from $c_{bb} = (n_{\text{gen}} - 1)\lambda$; the matching condition itself awaits analytical closure. The stratification matches §7.3 Cond 2: a Layer 1 representation-theoretic core ($c_{bb} = (n_{\text{gen}} - 1)\lambda$ here; the geometric projection structure there) plus a Layer 2(a)-tractable substrate-loop verification residual.

Residual gap. With Argument 1 understood as plausibility evidence rather than independent analytical derivation, the explicit analytical task is a fully explicit one-loop calculation within the OI lattice setup — computing the vertex-correction, fermion self-energy, and tadpole diagrams with the OI vertex $V_j = ig_0\mu T^\alpha \cos(k_j + p_j/2)$, combining them via the relevant Ward identity, and confirming the result reduces to a value consistent with $K = 1/(n_{\text{gen}} - 1)$ — which would close this gap definitively. Two intermediate routes are also available: (a) a Ward-identity argument establishing analytically that the residual $\cos(k_\mu)$ pieces from the vertex correction cancel against analogous pieces in the self-energy and tadpole, which would render the $\cos^2 \to 1/2$ symmetrization heuristic into an actual derivation; (b) tighter MC measurement to confirm $K = 1/2$ at sub-percent precision and bound the higher-order corrections explicitly.

The per-vertex $\cos^2$ algebraic identity $\cos^2(k/2) = (1 + \cos(k))/2$ is dimension-independent per direction; the 1/2 constant piece survives any dimensional change. Whether this 1/2 propagates cleanly through the full one-loop integral depends on the π-shift cancellation discussed under "Scope of the inheritance argument" above — a question of integral structure rather than dimensionality. Four bridge gaps in the SχPT inheritance are addressed below; their current status:

(i) *Substratum measure reduction.* **Level 2 closed via pseudofermion correspondence** (cf. §4.2 *Remark*). The standard MC code uses the bosonic pseudofermion representation, not Grassmann variables directly. The substratum (Gaussian on real scalar bijections at large L) and pseudofermion (bosonic Gaussian auxiliary fields) measures are in the same class; their covariances differ by a factor of 4 absorbable into normalization, with Theorem 2 ($D_{st}^2 = -\frac{1}{4}\Box_{\mathrm{lat}}$) providing the explicit match. The Level 3 residual is cycle ergodicity for OI's specific bijection orbit structure: pure linear lattice dynamics is integrable rather than chaotic (the frequencies $\omega(\vec{k}) = 2\sin(|\vec{k}|/2)$ are not linearly independent over $\mathbb{Q}$), so full-shell sampling requires either effective non-linearity from the discretization rule (the standard interpretation for lattice-substrate frameworks) or revision of the Gaussianization argument to sub-torus sampling. We adopt the former implicitly; rigorous treatment of OI's specific orbit structure is open work.

(ii) *3D operational MC vs 4D SχPT dimensional bridge.* **Decomposes into Part A (time-direction factorization, Level 1 closed) and Part B (measure identification, Level 2 closed via inheritance from gap (i)).** Part A: For ρ stationary under φ and T the framework's involution with $T\varphi T = \varphi^{-1}$ (§5), $(\rho \circ T) \circ \varphi = \rho \circ (T\varphi) = \rho \circ (\varphi^{-1}T) = \rho \circ T$, so $\rho \circ T$ is φ-invariant; by P-indivisibility (Main Theorem 4), the stationary distribution is unique, hence $\rho \circ T = \rho$. Static observables therefore factorize through the time direction independent of which time slice. Part B is the same investigation as gap (i) viewed operationally: closing gap (i) automatically closes gap (ii) Part B. With gap (i) Level 2 closed, gap (ii) is Level 2 closed framework-essentially.

(iii) and (iv) *Bridge theorem:* **Closed at Level 1 framework-essentially via the explicit theorem below.**

Bridge Theorem (3D-4D staggered correspondence). Let $\chi(x)$ be the staggered field on $\mathbb{Z}^3$ with bijection time-evolution $\varphi : S \to S$. Then:

(a) *Mode-space isomorphism.* The 16-dim space spanned by 3D BZ corners $\times$ sublattice grading is isomorphic to the 4D Dirac $\times$ taste space:

$$\{(\eta, s) : \eta \in \{0,1\}^3, s \in \{0,1\}\} \xrightarrow{\sim} V_{\mathrm{Dirac}}^{(4D)} \otimes V_{\mathrm{taste}}$$

via $(\eta_1, \eta_2, \eta_3, s) \mapsto (\eta_0 = s, \eta_1, \eta_2, \eta_3)$. Counting: $2^3 \times 2 = 16 = 4 \times 4$. (Level 1.)

(b) *Clifford algebra identification.* The 4D chirality operator corresponds to spatial sublattice grading: $\gamma_5 \leftrightarrow \varepsilon_{3D}$. The 4D time-direction Dirac matrix is derived from spatial structure: $\gamma_0 \leftrightarrow \pm\varepsilon_{3D} \cdot \gamma_1\gamma_2\gamma_3$. (Level 1 for γ_5 identification; Level 2 for γ_0 derivation, sign conventions not separately verified.)

(c) *Chirality equivalence.* For static observables in the bijection's stationary distribution, ε_{3D} and ε_{4D} classify modes equivalently. Their ratio is $\varepsilon_{4D}/\varepsilon_{3D} = (-1)^{x_0}$, constant on each time slice; chirality classification differs by overall sign convention only. (Level 1 for the algebraic identity; full operator-theoretic interpretation involving the taste structure remains at Level 3 but is not framework-essential — operational chirality assignment is the framework-essential claim.)

(d) *Symmetry compatibility.* Staggered $U(1)$ shift symmetry (underlying the two-loop structure of K_S) and cubic group O action are preserved by the isomorphism of (a). (Level 2.)

Proof, by part with explicit rigor levels.

Part (a) — Level 1. The map $(\eta_1, \eta_2, \eta_3, s) \mapsto (\eta_0 = s, \eta_1, \eta_2, \eta_3)$ is a bijection between $\{0,1\}^3 \times \{0,1\}$ and $\{0,1\}^4$ by direct counting ($2^3 \times 2 = 16 = 2^4$). The map respects the natural cubic-group action on spatial indices since the spatial coordinates η_1, η_2, η_3 are preserved unchanged, and $\eta_0 = s$ corresponds to the sublattice grading which is cubic-invariant.

Part (b), $\gamma_5 \leftrightarrow \varepsilon_{3D}$ — Level 1. Two properties to verify: $\varepsilon_{3D}^2 = 1$ and $\{\varepsilon_{3D}, \gamma_i\} = 0$ for $i = 1, 2, 3$. The first is direct: $\varepsilon_{3D}(x)^2 = ((-1)^{x_1+x_2+x_3})^2 = 1$. The second follows from $\varepsilon_{3D}(x \pm \hat{i}) = -\varepsilon_{3D}(x)$ via direct computation: $(\gamma_i \varepsilon_{3D} \chi)(x) = \eta_i(x)[\varepsilon_{3D}(x+\hat{i})\chi(x+\hat{i}) - \varepsilon_{3D}(x-\hat{i})\chi(x-\hat{i})]/2 = -\varepsilon_{3D}(x) \cdot \eta_i(x)[\chi(x+\hat{i}) - \chi(x-\hat{i})]/2 = -(\varepsilon_{3D}\gamma_i\chi)(x)$. Both γ_0 and γ_5 satisfy these two properties, so further identification is required: in chiral basis, γ_0 is off-diagonal in sublattice blocks while γ_5 is diagonal with opposite signs. Sublattice grading ε_{3D} acts diagonally with opposite signs on even/odd sublattices — that is the γ_5 pattern, not γ_0. Therefore $\gamma_5 \leftrightarrow \varepsilon_{3D}$.

Part (b), γ_0 derivation — Level 2. The 4D Clifford algebra contains the identity $\gamma_0 = \pm\gamma_5\gamma_1\gamma_2\gamma_3$ (up to a sign convention depending on the metric signature and the specific Clifford basis convention). Substituting the $\gamma_5 \leftrightarrow \varepsilon_{3D}$ identification gives $\gamma_0 \leftrightarrow \pm\varepsilon_{3D} \cdot \gamma_1\gamma_2\gamma_3$. The sign convention depends on the specific Clifford basis convention and is not separately verified here.

Part (c), chirality equivalence — Level 1. Direct algebraic identity: $\varepsilon_{4D}(x_0, \vec{x}) = (-1)^{x_0+x_1+x_2+x_3} = (-1)^{x_0} \cdot (-1)^{x_1+x_2+x_3} = (-1)^{x_0} \cdot \varepsilon_{3D}(\vec{x})$. For static observables (evaluated at fixed x_0), the prefactor $(-1)^{x_0}$ is a constant — observables computed via ε_{3D} and via ε_{4D} classify modes equivalently up to overall sign convention.

Part (d) — Level 2. The staggered $U(1)$ shift symmetry transforms $\chi(x) \to \omega(x)\chi(x)$ with sign factors depending on sublattice and BZ corner. Under the isomorphism of (a), this transformation acts on the corresponding Dirac $\times$ taste basis vectors compatibly — full verification with explicit taste-index tracking is at Level 2 and not written out here. The cubic group O action commutes with the isomorphism by part (a). $\square$

Operator-theoretic interpretation involving full taste structure (Level 3 — not framework-essential). The operational identification $\gamma_5 \leftrightarrow \varepsilon_{3D}$ is the framework-essential claim and is Level 1 closed in part (b). A full operator-theoretic interpretation involving the 4-dim taste rep ξ_5 and showing ε_{4D} corresponds to $\gamma_5 \otimes \xi_5$ in the spin-taste decomposition is at Level 3 (sketch only) and is not required for the framework's use of the bridge theorem.

Implication for the framework's $d = 3$ commitment. The bridge theorem reveals that the entire 4D Clifford structure emerges from 3D Clifford + sublattice grading, with γ_0 derived rather than independent. There is no input from time-direction structure; time enters only as bijection-evolution index x_0, and the Clifford-algebra role is played entirely by spatial structures. This means structural bandwidth quantities like $\theta_0 = C_2/d^2 = 2/9$ and $A^2 = (d-1)/d = 2/3$ must use spatial $d = 3$, not spacetime $d = 4$. The framework's commitment to $d = 3$ is structurally forced (not chosen) by the time-emergence picture.

The same gaps apply to the other §7 mass-cluster predictions (D11, m_τ, etc.) that share the SχPT machinery; the closure status above propagates to those predictions automatically. None of them changes the closure path for K_S, which remains the two-loop SχPT calculation specializing Panagopoulos–Spanoudes [Panagopoulos-Spanoudes 2017] to the OI cubic-group setting.

MC consistency check: the empirical matching scale extracted from the Monte Carlo data is $m_{\text{match}}^{\text{MC}} = 0.1234 \pm 0.0002$, which agrees with the predicted $\lambda \cdot g_0^2 = 0.1217$ within 1.4% (compatible with NLO truncation of the leading-order chain above). Alternative choices of coefficient — e.g., $K = 1$ giving $m_{\text{match}} = 2\lambda g_0^2 \approx 0.24$ — are strongly excluded by the MC data (would give $m_b/m_\tau \approx 3.17$ vs. observed 2.35).

The prediction.

$$\boxed{\frac{m_b}{m_\tau} = \frac{4.28}{Z_S(\lambda g_0^2)} = \frac{4.28}{1.813} = 2.361}$$

Observed: $m_b/m_\tau = 2.352 \pm 0.017$. The central-value match at $L = 32$ is 0.4% (0.5σ), with a realistic systematic uncertainty of $\sim$ 1–2% from finite volume, the regulator choice noted above, and higher-loop SM RGE corrections to $R = 4.28$. The $L \to \infty$ extrapolation in either lattice regulator gives $Z_S \approx 1.83$ and $m_b/m_\tau \approx 2.34$, consistent with observation at $\sim 0.7\sigma$. The $L = 32$ run (50 configs) gives $Z_S(0.122) = 1.813 \pm 0.001$ (statistical only) by cubic interpolation; finite-volume systematics are not exhaustively quantified in the released code (see `README.md` §Code, point 3). The inputs λ, g_0^2, and $R = 4.28$ are determined by the lattice structure; the matching-scale identification $m_{\text{match}} = \lambda \cdot g_0^2$ follows from power-counting in the taste-decomposed CW potential. The dimensional form $m_{\text{match}} \propto \lambda \cdot g_0^2$ is forced: λ controls inter-generation taste-breaking and g_0^2 controls the gauge-induced condensate, and these are the only two dimensionless framework primitives entering the leading-order CW potential. The scalar prefactor (the K coefficient) is the separate Layer 2 question addressed by Arguments 1–3 above and the Residual gap subsection.

7.6 Summary table of all predictions

Twenty-two quantitative observables from a $d = 3$ cubic lattice with spacing $\epsilon = 2\,l_P$, classified by input status: **S** = strictly parameter-free structural; **L** = layered (structural form combined with a Layer-2(b) substrate-specific coefficient or Layer-3 empirical input — see §8.3 four-layer framing); **M** = one-input mass/parameter chain; **E** = uses explicit empirical input; **R** = retrodiction; **P** = phenomenological/ansatz input.

Prediction	Formula	Value	Observed	Match	Input
$1/\alpha_1(M_Z)$	lattice + RGE	59.00	59.00	$< 0.1\%$	**E**[a] (fixes δ_0)
$1/\alpha_2(M_Z)$	lattice + RGE	29.57	29.57	$< 0.1\%$	**R**[a]
$1/\alpha_3(M_Z)$	lattice + RGE	8.47	8.47	$< 0.1\%$	**R**[a]
λ (Cabibbo)	$1/(\pi\sqrt{2})$	0.2251	0.2250 ± 0.0007	0.04%	**S**
A (Wolfenstein)	$\sqrt{2/3}$	0.8165	0.826 ± 0.012[h]	0.8σ	**S**
m_d/m_s	$1/(2\pi^2)$	0.0507	0.0503 ± 0.0007	0.56σ	**L** (structural λ; GST empirical)
$\lvert V_{cb}\rvert$	$\sqrt{2/3}/(2\pi^2)$	0.0414	0.0408 ± 0.0014[h]	0.4σ	**S**
J (Jarlskog)	$\eta/(12\pi^6)$	3.02×10^{-5}	$(3.08 \pm 0.13)\times 10^{-5}$[h]	0.5σ	**E** (empirical η)
Koide Q	$2/3$	0.66667	0.66666	$< 0.01\%$	**S**
Koide angle θ_0	$C_2/d^2 = 2/9$	0.22222	0.22227	0.02%	**S**
m_e (from m_τ)	$\theta_0 = 2/9$	0.51096 MeV	0.51100 MeV	0.007%	**M** (m_τ)
m_μ (from m_τ)	$\theta_0 = 2/9$	105.652 MeV	105.658 MeV	0.006%	**M** (m_τ)
Q_{down}	$(2/3)(1 + \alpha_s/\pi)$	0.7303	0.7314	0.15% (scheme-dep)	**P**
m_b (from m_s)	Koide(Q_{down})	4144 MeV	4180 ± 30	0.9%	**L**[d]
m_u/m_d	$\sqrt{2/9}$	0.4714	0.465 ± 0.024	0.27σ	**L**[e]

Prediction	Formula	Value	Observed	Match	Input
$\sin^2\theta_{12}$	$1/3 - 1/(4\pi^2)$	0.3080	0.3085 ± 0.0073 (Capozzi 2025 post-JUNO global)	0.07σ	$\mathbf{L}^{\mathrm{f}}$
$\sin^2\theta_{23}$	$1/2 + 1/(2\pi^2)$	0.5507	$0.561^{+0.012}_{-0.015}$ (NuFIT 6.0, NO)	0.74σ	$\mathbf{L}^{\mathrm{f}}$
$\sin^2\theta_{13}$	$4/(18\pi^2)$	0.02252	0.02195 ± 0.00056 (NuFIT 6.0, NO)	1.02σ	**S**
$\lambda(M_{\mathrm{Pl}}) = 0$	composite Higgs	0	-0.013 ± 0.020	0.6σ	**S**
m_H (from $\lambda(M_{\mathrm{Pl}}) = 0$, m_t)	RGE running	129–132 GeV	125.10 ± 0.14	m_t-consistent	**M** (m_t)
m_b/m_τ	$4.28/Z_S(\lambda g_0^2)$	2.361	2.352	0.5σ	$\mathbf{L}^{\mathrm{b}}$
m_t	$v/\sqrt{2}$ (weak-scale $y_t \approx 1$; mechanism open)	174.1 GeV	172.5 ± 0.3	0.9%	$\mathbf{R}^{\mathrm{g}}$

Footnotes:

- **a** The three SM gauge couplings at M_Z are reproduced with one empirical input in this sector (§6.3): the U(1) coupling fixes the universal threshold $\delta_0 = 10.02$, and the SU(2) and SU(3) entries are retrodictions using the C_2-dependent threshold parameterization $(A, B) = (8.3, 5.59)$ matched to the observed couplings. The structural input $1/\alpha_0 = 23.25$ is computed analytically from the one-loop staggered vacuum polarization. A 2-loop staggered VP rebuild indicates δ_0 is not reproduced by strict 2-loop PT and is accordingly treated as empirical input (see §6.2). The structural predictions in this sector are: (i) the specific value $1/\alpha_0 = 23.25$, (ii) the universality of δ_0 across gauge groups, and (iii) the 1-loop analytic computation of $A \cdot B$, whose extrapolated value is consistent with the fitted 46.4 under the $1/N^2$ leading-power assumption but is leading-power-dependent (range $\approx$ 48–62 across finite-size models; §6.2.1) and is inconclusive pending larger lattices. Deriving (A, B) individually from first-principles lattice perturbation theory would upgrade the SU(2) and SU(3) entries from retrodictions (**R**) to strict predictions (**S**).

- [b] *Layered prediction (m_b/m_τ).* The matching scale $m_{\text{match}} = \lambda \cdot g_0^2$ is partly structural in the taste-decomposed Coleman–Weinberg potential (§7.5): the b-quark is identified with the democratic eigenstate $\hat{h}$ of the cubic-symmetric taste-breaking operator (from §7.1, §7.2 representation theory), giving diagonal taste-breaking $c_{bb} = (n_{\text{gen}} - 1)\lambda = 2\lambda$ — Layer 1 algebra. Combining with the natural matching condition at condensate scale Σ and a loop kinematic coefficient K, the prediction $m_{\text{match}} = \lambda g_0^2$ requires $K = 1/(n_{\text{gen}} - 1) = 1/2$ — a structural identity that follows by definition *given* the matching condition; the matching condition itself is MC-verified to 1.4%, and analytical derivation of its specific dimensionless coefficient is the Layer 2(a)-tractable residual (an explicit one-loop calculation on the OI cubic lattice or a Ward-identity argument). Within $K \in \{1/4, 1/2, 1, 2\}$, $K = 1$ is excluded at 35% and $K = 1/4$ at 24%; see §7.5 for the structural detail. The MC-extracted value $m_{\text{match}}^{\text{MC}} = 0.1234 \pm 0.0002$ agrees with the derived $\lambda g_0^2 = 0.1217$ within 1.4% (NLO truncation). The central-value match $m_b/m_\tau = 0.5\sigma$ is at $L = 32$; systematic uncertainty including finite-volume and regulator effects is $\sim$ 1–2% (§7.5). Within the four-layer framing (§8.3), the prediction is MC-verified at the matching-condition level with the Layer 2(a)-tractable analytical closure pending.
- [d] *Layered conditional structural prediction (m_b via Q_down).* The structural form (Koide formula applied to the down-sector with input $Q_{\text{down}} + m_d/m_s + m_s = 93.4$ MeV) is Layer 1. The Q_{down} value ($= (2/3)(1 + \alpha_s/\pi) \approx 0.731$) is phenomenological per §7.2 — see the Q_{down} subsection for full discussion of why this is treated as empirical input rather than derived. The m_d/m_s input inherits §7.1's Cabibbo λ^2 via the GST relation. The m_s input is one M-chain mass scale. The empirical match at 0.9% is excellent given the layered structure.
- [g] *Retrodiction (m_t).* One-loop SM running gives an IR quasi-fixed point at $y_t \approx 1.25$, and the normalization $y_t(M_{\text{Pl}}) = 1$ flows to $m_t \approx 207$ GeV; the 0.9% match is the weak-scale $y_t(m_t) \approx 1$ coincidence, whose structural derivation is open (§7.2).
- [h] *Data source.* The observed values for A, $|V_{cb}|$, and J follow a single global-fit source per observable, quoted in §7.1 and used identically here. The observed Wolfenstein A is itself fit-family dependent (central values span approximately 0.79–0.83 across PDG, CKMfitter, and UTfit determinations, a spread larger than any single fit's quoted uncertainty); the prediction $\sqrt{2/3} = 0.8165$ lies inside that spread, so the honest match statement for A is consistency within the fit-family spread rather than a single-σ figure. The J entry additionally carries a parametrization caveat: η is extracted by global fits within the Wolfenstein parametrization using the fitted A and λ, so inserting the fitted η alongside the framework's structural A and λ mixes parametrizations; the J row tests the combination $A^2\lambda^6$ only to the extent the extraction of η is insensitive to that

mixing.

- [e] *Layered prediction (m_u/m_d), Layer 1 + Layer 2(b).* The structural form $m_u/m_d = \sqrt{\theta_0} = \sqrt{2/9}$ uses the same quadratic Casimir $C_2 = 2$ of the T_1 representation that produces the Koide angle $\theta_0 = 2/9$, with the square root following from amplitude-vs-eigenvalue PT scaling between intra-doublet and inter-generation channels (§7.2). Operator counting on the first-generation Yukawas shows that the relation requires alignment between SU(2)-doublet structure and cubic-group T_1 perturbation that is bijection-specific (Layer 2(b)). The empirical match at 0.05σ indicates the framework's bijection has the alignment property; promoting to Layer 2(a) requires explicit substrate-level Yukawa derivation that has not been performed.

- [f] *Layered prediction (sin²θ_12, sin²θ_23), Layer 1 + Layer 2(b).* The Layer 0 component (TBM zeroth-order value) and Layer 1 component (λ^2 scale) are unconditional structural. The values follow given Condition 2 of §7.3 (the relation $b_{23} = (4/3)(b_{12} + b_{13})$ among A_2 Wilson coefficients). The §7.3 structural decomposition closes the A_1 component at Layer 1 (the TBM sum rule is structurally protected against the A_1 perturbation by the $\cos^2\theta_{13}$ dressing alone) and reduces the A_2 component to the substrate ratio $b_{23}/(b_{12} + b_{13}) = 4/3 = 4(d-1)/(2d)$ for $d = 3$, with the structural reading from the squared off-democratic projection of vector representations. This ratio was Layer 2(a)-tractable until its defining substrate one-loop calculation was performed (§7.3, "Status of Cond 2") and excluded the second-order route, demoting it to Layer 2(b); the 7/6 sum rule is its direct experimental test, and only the recorded $O(\lambda^3)$ construction remains as a possible derivation. The natural Higgs-democratic spurion does not produce this ratio (the natural form gives $b_{12} : b_{13} = m_\tau/m_\mu \approx 17 : 1$ from Layer 1 mass-matrix algebra applied to a generation-democratic substrate Yukawa, per §7.3 "Positive support"). The empirical match ($\sin^2\theta_{12}$ at 0.07σ to the post-JUNO global fit; sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ at 0.6σ) tests the bijection's specific T_2-channel structure. See §7.3 "Status of Cond 2: structural decomposition" and "Positive support: explicit substrate calculation."

Parameter-accounting summary: Of the 22 observables tabulated above, **7 are unconditional parameter-free structural predictions** (**S**, Layer 0 or Layer 1: Cabibbo λ, Wolfenstein A, $|V_{cb}|$, Koide Q, Koide angle θ_0, $\sin^2\theta_{13}$, $\lambda(M_{\rm Pl}) = 0$). Two further parameter-free structural results — the induced coupling $1/\alpha_0 = 23.25$ and the $A \cdot B$ product — belong to the same class but are *not* observables in this table: they enter the chain as structural inputs to the coupling sector (footnote [a]), and $A \cdot B$ is in any case leading-power-dependent and inconclusive pending larger lattices. Earlier statements of this accounting counted those two among "nine of the twenty-two"; the observables and the inputs are counted separately here. m_d/m_s is classified **L** because its conversion from the structural λ runs through the Gatto–Sartori–Tonin relation, an external empirical relation whose status the chain inherits (§7.1); the top mass is classified **R** because one-loop SM running gives $m_t \approx 207$ GeV from $y_t(M_{\rm Pl}) = 1$, so the 0.9% match is a retrodiction of the weak-scale $y_t \approx 1$ coincidence with the mechanism open (§7.2); **6 are layered predictions** (**L**, Layer 1 form combined with one Layer-2 substrate-specific coefficient or Layer-3 empirical input: m_d/m_s via GST as above, $\sin^2\theta_{12}$ and $\sin^2\theta_{23}$ via Cond 2 (Layer 2(b) per the §7.3 routing computation), m_u/m_d via amplitude-vs-eigenvalue alignment (Layer 2(b)), m_b inheriting $Q_{\rm down}$'s phenomenological status, m_b/m_τ via $K = 1/(n_{\rm gen}-1)$ representation-theoretic identity (Layer 1 closed per §7.5) with Layer 2(a)-tractable substrate-loop verification residual); 3 are retrodictions (**R**: the SU(2) and SU(3) couplings, where (A, B) are matched to observation rather than computed, and the top mass per §7.2); 1 fixes the universal threshold δ_0 from the U(1) coupling (**E**); 3 form a one-input mass/parameter chain (**M**); 1 uses an explicit empirical input (**E**, the Jarlskog phase η). No parameters are fit to the predicted quantities in any of the **S** or **L** classes; the layered classifications reflect explicit dependence on stated structural inputs (Cond 2 of §7.3 — Layer 2(a)-tractable; the open derivations referenced in §7.2 and §7.5) or phenomenological inputs ($Q_{\rm down}$) rather than free parameter fitting. The C class (conditional structural pending derivation) is currently empty.

In total, 9 unconditional + 6 layered = 15 predictions in the structural-or-near-structural class.

In addition: SM gauge group, three generations, $\bar\theta = 0$, Majorana neutrinos, normal mass ordering — all consistent with data.

Remaining open: (i) m_s (the overall mass scale — requires taste-decomposed Coleman–Weinberg potential); (ii) m_c (no clean structural relation found; the up-sector hierarchy involves top Yukawa backreaction); (iii) CP-violating phases (solution-specific); (iv) neutrino masses (solution-specific, but structurally constrained: Majorana, normal ordering).

7.7 The coincidence bound

The unconditional structural (**S**) and layered (**L**) retrodictions of §7.6 — fifteen in total — are not independent draws: several derive from the same cubic-group quadratic Casimir $C_2(T_1) = 2$ (the Koide amplitude $A^2 = C_2$, the Koide angle $\theta_0 = C_2/d^2$, $\sin^2\theta_{13}$, and the m_e, m_μ chain from m_τ), and several derive from the same one-loop lattice integral structure (Cabibbo λ, Wolfenstein A, $|V_{cb}|$, m_d/m_s). Conservatively treated, the **S** set contains *two* structurally independent prediction families at the tightest experimental precision: the Casimir family — anchored by the Koide relation $Q = 2/3$ via $A^2 = C_2$ — and the Cabibbo angle $\lambda = 1/(\pi\sqrt{2})$ (one-loop lattice integral on $\mathbb{Z}^3$). The two are tested by structurally distinct features of the lattice (Casimir eigenvalue structure vs. spatial integral). Each family carries an identified open link: the Casimir family carries the dynamical-closure caveat of §7.2 (the amplitude $A^2 = C_2$ is structural and exact but not yet derived from the dynamics, which at one loop undershoots), and the Cabibbo family carries the propagator-identification link of §7.1 (the identification of the mixing element with the unit-normalized taste-changing propagator), whose structural level §7.1 discharges as an algebraic identity of the staggered spin-taste map, leaving the emergent-theory dressing as its residual open element. Neither pillar should be ranked above the other pending its respective closure.

Cubic-group T_1-Casimir consistency test. Beyond the two-independent-predictions framing, the cubic-group T_1-Casimir structure $C_2 = 2$ for vector reps on the $d = 3$ lattice produces a six-prediction simultaneous-match consistency check:

Prediction	Formula	Predicted	Observed	Match
Wolfenstein A^2	$C_2/d = (d-1)/d = 2/3$	0.6667	0.683	1.2%
$\sin^2\theta_{13}$	$\propto A^4 = (C_2/d)^2 = 4/9$	0.0225	0.0224	0.4%
Koide θ_0	$C_2/d^2 = 2/9$	0.2222	0.2222	0.02%
m_u/m_d	$\sqrt{C_2}/d = \sqrt{2}/3$	0.471	0.474	0.6%
$\lvert V_{cb}\rvert$	$A\lambda^2$	0.04136	0.0408	0.4σ
PMNS sum rule	$7/6 = 1 + 2A^2/3$ via Cond 2	1.167	1.178	0.6σ

All six derive from the structural identity $A^2 = C_2/d$ for vector representations on $d = 3$ lattice, and all six match observation at sub-percent or sub-σ precision simultaneously. This is one successful test of the framework's cubic-group structure with six observables — methodologically a single-test framing rather than six independent successes — but the simultaneous coherence at this precision provides strong evidence for the underlying C_2/d identity.

The framework's empirical case therefore rests on two independent structural ingredients passing simultaneous tests: (a) Cabibbo $\lambda = 1/(\pi\sqrt{2})$ from BZ geometry, tested directly and through its appearances in $|V_{cb}|$, $\sin^2\theta_{13}$, and m_d/m_s via GST; and (b) cubic-group T_1-Casimir $C_2/d = 2/3$, tested via the six-observable simultaneous match above. Both ingredients are Layer 1.

On quantifying the joint match, and why no probability is quoted. A joint null probability of the natural form — each observed value treated as an independent uniform draw against a fixed formula — answers the wrong question for this package. The observed values were fixed and known before the framework existed — the Cabibbo angle since 1963, the Koide relation since 1981 (a celebrated pre-existing empirical regularity that the framework explains rather than discovers) — and the structural expressions were identified against them. The correct null must therefore account for the multiplicity of candidate structural expressions available to any such search, a multiplicity this paper itself documents: the normalization analysis of §7.2 enumerates ten candidate rational structures of comparable naturalness, the K coefficient of §7.5 is selected within $\{1/4, 1/2, 1, 2\}$, and the $1/d$-versus-$1/(2d)$ and $d = 3$-versus-$d = 4$ forks each expanded the candidate set during the framework's development. With tens of natural closed-form candidates per observable, per-mille agreement of *some* candidate is not a per-mille-probability event, and multiplying per-observable probabilities is not licensed. No honest closed-form null is available without an explicit model of the expression-search space, which is not attempted here.

What the retrodictive package supports. Three statements carry the section's content. First, the matches are *concentrated*: the two tightest (λ at 0.04%, θ_0 at 0.02%) sit an order of magnitude or more inside the spacing of their own candidate families as enumerated in §7.2, so formula-multiplicity alone does not comfortably absorb them — though this is a qualitative observation, not a probability. Second, the six-observable C_2/d simultaneous-consistency table above is a genuine single structural test with no per-observable freedom, and it passes; its evidential weight is that of one test, as already stated. Third, and decisively, the framework's exact, freedom-free forms convert into *pre-registered forward* exposure — the JUNO design-lifetime precision on $\sin^2\theta_{12}$, the DESI Y5 ν measurement, the lattice thresholds of the A^2 computation — and it is these forward tests, not any retrodictive probability, that carry the framework's empirical case. A reader should weight the retrodictive package as a compression claim (the Standard Model's flavor structure is expressible in cubic-lattice invariants to the stated precisions) and reserve confirmation-strength language for the forward set.

7.8 Forward predictions and the retrodiction-vs-prediction distinction

The coincidence bound of §7.7 quantifies how concentrated the framework's *retrodictive* content is. It does not, and cannot, address a distinct and more fundamental concern: every quantity in §7.1–§7.5 is a known value, and a framework developed with knowledge of its targets will tend to reproduce them whether or not its underlying structure is correct, because the development process has access to the answers. No internal analysis — no coincidence bound, no audit, no re-derivation — distinguishes "the structure forced these values" from "the known values guided the construction of the structure." Only a prediction of a value *not known at construction time* breaks this symmetry. This subsection separates the framework's genuinely forward content from its retrodictions and states it as such, because the forward content is the only part that bears on the retrodiction-vs-prediction distinction.

The JUNO precedent (a realized forward prediction). The solar mixing angle prediction $\sin^2\theta_{12} = 1/3 - 1/(4\pi^2) = 0.3080$ is the framework's one realized forward prediction to date, and its status as *forward* rather than retrodictive is a matter of public record. The framework's value was fixed by the cubic-group structure independent of the neutrino data; the JUNO first oscillation result (November 2025) and the subsequent post-JUNO global fit (Capozzi et al.) tightened the experimental central value onto 0.3085 ± 0.0073 — a 0.07σ agreement — *after* the structural value was set. The relevant epistemic fact is not the smallness of the σ-distance but the temporal order: the prediction did not move to meet the data; the data moved to meet a prediction whose value was determined by representation theory. This is the one place where the post-hoc-selection concern is directly answered rather than argued.

Registered forward predictions (not yet measured to testing precision). The same construction forces the following, stated here as pre-registered commitments whose values are fixed by the framework before the measurements that will test them:

1. *The PMNS sum rule* $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$. Currently confirmed at 0.6σ but at the precision of present atmospheric-angle data. A direct measurement of the combination at the 0.005 level — within reach of DUNE and T2HK — tests this at a precision that no current fit reaches. The framework commits to 7/6 exactly; a measured value displaced from 7/6 by more than experimental error falsifies the structural relation.
2. *The atmospheric angle* $\sin^2\theta_{23} = 1/2 + 1/(2\pi^2) = 0.5507$. The framework commits to a specific octant ($\theta_{23} > 45°$) and a specific value. Present data ($\sim 1.1\sigma$) does not resolve the octant; DUNE/T2HK will. An ultimate measurement in the lower octant, or displaced from 0.5507 beyond error, falsifies the prediction.
3. *The reactor angle* $\sin^2\theta_{13} = 4/(18\pi^2) = 0.02252$, unconditional structural (Layer 1). Already consistent with current data; tightening reactor measurements provide a continuing test against a value the framework cannot adjust.
4. *Continued JUNO precision on* $\sin^2\theta_{12}$. JUNO's 30-year design lifetime projects ± 0.0014 on $\sin^2\theta_{12}$, a $6\times$ improvement. The framework's 0.30798 is tested at sub-percent precision against a value it committed to before this precision exists; a 2σ ultimate disagreement falsifies it.

Distinction from a parameter-fitted account. A framework that accommodated the known mixing angles by fitting would retain latitude to absorb a future measurement that moved against it. The four predictions above have no such latitude: they are forced by the same cubic-group structure that produces the retrodictions, with no free parameter to adjust if the data moves. The retrodiction-vs-prediction concern of §7.7 is therefore not resolved by the retrodictions themselves; it is resolved, to the extent it can be resolved, by the JUNO precedent already realized and by the registered predictions above as they are measured. The framework's correctness is not internally decidable from the known-value matches; it is decidable, in either direction, by the forward predictions, which are stated in advance and carry fixed values.

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8. Discussion

The cosmological application — including the derivation of $\hbar$, the Bekenstein-Hawking area law, the dissolution of the cosmological constant problem, and the dynamical dark energy prediction — is developed in [GR]. The substratum-level reconstruction theorem and the synthesis claim that quantum mechanics, general relativity, and the arrow of time descend as three projections of the same (S, φ) are developed in [Substratum].

8.1 Structural realism

The structural reading aligns with ontic structural realism but does not require it. What the framework proves is that (S, φ) is a *complete description* of reality up to gauge equivalence (the reconstruction theorem, developed in [Substratum]). Whether the structure *is* reality (the OSR commitment) or *describes* a reality that exists independently is a question the framework identifies as gauge for physics — provably undecidable by any measurement an embedded observer can perform — while remaining potentially substantive in mathematical foundations or other modes of inquiry not bound by embedded physical observation. The "stuff" of the universe, in any reading, is fully characterized by the abstract structure of a bijection on a finite set with bounded coupling.

The structural-realism stance articulated here operates at one specific level of the framework's hierarchical structure: the equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$ within OI's specific universality class, identified in [Structure §2.3] as Level D $\times$ Level G3. The framework's broader structural realism extends to other levels — the universality-class level (Level C $\times$ Level G4) where partial-trace observational features are universal across structural classes, and the foundational levels (A and B) where the observation axiom and observer-admission criterion apply. The complete-description claim of this paper is class-specific: it asserts that within OI's universality class, (S, φ) modulo $\mathcal{G}_{\text{sub}}$ is the complete description. The full multi-level articulation is in [Structure §2.3].

8.2 Observer genericity

Theorem 22 (Genericity of observers). *Let φ be the wave equation (or any energy-conserving dynamics) on a connected bounded-degree coupling graph with diameter $D \geq 4$. Then for any connected subgraph V with $|V| \leq N/3$, the partition (V, H) satisfies C1–C3.*

Proof. C1: G_φ is connected and V is a proper subgraph, so ∂V $\neq \emptyset$. The wave equation couples nearest neighbors; any edge in ∂V produces non-trivial dynamical coupling.

C2: the wave equation is a Hamiltonian system — it conserves energy and preserves phase-space volume. The hidden sector has spectral gap $\Delta = 0$: correlations persist indefinitely, not just until some relaxation time. The C2 necessity proof [Main, §3.4] establishes that $\Delta\tau_S \gg 1$ drives the system to Markovianity;

for any Hamiltonian hidden sector, $\Delta\tau_S = 0 \ll 1$ — maximally in the slow bath regime ($\tau_B \to \infty$, the strongest possible satisfaction of $\tau_S \ll \tau_B$), regardless of partition geometry.

C3: $|H| = N - |V| \geq 2N/3$. The hidden-sector configuration space has $q^{|H|}$ states; the non-Markovian mutual information is bounded by $|H| \log_2 q$ bits [Main, §3.4]. The capacity ratio $q^{|H|-|V|} \geq q^{N/3}$ is exponentially large. No visible-sector process of sub-exponential duration saturates the hidden sector's memory. □

Remark (validity window). While C2 is satisfied maximally, the emergent Hamiltonian has a finite *stationarity window*: the time before information deposited at the V–H boundary propagates through H and returns to V at a different boundary point. The wave equation has an exact light cone (no exponential tails), so the minimum return time is $\tau_{\text{return}} = \min_{b,b' \in \partial V} \text{dist}_H(b, b')$. For core-shell partitions with core radius r on a lattice of side L: $\tau_{\text{return}} = (L - 2r)/v$. After τ_{return}, the emergent description remains quantum mechanical (C2 is still satisfied), but the emergent Hamiltonian becomes time-dependent. Numerical verification: for L = 20–320, τ_{return} ranges from 6 to 318 steps, confirming the light-cone bound.

Corollary. Any finite deterministic system with energy-conserving dynamics and bounded-degree coupling necessarily contains subsystems whose internal description is quantum mechanics. The observer is not postulated — it is a mathematical consequence.

8.3 What the framework determines and what the specific bijection determines

The framework's predictions fall into two categories, and the distinction matters for assessing their status.

Structural predictions are properties of the equivalence class $[(\text{S}, \varphi)]/\mathcal{G}_{\text{sub}}$ — they hold for every bijection satisfying the six structural properties. These include: SU(3) × SU(2) × U(1) with multiplicities (3, 2, 1), three chiral generations with identical gauge quantum numbers, one Higgs doublet (1, 2, +1/2), unique hypercharges from anomaly cancellation, $\hbar = \text{c}^3\varepsilon^2/(4\text{G})$, the Bekenstein-Hawking formula, and the universal induced coupling $1/\alpha_0 = 23.25$ (§6.1). These are theorems. They do not depend on which specific φ describes our universe.

Solution-specific properties are determined by the particular bijection φ within the equivalence class. These include: gauge couplings at M_Z (set by the eigenvalues μ_c, μ_w of M, the threshold δ_0, and the resummation parameters A, B — see §6); fermion mass *scales* (set by taste-breaking in φ at $\mathcal{O}(\epsilon^2)$ — though many mass *ratios*, including the Koide chain $m_e/m_\mu/m_\tau$, m_d/m_s, and m_u/m_d, are structural, see §7.2); basis-alignment unitaries between up-type and down-type Yukawa eigenstates (and hence the Jarlskog invariant, δ_{CKM} via the empirical η, and δ_{PMNS}); and individual Wilson coefficients of the cubic-invariant operators contributing to the Yukawa sector beyond whatever structural relations (such as Condition 2 of §7.3, a stated structural input whose second-order derivation route is excluded by the monomial-orthogonality result there, and which the 7/6 sum rule tests directly) constrain them. The CP phases are explicitly compatible with Theorem 21: the theorem establishes $\bar{\theta} = 0$ (the determinant-phase combination) but does not constrain the basis-alignment structure that produces the Jarlskog invariant (see §5.4 for the full scope statement).

The framework's predictions stratify into four layers, made precise here for downstream reference:

Layer 0 — Gauge / discrete structure. Properties of the equivalence class $[(S, \varphi)]/\mathcal{G}_{\rm sub}$ that follow from C1–C3 + cubic-group representation theory (§4.5–§4.7). Hold for every bijection. Examples: the gauge group $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ with multiplicities $(3, 2, 1)$, three chiral generations, anomaly-cancelling hypercharges, $\bar{\theta} = 0$, Majorana neutrinos with normal mass ordering, the TBM zeroth-order pattern ($\sin^2\theta_{12} = 1/3$ etc.), the Higgs as A_1 taste, the universal induced coupling $1/\alpha_0 = 23.25$ from $N_f = 6$ and $T(R) = 1/2$.

Layer 1 — Structural form of corrections. Properties of the *form* of the perturbative expansion around the zeroth-order structural prediction. Follow from cubic-group representation theory + uniqueness constraints. Hold for every bijection. Examples: the Cabibbo angle scale $\lambda^2 = 1/(2\pi^2)$ (Brillouin geometry, §7.1), the Wolfenstein $A = \sqrt{2/3}$ (off-democratic projection), the Koide angle $\theta_0 = C_2/d^2 = 2/9$ (cubic-group quadratic Casimir over bandwidth squared), the form $V_\ell = \exp(\lambda A_1 + \lambda^2 A_2 + \cdots)$, A_1's unique determination, $\sin^2\theta_{13} = (4/9)\lambda^2$ (independent of A_2 at $O(\lambda^2)$), the Higgs quartic boundary condition $\lambda(M_{\rm Pl}) = 0$, the universality of δ_0 across gauge groups (§6.3).

Layer 2 — Wilson-coefficient relations. Relations among the Wilson coefficients of cubic-invariant operators in the substrate Lagrangian. Three sub-cases:

- *Layer 2(a) — operator-level structural.* A finite family of cubic-invariant operators produces coefficient relations that hold for every bijection.
- *Layer 2(a)-tractable — sharply specified residual question.* A coefficient ratio reduces to a sharply specified structural quantity (with structural reading from representation theory) not yet derived from substrate dynamics, but with concrete substrate calculations available to test it. Promotes to 2(a) once the calculation is performed and confirms the ratio. The intermediate case between 2(a) and 2(b) (see "Diagnostic" below).
- *Layer 2(b) — solution-specific.* Coefficient values are free parameters of the bijection.

Cond 2 ($b_{23} = (4/3)(b_{12} + b_{13})$) admits the structural decomposition of §7.3: an A_1 component closed at Layer 1 (sum rule preserved by A_1 alone via $\cos^2\theta_{13}$ dressing) and an A_2 component whose ratio $b_{23}/(b_{12}+b_{13}) = 4(d-1)/(2d) = 4/3$

now sits at Layer 2(b): the sharply specified substrate calculation that defined its 2(a)-tractable period has been performed (§7.3) and excludes the second-order route, so the ratio is a solution-specific input unless the recorded $O(\lambda^3)$ construction is some day pursued and succeeds. The natural Higgs-democratic spurion does not produce this ratio (the natural form gives $b_{12} : b_{13} = m_\tau/m_\mu \approx 17 : 1$ from Layer 1 mass-matrix algebra applied to a generation-democratic substrate Yukawa, per §7.3 "Positive support"); the closure path is a one-loop staggered substrate calculation (§7.3 "Status of Cond 2"). Similarly, the intra-doublet vs inter-generation channel decomposition for $m_u/m_d = \sqrt{\theta_0}$ (§7.2) requires alignment between SU(2) Yukawa structure and cubic-group T_1 perturbation that is bijection-specific (Layer 2(b)). The $K = 1/2$ coefficient in the m_b/m_τ derivation (§7.5) follows by structural identity $K = 1/(n_{\text{gen}} - 1)$ *given* the matching condition $m_{\text{match}} = \lambda g_0^2$ (which is MC-verified to 1.4%); the residual Layer 2(a)-tractable question is whether the OI's specific taste-decomposed Coleman-Weinberg one-loop calculation derives this matching condition's specific dimensionless coefficient from first principles, rather than relying on MC verification of the matching scale.

Diagnostic for distinguishing Layer 2(a) from 2(b): When a Layer-2-looking ratio reduces to a structural identity for the relevant representation, it can promote to Layer 1. For example, $A^2 = (d-1)/d = C_2/d$ is a representation-theoretic identity for vector reps (§7.1), not a numerical coincidence; the off-democratic projection-squared and the Casimir-over-dimension ratio are the same quantity. This identity is what makes $\sin^2\theta_{13} = A^4\lambda^2$ Layer 1 structural, even though the calculation goes through PMNS matrix-element products. Conversely, when a Layer-2 ratio requires alignment between distinct operators or transfer between sectors, the natural classification is Layer 2(b) until the alignment is shown to be forced by substrate structure. The intermediate case — a ratio that reduces to a sharply specified structural quantity not yet derived from substrate dynamics, but with concrete substrate calculations available to test it — is Layer 2(a)-tractable. The Cond 2 ratio $b_{23}/(b_{12} + b_{13}) = 4(d-1)/(2d)$ was the canonical example until its defining calculation was performed (§7.3) and excluded the second-order route — demoting it to 2(b) and demonstrating the class's exit rule in the negative direction; the current 2(a)-tractable residents are the D11 mechanism, the D10 K-value, and the δ_0 threshold questions.

Sub-classifications of Layer 2(b) by mechanism. Within Layer 2(b), the framework distinguishes three mechanisms by which a coefficient becomes bijection-specific: (i) *single-parameter direction selection* — a single direction in some internal space is selected by the bijection (D11: which V_2-axis the Higgs condenses along determines m_u/m_d); (ii) *multi-parameter operator alignment* — multiple Wilson coefficients in distinct operator channels align in a specific pattern (Cond 2 was the canonical example before its §7.3 structural decomposition; the decomposition's residual ratio, briefly Layer 2(a)-tractable, has since been computed and demoted to 2(b); sub-classification (ii) currently has no Layer-2(b)-resident example); (iii) *cross-layer transfer* — a structural relation transfers between distinct group-theoretic layers (spatial vs internal T_1). The mechanism specificity is informative: single-parameter selection constrains the bijection more tightly than generic Wilson-coefficient ambiguity, and the empirical match becomes a tighter test of the bijection's structure.

Layer 2 values are discretization-determined, not free Wilson coefficients. The bijection φ is constrained by linear dynamics, T-invariance, P-indivisibility, and cycle ergodicity (cf. §4.2 *Remark*, §7.5). Within this constrained class, the bijection space is parametrized by *discretization-scheme choices* (field quantization, lattice size, timestep, boundary conditions, rounding rule), not by abstract permutation freedom. Once the discretization scheme is fixed, the bijection is essentially unique. As a consequence, Layer 2 coefficients (Cond 2 ratio at 2(b) status, D11 mechanism, D10 K-value, δ_0 threshold — all at Layer 2(a)-tractable or 2(b) status) are *all* functions of the same underlying discretization choices — they are not independent. This sharpens what Layer 2 means: the framework's empirical matches at sub-percent / sub-σ precision (Cond 2 at 0.6σ on the sum rule, m_u/m_d at 0.6%) are tighter tests of discretization scheme than they appear as tests of "generic bijection alignment." Cross-prediction consistency is therefore a meaningful test within structural clusters; the cubic-group T_1-Casimir cluster of §7.7 is one such successful test.

Selection problem. The framework characterizes a constrained class of substratum bijections (above) and derives empirical predictions from any element of this class. *Selection within the class* — which specific discretization scheme realizes our universe — is not addressed. This is the framework's analog of "why these laws of physics?" in standard physics, but the question is more pointed because the framework explicitly characterizes a meta-space (discretization-scheme space) within which the selection occurs. We treat this as an open question: the framework is a theory of what follows from a given discretization scheme, not a theory of which scheme is realized.

Derivation status: the three tiers. A recurring question about any framework that "derives the Standard Model from one object" is whether the derivation is genuine or whether the answer was quietly inserted in the construction — the tautology objection (stated in full at [Substratum §3, Remark on substantive content]). The objection does not admit a single yes/no answer, because the framework's results do not all have the same logical status. They separate cleanly into three tiers, and naming the tier is the honest form of the claim.

Tier 1 — forced by representation theory (genuinely non-tautological). Given the simple-cubic lattice and the wave equation, the following follow with no further freedom, by the character table of the octahedral group and standard anomaly analysis: the eigenvalue multiplicities $(3, 2, 1)$ and hence the gauge factors $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ (§4.5–4.6); the absolute generation count of three from the $K = 2d = 6$ minimizer (§4.7); the unique anomaly-free hypercharge assignment (§4.9); $\bar{\theta} = 0$ from T-invariance (§5); and the Cabibbo angle $\lambda = 1/(\pi\sqrt{2})$ from Brillouin-zone geometry (§7.1). These could *a priori* have come out otherwise — a different point group, a different multiplicity split, a different generation count — and the content of the derivation is that they do not. This tier is the framework's real non-tautological residue, and it is substantial. The Cabibbo prediction is the sharpest illustration: it is quantitative, it matches to 0.04%, and the competing cubic Bravais lattices (BCC, FCC) miss by 42% and 29% — a result that could have falsified the cubic commitment and did not.

Tier 2 — empirically selected (the inputs that do the forcing). Several quantities that the chain treats as fixed are in fact pinned by observation, not by the substratum: the spatial dimension $d = 3$ (argued at self-consistency strength by filters E5–E7 of §3.2 — propagating gravity requires $d \geq 3$ and stable matter requires $d \leq 3$, with the boundary-entropy concordance an independent, non-anthropic filter that also excludes $d \geq 4$ by equalling unity only at $d = 3$, and so substitutes for the anthropic stable-matter step in the circularity rebuttal of §3.2; the argument's declared exposure is that the filters consume d-dependent emergent physics, so this is a self-consistency selection rather than an axiomatic derivation, independently anchored empirically by the observed gauge group); the simple-cubic Bravais structure itself (Theorem 7b selects it *because* an SU(3) factor is observed; §4.6); the universal threshold δ_0 (fixed by the U(1) row, §6.2); and the absolute mass scales (Tier 3 / solution-specific inputs to the mass chains). For these the direction of inference runs from the observed world back to the commitment. In particular, "the Standard Model from a cubic lattice" and "the lattice is cubic because SU(3) is observed" are the same fact stated in two directions; the framework treats the cubic structure as a commitment forced by the observed gauge group rather than a free choice (§4.6), and that candor belongs in the headline, not only in a remark.

Tier 3 — definitional stipulation. Some load-bearing moves are choices about what counts as physical structure rather than facts the bijection forces: coupling-degree minimization as the *definition* of "site" and of the internal factorization (§2.4, §4.5 — a system with a different K corresponds to a different bijection, not to the same one factored differently, so the definition is principled but is still a definition); the nearest-neighbor restriction (flagged as an assumption beyond bounded coupling degree, §4.1 *Remark (scope)*); and, at the foundation, the C1–C3 architecture as the model of observational limitation ([Main §1.3]).

The upshot is that the framework is best described as a *maximally tight compression* of the Standard Model onto the cubic-lattice commitment: Tier 1 shows that, once the commitment is granted, the SM's discrete structure is largely not independent — a single representation-theoretic object — which is a genuine and unusual claim about the unity of physics; while Tiers 2–3 show that the commitment itself is empirically anchored and partly stipulated, so the framework is not deriving physics from a parameter-free first principle. The tautology question is therefore not "open pending computation" — no lattice calculation moves it, because the status of each link is already determined by the construction as written. It is resolved by this classification: non-tautological at Tier 1, tautological-in-the-relevant-sense at Tier 2, definitional at Tier 3.

Organizing principle behind the tiers: constitution versus consequence. The split between what lands in Tier 1 (derived) and what lands in Tiers 2–3 (input or stipulated) is not arbitrary; it tracks the distinction between the substratum's *constitution* and its *consequences*. Tier 1 is exactly the set of consequences that follow, by representation theory of the cubic group on the link directions together with the trace-out, once the substratum is fixed. Tiers 2–3 are the constitution itself — the spatial dimension, the Bravais type, the rotational isotropy, the periodicity, the definition of a site. The rule is that the framework derives the consequences of a constitution but not the constitution, with one qualification on tractability grounds: a consequence the framework cannot yet compute is also booked as input (the threshold δ_0 of §6.2 is a consequence of the coupling-running held as input only because it is not yet derived, not a constitutional feature). So the inputs comprise the constitution together with any not-yet-tractable consequences, and the derived results are the tractable consequences. The structural reason an embedded observer can derive consequences but not constitution — and the reason the boost sector of §3.1 is more tractable than the rotational sector — is developed in the book's open-problems chapter (Ch. 19 §19.3.7): the observation map is realized at a horizon that selects a rest frame, so it makes contact with the time-distinguishing (boost) structure but not with any spatial-direction-distinguishing (rotational) structure, the latter being pure constitution. This is an organizing observation about the derivability boundary, not a formal theorem; notably, the representation-theoretic Tier-1 results would follow for any analyst of the fixed lattice without reference to an observer, so derivability is not claimed to coincide with any single relational property.

Formal statement of the determination (Proposition). The contentful half of the organizing principle can be stated precisely. **Proposition.** *Fix the substratum constitution C — the spatial dimension, the Bravais type, the wave-equation dynamics with its nearest-neighbor and linearity stipulations, and the alphabet — and the trace-out map T. Then the emergent gauge group, the fermion generation count, the anomaly-free hypercharge assignment, the vanishing of $\bar{\theta}$, and the Cabibbo angle are each uniquely determined as functions of (C, T)* — by the octahedral representation theory of §4.5–4.7 together with the chiral/taste structure of §5 and §7. This is the conjunction of the structural results already proved above, restated as a single determination claim; it carries no new proof burden. What it deliberately does *not* assert is that C is underivable or that "derivable $\Leftrightarrow$ consequence": it states only that the listed Tier-1 quantities are functions of (C, T). The epistemic observation that C is *input* to the framework is separate, and is not part of the proposition.

Why the constitution/consequence split is a definition with a census, not a theorem. The biconditional "derivable $\Leftrightarrow$ consequence-of-constitution" is not a theorem in either reading: under the natural definition of "consequence" (that which is derivable from C) it is tautological and contentless, and under any observer-independent definition of "consequence" it is false — the threshold δ_0 is an independently characterizable consequence of the coupling-running that is nonetheless held as input, on tractability grounds. The contentful objects are therefore (i) the *census* — the explicit assignment of each quantity to constitution or consequence, with the intractable-consequence exceptions (δ_0, the absolute mass scales) flagged — and (ii) the rest-frame-selection lemma of [Main §3.5], which explains the one structurally interesting case, the boost/rotation asymmetry, via the fact that the observation map selects a timelike direction but no spatial one. The honest summary is that the framework derives the tractable consequences of a constitution it must otherwise take as given, and that the boundary between the two is organized — not by a single formal property, but by this constitution/consequence distinction together with the rest-frame lemma that explains its one non-obvious instance.

Relation to the framework's hierarchy of structural realism. The Layer 0–Layer 3 classification refines the within-paper structure of which predictions are derivable from the cubic-lattice substratum and which are bijection-specific. This is a finer-grained classification than the framework's broader observation $\times$ gauge hierarchy (per [Structure §2]). In hierarchy terms, all SM predictions live at Level D $\times$ Levels G1–G2 — they are predictions of OI's specific universality-class representative on the emergent QFT. Within this single intersection, Layer 0–Layer 3 distinguishes which predictions are class-specific structural theorems (Layer 0 / Layer 1) vs. which depend on the specific representative within the class (Layer 2 / Layer 3). The hierarchy framing locates this paper's predictions within the framework's overall geography; the layered framing distinguishes the derivability status of each prediction within that location.

Rigor-level discipline. Closure claims in the framework are accompanied by explicit rigor levels: Level 1 (verified at the level of an explicit chain of equations, suitable for referee submission), Level 2 (verified up to standard moves of the field, where each step is "obvious to anyone in the field" without being written out), or Level 3 (sketch identifying the right approach, *not closure*). Multi-part theorems may have different rigor levels for different parts. The bridge theorem of §7.5 is an example: parts (a) and (c)-direct are Level 1 closed; parts (b)-γ_5 and the framework-essential γ_5-chirality identification are Level 1 closed; parts (b)-γ_0-derivation and (d) are Level 2; the full operator-theoretic interpretation involving taste structure is Level 3 but is not framework-essential. Honest annotation of rigor level is part of the framework's methodology.

Layer 3 — Solution-specific properties. Properties that depend on the particular bijection φ. Examples: absolute fermion mass scales (e.g., m_τ, m_t, m_s as input scales in the M-chain), basis-alignment unitaries, CP-violating phases (δ_{CKM} via empirical η, δ_{PMNS} identified with $b_{12} - b_{13}$ in §7.3), eigenvalues μ_c, μ_w of the coupling matrix M that set the gauge couplings.

The §7.6 summary table classifies each prediction by its layered status: items marked **S** are unconditional structural (Layer 0 or Layer 1, fully derived); items marked **L** are layered (combine multiple layers, typically Layer 0 or 1 + Layer 2(b) substrate-specific coefficient or Layer 3 empirical input). Items marked **R**, **P**, **M**, **E** denote retrodictions, phenomenological inputs, one-input mass/parameter chains, and explicit empirical inputs respectively, as detailed in §7.6. The C class (conditional structural pending derivation) is currently empty:

Cond 2 — the only candidate for C in v1.9.5 — has been classified Layer 2(b) following operator counting, moving the dependent predictions to L; the §7.3 routing computation subsequently confirmed this classification by excluding the second-order derivation route.

These are analogous to the mass M in the Schwarzschild solution: GR derives the functional form of the metric but does not predict which M describes a particular black hole. The OI framework derives the SM's structure but does not predict which φ describes our universe.

The "mass hierarchy" — the five-order-of-magnitude spread of fermion masses — falls in the second category. It is not a gap in the framework; it is a property of the specific solution. The framework makes the correct structural prediction: three generations with *identical* gauge quantum numbers (from cubic symmetry) but *non-degenerate* masses (because the specific φ breaks the cubic symmetry through its Yukawa structure). The hierarchy problem proper — why the Higgs mass is so much lighter than the Planck scale — is dissolved: the Higgs mass is an eigenvalue of M (a property of φ), not a radiative correction within the emergent QFT (§4.11).

Remaining open problems. (i) *Lattice-continuum comparison* — the structural predictions are exact theorems at the lattice level and do not require a continuum limit. The lattice is the fundamental description ($\epsilon = 2l_p$ is physical, not a regulator), so there is no $\epsilon \to 0$ limit to take. The comparison to experiment — which is interpreted through continuum perturbation theory — introduces corrections of order $(E/M_{\rm Pl})^2 \sim 10^{-32}$ at LHC energies, far below any foreseeable experimental sensitivity. The Yang-Mills mass gap problem (proving rigorous existence of the continuum limit for Yang-Mills theory) is a famous open problem in mathematics, but it does not affect any prediction of the OI framework: the lattice theory is complete as stated, and the lattice is the fundamental description, not a regularization. The same $(E/M_{\rm Pl})^2$ suppression sets the framework's structural commitment on Lorentz-invariance violation: linear LIV is forbidden at the substrate level (the cubic-lattice dispersion relation admits no $\mathcal{O}(k^3)$ term), and the gauge-invariant quadratic coefficient evaluates to $\omega^2 = k^2[1-(4/45)(E/M_{\rm Pl})^2+...]$ — subluminal, with the specific 4/45 fixed by cubic coordination and the statistical-isotropy gauge average. Pre-registered as Class A in [Substratum, §6.4].

8.4 Neutrino masses from taste-breaking

The staggered fermion construction gives 4 tastes decomposing as $1(A_1)+3(T_1)$ under the cubic group O (§4.7). The singlet taste is identified as the Higgs sector — not a right-handed neutrino. Neutrinos therefore acquire mass only through the dimension-5 Weinberg operator $(LLHH)/\Lambda$, making them Majorana particles.

The mass matrix. The neutrino mass matrix in the generation basis $\{e_x, e_y, e_z\}$ (the T_1 representation) is $M_\nu = m_0 I_3 + \delta M$, where δM is the taste-breaking correction. Under the cubic group, the symmetric matrix δM decomposes as $\mathrm{Sym}^2(T_1) = A_1 \oplus E \oplus T_2$. The A_1 part shifts all masses equally (absorbed into m_0). The E part (2-dimensional) controls the mass splittings via two parameters (a, b). The T_2 part (3-dimensional) controls the mixing angles.

Structural predictions (Tier 2). Four predictions follow from the representation theory alone, independent of the specific φ:

(i) *Majorana neutrinos.* No right-handed neutrino exists in the spectrum — the singlet taste is the Higgs. Neutrinoless double beta decay ($0\nu\beta\beta$) should be observed.

(ii) *Normal mass ordering.* The taste-breaking in T_1 generically gives the largest correction to the state most coupled to all three lattice directions — the [111] direction in taste space. This corresponds to $m_3 > m_2 > m_1$.

(iii) *Large PMNS mixing angles.* The charged lepton mass matrix (Dirac, general 3×3) and the neutrino mass matrix (Majorana, symmetric 3×3) have different symmetry structures under the cubic group. The structural mismatch produces large angles — unlike the quark sector, where both up-type and down-type matrices are general 3×3 with similar dominant structure, giving small CKM angles.

(iv) *Hierarchical spectrum.* The taste-breaking splittings are the same order as the common mass m_0. A quasi-degenerate spectrum ($m_1 \approx m_2 \approx m_3 \gg \Delta m$) requires fine cancellation between m_0 and the splittings, which is not generic. The framework naturally produces Σm_ν near the minimum allowed by the mass splittings.

Observational status. DESI DR2 combined with CMB data gives $\Sigma m_\nu < 0.064$ eV with preference for normal ordering and $m_{\mathrm{lightest}} < 0.023$ eV — consistent with all four structural predictions. The minimum sum for normal ordering is $\Sigma m_\nu^{\mathrm{min}} \approx 0.059$ eV; the DESI+CMB bound is only 8% above this minimum, consistent with the hierarchical prediction. The observed PMNS angles ($\theta_{12} \approx 34°$, $\theta_{23} \approx 49°$, $\theta_{13} \approx 9°$) are all large compared to CKM angles, as predicted.

Solution-specific properties (Tier 3). The absolute mass scale (set by $\Lambda_{\mathrm{seesaw}}$), the ratio $\Delta m^2_{32}/\Delta m^2_{21} \approx 33$ (set by the relative magnitudes of the E-component parameters a/b), and the CP-violating phase δ_{CP} (which corresponds to the unconstrained free parameter $b_{12} - b_{13}$ in the $O(\lambda^2)$ charged-lepton structure of §7.3) are properties of the particular φ. The PMNS *angles* themselves are structural (§7.3): TBM from cubic-group representation theory plus U-parity-graded corrections from the charged-lepton sector, with the Cabibbo angle λ as the structural scale.

8.5 Position relative to mainstream observer-essentiality programs

The Standard Model derivation chain developed in this paper sits within a broader framework whose foundational claim — that observation cannot be treated as an external operation on physics — has been independently arrived at by several mainstream programs since 2022. Chandrasekaran-Longo-Penington-Witten [CLPW 2022, JHEP 02 (2023) 082] showed that incorporating an observer in QFT in a gravitational subregion promotes the von Neumann algebra of observables from type III to type II_∞. Maldacena's recent dS observer cancellation [arXiv:2412.14014] removes the unphysical phase of the de Sitter sphere partition function exactly when an observer-with-clock is properly incorporated. Harlow-Usatyuk-Zhao [JHEP 02 (2026) 108] and the AAIL construction [arXiv:2501.04305] show that the closed-universe Hilbert-space dimension is determined by the observer's degrees of freedom, with refined scaling $d_{\mathrm{Ob}} \cdot e^{2S_0}$. Slagle-Preskill [Phys. Rev. A 108, 012217 (2023)] demonstrated emergent quantum mechanics at the boundary of a classical lattice model, establishing the logical possibility of QM-from-classical at the cost of an exponentially long extra spatial dimension and explicit stochasticity. The present framework is the unique member of this convergence in which the observer-essentiality content is derived from a finite *deterministic* substratum — no extra spatial dimension, no fundamental stochasticity — and in which the resulting characterization theorem produces an explicit empirical record (the twenty-two structural predictions of §7) that the comparison programs do not currently match. The convergence positions the framework's foundational stance within mainstream observer-emergent physics; the SM derivation chain provides the specific empirical content that distinguishes it. The full positioning relative to this body of work is developed in [Main, §4.4].

8.6 Pattern vs. output: what transports across SM-reproducing classes

The derivation in §4 establishes the SM gauge group $SU(3) \times SU(2) \times U(1)$ as the unique commutant of the cubic-group action on the $K = 6$ internal components, with multiplicities $(3, 2, 1)$ from the cubic-group representation theory (Theorem 7) and background independence (A6) promoting the global commutant symmetry to local gauge invariance. The derivation has two structural layers worth distinguishing for cross-framework purposes.

The *commutant gauge-invariance pattern* — that the visible-sector gauge group is the commutant of a substratum-level symmetry under the trace-out — operates at the universality-class level (Level G4 $\times$ Level C of [Structure §2]). Per the four-feature audit in [Structure §14.4], this pattern transports to other SM-reproducing structural classes through different machinery: heterotic string compactifications realize it through E_8 broken by Calabi-Yau holonomy and bundle data; Type II / D-brane compactifications realize it through D-brane stack gauge groups with chiral matter from intersection or magnetization; F-theory realizes it through singularity structure of elliptic fibrations with hypercharge from flux configurations. The pattern is shared across all SM-reproducing universality-class members; the framework-specific machinery differs.

The *specific output* — the SM gauge group precisely, with the multiplicities $(3, 2, 1)$, the unique anomaly-free hypercharges, three chiral generations, and one Higgs doublet — is OI-specific (Level D $\times$ Level G2 of [Structure §2]). This paper's derivation forces the specific output uniquely from the cubic-group commutant on $K = 6$; in other universality-class members, the same output is produced through different group-theoretic structure, with the uniqueness depending on framework-specific structural conditions analogous to OI's A1–A6.

The distinction matters for what falsification of the framework would refute. Falsifying the commutant gauge-invariance pattern (e.g., demonstrating an SM-reproducing substratum where the gauge group does *not* arise as a commutant of substratum symmetry) would refute the partial-trace structural-realism claim that the pattern is universality-class-level. Falsifying the specific cubic-group derivation while preserving the commutant pattern (e.g., demonstrating that the SM gauge group can be reached through structurally simpler routes than cubic-group commutant on $K = 6$) would refute OI's specific representative within the SM-reproducing universality class without refuting the pattern. The twenty-two quantitative predictions of §7 are at the second level — they depend on OI's specific cubic-group structure, not on the commutant pattern alone.

8.7 Substratum-level baryon-number conservation and the substratum-emergent operator distinction

The framework's matter sector (§4.7) places fermions on links with carrier $V_K \otimes V_K^*$, where the V_3 factor distinguishes quarks (with a V_3 component) from leptons (without). At the substratum level, baryon number B is essentially the count of V_3 factors weighted by $1/3$. This subsection shows that B is exactly conserved by the substratum dynamics, derives the consequences for the emergent EFT, and articulates the substratum-emergent operator distinction that organizes the framework's treatment of non-perturbative SM features.

Substratum-level B conservation. Two structural features of the substratum dynamics force exact B conservation:

(i) *Bilinear coupling structure.* The lattice wave equation couples each link's field $\phi(\mathbf{n})$ to its neighbor $\phi(\mathbf{n}+\hat{e}_j)$ via the bilinear $\bar{\phi}(\mathbf{n})M(\mathbf{n}, \hat{e}_j)\phi(\mathbf{n}+\hat{e}_j)$. Bilinear couplings can move excitations between sites and produce particle-antiparticle pairs (preserving B) but cannot change the *count* of fermions in each gauge-charge sector through unilateral creation or annihilation.

(ii) *Block-diagonal structure of M.* By Theorem 7, the coupling matrix M is block-diagonal in the $V_3 \oplus V_2 \oplus V_1$ decomposition: $M = \mathrm{diag}(\mu_c I_3, \mu_w I_2, \mu_y)$. Off-block matrix elements that would mix V_3 with V_1 are absent; the dynamics therefore never converts a quark-like state (V_3 component) into a lepton-like state (V_1 component) directly.

These two features jointly imply that the substratum dynamics preserves the count of V_3-fermion factors modulo pair production, which is the substratum-level statement that B is exactly conserved *in the gauge/kinetic sector*. Baryon violation through non-gauge operators — the condensate and Yukawa sectors, and higher-dimension operators — is a separate accounting item, not excluded here.

The substratum-emergent operator distinction. The Standard Model has $B + L$ violation through non-perturbative sphaleron processes, finite at high temperatures and exponentially suppressed at zero temperature [9]. This appears to conflict with substratum-level exact B conservation. The resolution lies in recognizing that B at the substratum level and B at the emergent EFT level are distinct operators related by the trace-out:

(a) *Substratum-level* B is the count of V_3-fermion factors on the cubic lattice. This is exactly conserved by the bilinear, block-diagonal substratum dynamics, as shown above.

(b) *Emergent-level* B is the standard SM operator on coarse-grained fermions, which is acted on by sphaleron processes in the emergent quantum field theory. Sphalerons are emergent classical solutions of the electroweak Lagrangian; they exist as non-perturbative features of the emergent EFT but have no direct substratum-level analog.

The trace-out from substratum to emergent EFT introduces non-perturbative quantum-mechanical phenomena (instantons, sphalerons, θ-vacuum structure) that the deterministic substratum dynamics does not have. The induced gauge action $S_{\text{eff}}[U] = -N_f \ln\det(D^\dagger D)$ inherits topological structure from the fermion determinant via the Atiyah-Singer index theorem on the lattice; configurations with non-trivial Chern-Simons number produce the standard sphaleron and instanton physics in the emergent EFT.

Substratum-level B conservation and emergent-level $B + L$ violation therefore coexist consistently: the trace-out maps the conserved substratum operator onto an emergent operator that is conserved at the bilinear renormalizable level but violated by emergent non-perturbative effects.

Generalization to other non-perturbative SM features. The substratum-emergent operator distinction generalizes to a systematic mapping. The substratum has exact discrete symmetries (T-invariance, gauge-invariance, block-diagonal coupling, bipartite checkerboard structure) at the bilinear level. The trace-out introduces emergent non-perturbative effects (sphalerons, instantons, chiral anomalies, θ-vacuum structure) through standard mechanisms operating in the emergent QFT. The mapping has the following pattern across the framework:

Substratum-level structural precursor	Emergent-level non-perturbative feature	Trace-out role
T-invariance of the wave equation (§5)	$\theta = 0$ in QCD	Forces parameter value
Topology of induced SU(2) gauge action (via fermion-determinant index)	Sphalerons, $B + L$ violation	Permits standard mechanism
Bipartite checkerboard (Theorem 12)	Chiral fermions, anomaly structure	Produces emergent structure
Block-diagonal M (Theorem 7)	Substratum B conservation	Inherits to emergent B at bilinear level
Cubic-commutant gauge structure + matter content (§4.7.1.2)	Anomaly cancellation	Forced by structural derivation
Chiral structure of substratum + commutant matter content (Theorems 13, 15)	't Hooft anomaly matching	Structurally automatic
Fermion determinant + induced gauge action (§6.1)	Anomalous dimensions, non-perturbative renormalization	Determined by fermion-determinant structure (matches standard QFT at one- and two-loop; higher-order open)
Induced gauge action with running coupling (§6.1)	Conformal/trace anomaly $T^\mu_\mu \propto \beta(g) F^{\mu\nu} F_{\mu\nu}$	Standard form, emerges automatically from induced gauge structure
Topology of induced SU(3) gauge action (instantons)	't Hooft determinant interaction; η' mass via Witten-Veneziano	Standard mechanism (open derivation in framework)

The framework's treatment of the strong CP problem (§5: substratum T-invariance forces $\theta = 0$) is one instance of this pattern: substratum-level symmetry forces a value of an emergent parameter that would otherwise be free. The substratum-level B conservation derived above is another instance: substratum-level structural feature provides exact conservation that maps to bilinear-renormalizable conservation at the emergent level, with non-perturbative violations through emergent processes only.

Empirical implications. The composite picture supports the framework's existing predictions and clarifies their structural origins:

(i) *Proton decay rate.* With substratum-level B conservation and emergent-level B violation only through Planck-suppressed dimension-6 operators (since OI has no GUT scale per §6.7), proton lifetime is $\tau_p \sim M_{\rm Pl}^4/m_p^5 \sim 10^{45}$–$10^{46}$ years.

(ii) *Baryogenesis mechanism is open.* Sphaleron-mediated $B + L$ violation operates normally in the emergent EFT. But the framework's spectrum contains **no right-handed neutrino** (§4.9, §8.4: the singlet taste is the Higgs, and anomaly cancellation forces exactly five representations per generation), so standard type-I thermal leptogenesis — which requires heavy right-handed Majorana neutrinos decaying out of equilibrium — does not directly apply. The lepton asymmetry must instead arise from whatever ultraviolet physics generates the dimension-5 Weinberg operator (§8.4), which the framework does not currently fix. The full quantitative baryogenesis prediction — connecting that scale to the observed baryon asymmetry $\eta = n_B/n_\gamma \approx 6 \times 10^{-10}$ — is open work.

(iii) *Cleanness of anomaly structure.* The substratum-derived matter content (§4.7.1.2 + Theorem 15) automatically satisfies all SM anomaly conditions; anomaly cancellation is a derived consequence rather than an imposed consistency condition.

(iv) *Strong CP, sphalerons, instantons coexist.* The framework permits all standard non-perturbative SM phenomena while retaining exact substratum-level discrete symmetries; the trace-out is what allows this to be consistent.

(v) *'t Hooft anomaly matching.* The framework's chiral structure derivation (Theorem 13) and anomaly cancellation derivation (Theorem 15) together imply 't Hooft anomaly matching automatically: the chiral anomalies of the emergent SM fermions match those required by the substratum's bipartite-lattice plus commutant structure. This is not an added constraint but a structurally automatic feature, consistent with the framework's broader pattern of deriving anomaly cancellation rather than imposing it.

(vi) *Open work in the non-perturbative sector.* Three features in the extended table are flagged as open: higher-loop anomalous dimensions matching to all orders (the framework's matching at M_Z works to few-percent at one and two loop; higher-order non-perturbative effects could differ from standard QFT); the trace anomaly's explicit derivation from induced gauge action (likely automatic but not verified); the η' mass via 't Hooft determinant interaction in OI's induced theory (standard mechanism applies but the explicit derivation has not been worked through in the framework).

The substratum-level discrete symmetries (T-invariance, gauge invariance, block-diagonal coupling, bipartite checkerboard structure) and the emergent-level non-perturbative effects (sphalerons, instantons, chiral anomalies, θ-vacuum structure) coexist consistently under this mapping. The substratum operators are exactly conserved at the bilinear level; the emergent operators inherit the conservation modulo non-perturbative violations that follow from standard QFT mechanisms operating in the emergent EFT.

8.8 Composite-Higgs chiral protection and the unified naturalness pattern

The framework's treatment of three open problems of the Standard Model — strong CP, the hierarchy problem, the cosmological constant — exhibits a common structural pattern: each is dissolved by a substratum-level structural feature that forces an emergent parameter value which would otherwise be naturally at the cutoff. This subsection articulates the pattern and develops its hierarchy-problem instance (composite-Higgs chiral protection) which extends the framework's existing treatment of strong CP (§5) and the cosmological constant ([GR §6]).

The unified naturalness pattern. Standard quantum field theory has three classical naturalness problems where dimensionless parameters of the SM should naturally be $O(1)$ at the cutoff but are observed to be many orders of magnitude smaller: (i) the strong CP angle $\bar{\theta}$ is observed to be $< 10^{-10}$ but the natural cutoff value is $O(1)$; (ii) the Higgs mass-squared ratio m_H^2/M_{Pl}^2 is observed to be $\sim 10^{-34}$ but the natural cutoff value is $O(1)$; (iii) the cosmological constant ratio $\Lambda/M_{\mathrm{Pl}}^4$ is observed to be $\sim 10^{-122}$ but the natural cutoff value is $O(1)$. In standard SM, each requires either fine-tuning or anthropic selection or new physics (axion, supersymmetry, dark energy mechanism) to resolve.

OI's framework dissolves all three through the same mechanism: a substratum-level structural feature (discrete symmetry, conservation law, or geometric constraint) forces the emergent parameter to a specific value that escapes the naturalness puzzle. The mechanism is one instance per problem:

(i) *Strong CP* (§5, Theorems 17–21): substratum-level T-invariance of the wave equation combined with detailed balance in the emergent Hamiltonian forces $\bar{\theta} = 0$ at the kinematic level of the construction. The structural feature is T-invariance; the protected emergent parameter is $\bar{\theta}$. This instance of the pattern is conditional in the sense of §5.5, and is the weakest of the three.

(ii) *Hierarchy problem* (developed below): the staggered-taste chiral symmetry pairing $\mu = +1$ singlet with $\mu = -1$ singlet provides composite-Higgs-style chiral protection. The structural feature is the singlet-taste chiral symmetry; the protected emergent parameter is m_H/f where f is the compositeness scale.

(iii) *Cosmological constant* ([GR §6]): the 10^{122} discrepancy between Λ_{obs} and M_{Pl}^4 is dissolved as the trace-out's information compression ratio between the substratum's full information content and the observer-accessible coarse-grained projection. The structural feature is the cosmological-horizon partition; the protected emergent parameter is

$\Lambda/M_{\rm Pl}^4$.

The three are not three separate fine-tuning miracles but three instances of the same pattern: *substratum-level structural symmetries dissolve naturalness problems by forcing emergent parameter values that would otherwise be free.* This is a substantive structural feature of the framework that has not been articulated as a unified theme.

Hierarchy-problem instance: composite-Higgs chiral protection. The framework predicts the Higgs as a composite scalar in the singlet staggered taste ([Theorem 11]). Standard composite-Higgs scenarios in beyond-the-SM phenomenology (Kaplan-Georgi 1984; Agashe-Contino-Pomarol 2005) dissolve the hierarchy problem by realizing the Higgs as a pseudo-Nambu-Goldstone boson of an approximate global symmetry, with mass protected by Goldstone's theorem against cutoff-scale corrections. OI's framework structurally implements an analog of this mechanism.

The relevant symmetry is the *singlet-taste chiral symmetry*. The framework's staggered-fermion construction (§4.8) produces four staggered tastes labeled by the lattice momentum eigenvalues $\mu \in \{+1, +1/3, -1/3, -1\}$. The chirality structure (Theorem 13) pairs the $\mu = +1$ singlet taste with the $\mu = -1$ singlet taste through the bipartite checkerboard. At the substratum level, the symmetry pairing these tastes is *exact* — a discrete chiral symmetry of the wave equation acting on the singlet-taste sector. In the emergent EFT, this symmetry is *spontaneously broken* by the Higgs vev, which provides asymmetric mass contributions to the $\mu = +1$ and $\mu = -1$ singlet tastes.

The Higgs is the order parameter of this spontaneous breaking. Concretely, the Higgs lives in the E projection of the singlet taste ([Theorem 11]), which is a 2-dimensional complex irrep of the cubic group. Under the chiral symmetry, the E projection transforms in a specific way that makes the Higgs's vev the natural order parameter for breaking the chiral pairing.

By Goldstone's theorem, the Higgs mass is bounded by the *explicit* breaking of the chiral symmetry — that is, by gauge couplings and Yukawa couplings squared times the chiral-breaking scale, not by the cutoff $M_{\rm Pl}$. The hierarchy problem is dissolved at the level of *cutoff sensitivity*: the Higgs mass-squared does not receive $O(M_{\rm Pl}^2)$ corrections, only $O(g^2 f^2/16\pi^2)$ corrections where g is a gauge or Yukawa coupling and f is the chiral-symmetry-breaking scale.

Scope and caveats. Chiral protection bounds the Higgs mass by explicit breaking but does not automatically explain why the Higgs vev v sits 17 orders of magnitude below the compositeness scale f. In OI, f is set by the substratum lattice scale and is effectively $M_{\rm Pl}$, so the ratio $v/f \sim 10^{-17}$ remains. Standard composite-Higgs scenarios put f at $\sim$ TeV to resolve this; OI does not have an analogous intermediate scale (per [Structure §13.3] Q10). The full hierarchy resolution therefore consists of two parts: (a) chiral protection against cutoff corrections (developed here), and (b) the structural origin of $v \ll f$ (open).

Honest characterization: the framework structurally implements (a). For (b), the framework currently treats v as solution-specific (Tier 3) input, with the small value $v \sim 246$ GeV being a feature of the specific bijection φ rather than a derived structural constant. This is consistent with the framework's Tier 3 honesty about parameters not forced by structure ([Structure §13.3] Q10's negative resolution). Possible structural origins for (b) — including trace-out-induced fractional-power suppression connecting the Higgs hierarchy to the cosmological-constant compression — are speculative directions for future investigation; the framework does not currently commit to them.

Linkage to the seesaw scale. The same singlet-taste chiral symmetry breaking that gives the Higgs its mass also breaks the lepton-number-like pairing in the singlet sector. The emergent Weinberg operator $(LH)(LH)/\Lambda_{\text{seesaw}}$ that generates Majorana neutrino masses (§8.4) has its suppression scale Λ_{seesaw} structurally linked to the same chiral-breaking scale. The seesaw hierarchy ($\Lambda_{\text{seesaw}}/M_{\text{Pl}} \sim 10^{-3}$) and the Higgs hierarchy ($v/M_{\text{Pl}} \sim 10^{-17}$) are therefore not independent — both should be derivable from the same trace-out-induced mechanism, with the dimension-dependent suppression giving different effective scales for different operators (dimension-3 Majorana, dimension-4 Higgs, dimension-5 Weinberg). The quantitative relationship is the generic Weinberg-operator estimate $\Lambda_{\text{seesaw}} \sim v^2/m_\nu \sim 10^{16}$ GeV — *not* a type-I seesaw, since the framework's spectrum contains no right-handed neutrino to integrate out (§4.9, §8.4); Λ_{seesaw} is the suppression scale of the dimension-5 Weinberg operator, whose ultraviolet completion the framework does not fix. Whether this scale is structurally linked to the Higgs sector's chiral-breaking scale or is a solution-specific (Tier 3) input is not settled: [Structure §13.3] Q10 finds no derived intermediate scale, so it is currently treated as a Tier-3 input (§8.4).

Empirical signatures. Standard composite-Higgs scenarios predict modified Higgs couplings to SM particles by factors of $\sqrt{1 - v^2/f^2}$ where f is the compositeness scale, with deviations from SM scaling as v^2/f^2. In OI, with $f \sim M_{\text{Pl}}$, $v^2/f^2 \sim 10^{-34}$ — far below any conceivable experimental sensitivity. The framework's prediction is therefore no observable deviations from SM Higgs couplings, consistent with current LHC data. This is not a sharp empirical discriminator from standard SM phenomenology, which also predicts SM-like Higgs couplings within current sensitivities.

Status. The composite-Higgs chiral protection mechanism is *structurally identified* in OI's framework through the singlet-taste chiral symmetry breaking. A fully rigorous derivation — explicit identification of the broken global symmetry's generators, identification of the Higgs as a specific Goldstone direction in the coset, computation of the explicit-breaking corrections to the Higgs mass — is open work. The structural pattern is identified; the theorem-level derivation is deferred.

Derived components and Tier 3 inputs. The framework derives: the singlet-taste chiral symmetry at the substratum level (Theorem 13); the Higgs as the E-projection order parameter of the singlet taste (Theorem 11); spontaneous breaking of the chiral pairing by the Higgs vev in the emergent EFT; the Goldstone bound $\delta m_H^2 \lesssim g^2 f^2/(16\pi^2)$ per coupling; cutoff insensitivity of m_H^2, with no $O(M_{\rm Pl}^2)$ corrections; and the compositeness scale $f \sim M_{\rm Pl}$ (no intermediate scale exists structurally, per [Structure §13.3]). The framework does not derive the Higgs vev $v \sim 246$ GeV itself, which is Tier 3 (solution-specific to the bijection φ). The Higgs mass value $m_H \sim 125$ GeV given v follows from $\lambda(M_{\rm Pl}) = 0$ and RG running (§7.4).

The derived/open boundary is therefore: *the framework derives why m_H is not at the cutoff* — chiral protection ensures cutoff-insensitivity — but *does not derive why v is at the electroweak scale rather than at the compositeness scale.* The first is the substantive content of the hierarchy problem in standard SM phenomenology: why do quantum corrections not push m_H to the cutoff despite the absence of a protecting symmetry? The framework provides a structural answer. The second is a residual question about the Higgs vev's specific magnitude, which the framework treats as Tier 3 alongside other solution-specific parameters such as CP phases and generation-specific Yukawa magnitudes.

Quantitative content of the partial resolution. Standard SM without chiral protection predicts $\delta m_H^2 \sim M_{\rm Pl}^2/(16\pi^2) \sim 10^{37}$ GeV^2 from quadratically-divergent loop corrections, requiring fine-tuning of the bare Higgs mass to ~ 34 decimal places to obtain $m_H^2 \sim (125\,\text{GeV})^2$. The framework's chiral protection mechanism predicts $\delta m_H^2 \sim g^2 f^2/(16\pi^2)$ from explicit-breaking corrections only, with no quadratic-divergence dependence on the cutoff. The two predictions differ by a factor of $\sim (f/v)^2 g^{-2} \sim 10^{34}$ in the size of the required fine-tuning. The framework's claim is that m_H does not require 34 decimal places of fine-tuning to be at the electroweak scale; it requires only as much fine-tuning as the Higgs vev's specific value itself. The factor of 10^{34} is the substantive empirical content of "partial resolution" versus "no resolution," even when v is taken as solution-specific input.

Status of v as solution-specific. Some parameters must be solution-specific in any finite-bijection framework — the bijection φ has freedom that is not fixed by structural constraints, and the values realized in nature are features of the specific φ rather than of the structural class. The framework identifies which parameters are structural and which are solution-specific. The Higgs vev $v \sim 246$ GeV sits in the solution-specific category alongside CP phases, generation-specific Yukawa magnitudes, and other parameters whose values are not forced by the derivation chain in §§3–6.

Custodial symmetry is not required at $f \sim M_{\rm Pl}$. Conventional composite-Higgs models at $f \sim$ TeV require an engineered custodial $SU(2)_C$ symmetry to suppress T-parameter / ρ-parameter deviations that scale as $v^2/f^2 \sim 10^{-2}$. At $f \sim M_{\rm Pl}$, $v^2/f^2 \sim 10^{-34}$ and all such deviations are automatically negligible regardless of whether custodial symmetry is present in the framework's structure. The framework therefore predicts no observable composite-Higgs precision-electroweak deviations from SM, consistent with current LEP and LHC data to the 0.1% level, without needing to engineer custodial structure.

Empirical signatures and discriminating from conventional composite Higgs. Standard composite-Higgs scenarios at $f \sim$ TeV predict Higgs-coupling deviations from SM scaling as $v^2/f^2 \sim 10^{-2}$ — potentially detectable at the HL-LHC. The framework's $f \sim M_{\mathrm{Pl}}$ predicts deviations $v^2/f^2 \sim 10^{-34}$, structurally identical to SM at all practical experimental precision. Future Higgs-trilinear coupling measurements at HL-LHC (15% precision, [41]) and FCC-hh (4–8% precision, [41]) will progressively constrain conventional composite-Higgs models. If no deviations are observed up to FCC-hh sensitivity, the framework's high-f stance is the surviving regime; the conventional low-f composite-Higgs hypothesis would be progressively disfavored. The framework's prediction for the Higgs trilinear coupling $\kappa_3 \equiv \lambda_3/\lambda_{3,\mathrm{SM}} = 1 - O(10^{-34})$ is the same as SM to all relevant precision, and this prediction will be tested experimentally over the next 20 years.

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9. Conclusion

This paper has shown that the Standard Model's gauge group, matter content, and discrete symmetries are determined by a finite bijection on a $d = 3$ cubic lattice — itself empirically anchored (the dimension fixed by the §3.2 filters and the cubic Bravais structure forced by the observed gauge group, §4.6; classified as Tier-2 input in §8.3, not derived) — through the lattice wave equation and a chain of theorems running from center independence to anomaly cancellation. The chain is proved end-to-end. Where it makes contact with measured numbers, the agreement is striking: the Cabibbo angle from a single Brillouin-zone distance to 0.04%, the Koide angle from the cubic-group quadratic Casimir to 0.02%, the down-to-strange mass ratio $m_d/m_s = 1/(2\pi^2)$ at 0.56σ, all three PMNS angles within 1.1σ — with $\sin^2\theta_{12}$ matching the November-2025 JUNO measurement at 0.14σ (a retrodiction) — six fermion masses from a single empirical input to better than 1%, and the SM gauge couplings at M_Z to 0.1% and 0.3% (the SU(2) and SU(3) values being retrodictions matched through the threshold parameters, not first-principles predictions; §6.3). For strong CP the paper derives $\bar{\theta} = 0$ within the lattice construction, with no axion required, and then shows in §5.5 that the derivation cannot be carried to the Standard Model without also removing CKM CP violation — so the strong CP problem is not dissolved here, pending the four conditions of §5.5.

The framework is sharply falsifiable by experiments now in progress or under construction. JUNO's first results (November 2025) are matched at 0.14σ, and the subsequent post-JUNO global fit at 0.07σ, by the parameter-free value $\sin^2\theta_{12} = 1/3 - 1/(4\pi^2)$ — a retrodiction; sub-percent precision over the 30-year design lifetime provides the pre-registered forward test at the level required to distinguish it from competing tribimaximal variants. The n2EDM experiment at PSI is targeting neutron EDM sensitivity at $10^{-27}\, e \cdot \mathrm{cm}$; the framework predicts exactly zero. Neutrinoless double beta decay searches will test the Majorana-neutrino prediction. The continuing absence of any new gauge bosons, fundamental scalars, or supersymmetric partners at the LHC and its successors corroborates structural predictions of the framework, which forbids all such states.

The cosmological application of the same framework — including the derivation of $\hbar$ from the cosmological horizon, the Bekenstein-Hawking area law including the 1/4 coefficient, the dissolution of the cosmological constant problem, and the prediction of dynamical dark energy in Type II running-vacuum functional form (structural prediction; magnitude not derived at theorem level but consistent with the Bertini et al. 2025 direct DESI-DR2-era fit at 2σ) — is developed in [GR]. The substratum-level reconstruction theorem, the substratum gauge group, and the synthesis claim that quantum mechanics, general relativity, and the arrow of time descend as three projections of the same (S, φ) are developed in [Substratum]. Together, this four-paper synthesis ([Main], [SM], [GR], [Substratum]) makes a single claim: the Standard Model, general relativity, and the arrow of time are three projections of one finite deterministic construction — the visible-sector projection developed here, the boundary-level thermodynamic limit developed in [GR], and the horizon-clock-plus-observer-selection projection developed in [Substratum §5.4]. The fifth core paper [Structure] articulates the framework's two-dimensional hierarchical structural realism and develops its relationship to other unification programs through universality classes of embedded observers.

Appendix A: The Trace-Out as a Jordan-Chevalley Projection

This appendix proves that the OI trace-out extracts the semisimple part of the evolution matrix's Jordan-Chevalley decomposition, erasing the nilpotent monodromy. Results are stated for the scalar wave equation on a ring of L sites over $\mathbb{F}_q$; the multi-component extension incorporates the gauge group.

Setup

The wave equation $x(n, t+1) = x(n-1, t) + x(n+1, t) - x(n, t-1) \pmod q$ has phase space $(\mathbb{Z}/q\mathbb{Z})^{2L}$ and evolution matrix $F = \left(\begin{smallmatrix} 0 & I \\ -I & A \end{smallmatrix}\right)$ where A is the circulant adjacency matrix.

Theorem A.1 (Period formula)

For q prime with $\gcd(L, q) = 1$*:* $\mathrm{ord}(F \bmod q) = qL$ *if q is odd, and L if q = 2.*

Proof. For each Fourier mode k, the 2×2 block B_k has characteristic polynomial $t^2 - \lambda_k t + 1$. At parabolic modes ($\lambda_k = \pm 2$), the Jordan form gives $B_k^n = \alpha^n \left(\begin{smallmatrix} 1 & n\alpha^{-1} \\ 0 & 1 \end{smallmatrix}\right)$, contributing order q (or 2q). The diagonalizable eigenvalues contribute orders dividing L. Therefore $\mathrm{ord}(F) = qL$. For q = 2: the nilpotent part is automatically killed. □

Theorem A.2 (Jordan-Chevalley decomposition)

Define $N = (F^L - I)/L \bmod q$. Then: (i) N is nilpotent with $N^2 = 0$ and rank(N) = 2. (ii) $F_u = I + N$ is unipotent with $F_u^q = I$. (iii) $F_{\text{ss}} = F \cdot (I - N)$ is semisimple with $F_{\text{ss}}^L = I$. (iv) $F = F_{\text{ss}} \cdot F_u$ and $[F_{\text{ss}}, N] = 0$.

Proof. F^L has the form $I + LN'$ where N' arises from the Jordan blocks: $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)^L = \left(\begin{smallmatrix}1&L\\0&1\end{smallmatrix}\right)$ contributes a rank-1 nilpotent, and similarly for $\alpha = -1$ (since $(-1)^L = 1$ for even L). So $N = N'$ has $N^2 = 0$ (each block is rank-1 nilpotent on a 2D subspace) and $\text{rank}(N) = 2$ (two parabolic modes). Since $N^2 = 0$: $(I + N)^{-1} = I - N$, giving $F_{\text{ss}} = F(I - N)$. Then $F_{\text{ss}}^L = F^L(I - N)^L = (I + LN)(I - LN + ...) = I \bmod q$ since terms involving LN cancel modulo q (using $N^2 = 0$). Commutativity follows from N being supported on the parabolic eigenspaces, which are F-invariant. □

Verified computationally for L = 4, 6, 8, 10, 12 and q = 3, 5, 7, 11, 13.

Theorem A.3 (q-independence of the Weil-Deligne conductor)

The Weil-Deligne conductor $\mathfrak{f}_{\text{WD}} = \mathfrak{f}_{\text{ss}}(L) + \text{rank}(N) = \mathfrak{f}_{\text{ss}}(L) + 2$ is q-independent when $\gcd(L, q) = 1$, where $\mathfrak{f}_{\text{ss}}(L) = \sum_\alpha (\text{ord}(\alpha) - 1)$ is computed from the eigenvalue orders of F_{ss}, all dividing L.

Proof. F_{ss} has order L; its eigenvalues are L-th roots of unity. For $\gcd(L, q) = 1$, the L-th roots in $\bar{\mathbb{F}}_q$ are isomorphic to those in $\mathbb{C}$, so their orders match. □

L	$\mathfrak{f}_{\text{ss}}(L)$	$\mathfrak{f}_{\text{WD}}$	$\text{NM}^2 = 3L/4$
4	14	16	3.00
6	30	32	4.50
8	70	72	6.00
10	106	108	7.50
12	130	132	9.00

Both $\mathfrak{f}_{\text{ss}}$ and NM^2 are q-independent encodings of the same semisimple eigenvalue data — one via multiplicative orders (integers), one via fourth moments of magnitudes (reals).

Theorem A.4 (Additive decomposition over gauge irreps)

For the K-component wave equation with coupling matrix $M = \text{diag}(\mu_1 I_{n_1}, \ldots, \mu_r I_{n_r})$:

$$\mathfrak{f}_{\text{WD}}(M) = \sum_{i=1}^{r} n_i \cdot \mathfrak{f}_{\text{WD}}(\mu_i)$$

In particular, for $K = 6$ with $M = \mathrm{diag}(\mu_c I_3, \mu_w I_2, 1)$: $\mathfrak{f}_{\mathrm{WD}} = 3\mathfrak{f}_{\mathrm{WD}}(\mu_c) + 2\mathfrak{f}_{\mathrm{WD}}(\mu_w) + \mathfrak{f}_{\mathrm{WD}}(1)$, *with the same multiplicities* $(3, 2, 1)$ *that determine* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$.

Proof. The multi-component evolution matrix is block-diagonal: the a-th component has its own $2L \times 2L$ block F_{μ_a} with eigenvalues depending only on μ_a. Both F_{ss} and N inherit the block-diagonal structure, so $\mathfrak{f}_{\mathrm{ss}}$ and $\mathrm{rank}(N)$ decompose additively. □

Verified for all $(\mu_c, \mu_w) \in \{1, \ldots, q-1\}^2$ at $q = 3, 5, 7$ with $L = 4$. Every case matches exactly.

The projection

The OI trace-out (marginalization over the hidden sector) produces the emergent quantum description, which depends on the coupling eigenvalues μ_k via the NM formula $\mathrm{NM}^2 = 3\langle\mu^4\rangle$ (a consequence of the stochastic-quantum correspondence applied to the wave equation's Fourier decomposition). These μ_k are properties of F_{ss} — the semisimple part of the dynamics. The nilpotent monodromy N contributes nothing to the emergent description: it affects only the off-diagonal Jordan block entries, which are erased by the coarse-graining over the hidden sector.

The trace-out therefore performs the Jordan-Chevalley projection $(F_{\mathrm{ss}}, N) \mapsto F_{\mathrm{ss}} \mapsto \{\mu_k\} \mapsto \mathrm{NM}^2$: it extracts the semisimple part, encodes it via magnitudes rather than orders, and organizes it by the representation theory of the partition. The nilpotent monodromy — genuine mathematical structure present in the full dynamics — is invisible to the embedded observer. This is the precise sense in which physics is the semisimple shadow of mathematics: the observer sees only the diagonalizable spectral data, projected by the trace-out and organized by the gauge group's representation structure.

Code and Data Availability

The lattice Monte Carlo computations reported in this paper were performed using a custom C implementation. Specifically:

- The scalar density renormalization $Z_S(m)$ in §7.5 was computed on SU(3) gauge backgrounds at $\beta_3 = 11.1$, scanning 30 bare masses from $m = 0.005$ to 0.50. Volumes $L = 16$ (30 configurations) and $L = 32$ (50 configurations) were the primary ensembles, with $L = 64$ used for convergence checks. The reported value $Z_S(0.122) = 1.813 \pm 0.001$ is the cubic interpolant of the $L = 32$ measurements. The chiral condensate $\Sigma \approx 0.20$ is the linear extrapolation in the volume-converged region ($mL \gtrsim 3$).

The source code under `papers/oi_lattice_code/`, together with the run drivers and analysis scripts, is archived on Zenodo with concept DOI

10.5281/zenodo.19060318, which resolves to the latest version; specific per-release DOIs are minted at release time. Paths are relative to the repository root, which is also the root of the archived snapshot. The code is also available on GitHub at https://github.com/amaybaum/incompleteness and is released under the MIT License. Gauge configurations at the OI coupling $\beta_3 = 11.1$ and the raw $Z_S(m)$ measurement outputs underlying the §7.5 prediction are available from the author on request.

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